



DESIGN (E) 344 TECHNICAL REPORT

Analogue Signal Generator

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[24/10/2025]

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Abstract

This report presents the design and implementations of an analog signal generator which can produce a sine, square and triangle waveforms. These waveforms should be able to frequency, Amplitude, skew, duty cycle and level adjust. It will only have a single supply voltage, so a powerful stable virtual ground is required to help sink/source current from a low resistive load. The circuit was validated using simulations, custom software as well as laboratory bench measurements. This revealed key practical limitations of the system however the final system did provide flexible accurate waveforms which can be used for laboratory needs.

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List of Abbreviations

- POT** potentiometer
- THD** Total Harmonic Distortion
- TRIM POT** trimming potentiometer
- VG** Virtual Ground
- Power VG** Power virtual ground
- KVL** Kirchoff's voltage law
- Op-amp** Operational Amplifier
- DPDT SW** Double pole, double throw switch
- CSV** comma separated values

List of Symbols

- A* gain
- f_s* sample frequency
- V_{cc}* power supply voltage
- R* resistor
- C* capacitor
- L* inductor
- W* Watt
- Hz* Hertz
- V* voltage
- A* Ampere
- k* kilo =10e3

1 Introduction

A signal generator is a piece of electrical equipment usually used in labs to produce varying periodic waveforms with different shapes, Amplitudes, frequencies and phases [1]. These waveforms are used to test electronic circuits as well as control systems. Among these waveforms the most common three are sine, square and triangle waveforms. Signal generators are also used for calibration, and testing audio equipment. This project aims to design and construct an all-analogue signal generator, which can produce a sine, square and triangle waveform at 2 frequency bounds 100Hz-1kHz and 1kHz-10kHz. The amplitude, up to 20Vpk-pk over a 27 Ω load, should be independent of frequency as well as the duty cycle, 10%-90% and the skew, 0.1-10 as well as level adjusted from -5V to +5V.

It will only be powered by a single 30V supply voltage and will need to make use of a virtual ground to bias the signals. A power amplifier circuit is also needed to drive a very low resistive load of 27 Ω . Specifications and equations are derived from online resources as well as textbook foundations. The Schmitt trigger/comparator, integrator and Wien-bridge oscillator are the primary oscillators in this system, were LTspice simulations as well as practical measurements of final system are done to validate the design.

A custom oscilloscope software was built using Python and it was used as a validation technique on all waveforms. Using simulated, measured and calculated results. The system was proven to work. Limitations arise at high frequencies due to slew rate of op-amps these findings are well documented and discussed. The complete system demonstrated effective signal quality and independent amplitude and frequency control. It also offers more practical insight where simulations fell short. This brought insight on the practical limitations and solutions needed to build a quality signal generator.

2 System description

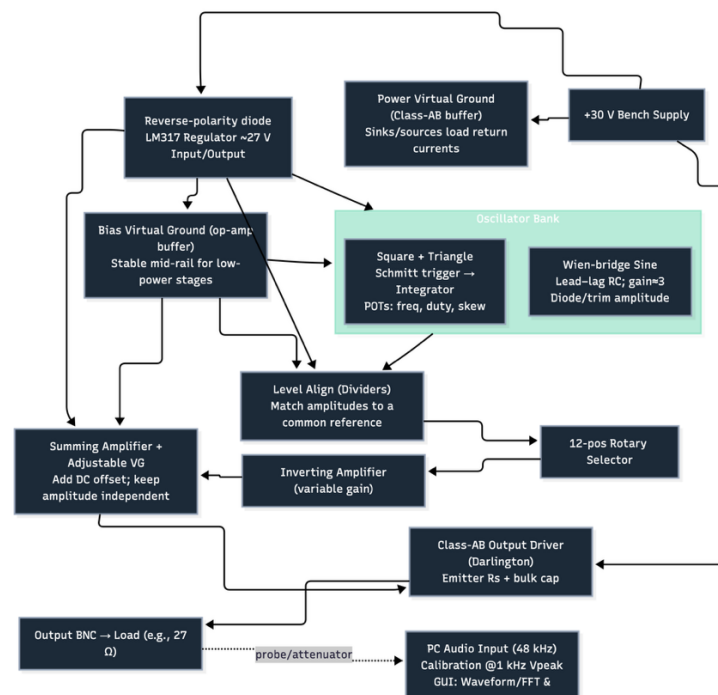


Figure 1: High level block diagram of system.

Power and protection:

- External bench power supply: +30V from the back Banana sockets
- Reverse-polarity diode into a LM317 regulator which generate a +27V rail.
- Local decoupling capacitors to decrease transients for dynamic loads.

Reference rails (virtual grounds):

- Bias virtual ground uses a mid-rail as reference for low-power stages (oscillators and pre-amplifiers).
- Power VG: mid rail reference that can sink/source load return currents.

Oscillator circuits:

- Square and triangle: Uses a Schmitt trigger that drives an integrator. The potentiometers set frequency, duty cycle and skew. DPDT switch swaps frequency range from 100Hz-1kHz and 1kHz-10kHz.
- Sinewave (wien-bridge): RC lead-lag network, with the adjustable gain trimming it to 3. It also includes a diode limiter and another trimmer for amplitude control.

Signal selection & level alignment:

- Inverting amplifier sets overall amplitude range.
- Summing amplifier and adjustable virtual ground adds a DC offset so the single supply system can level adjust to any point in voltage range.

Output driver:

- Class-AB push-pull (Darlington pair) stage provides up to 400mA through a 27Ω load, uses emitter resistors to stabilize output in output stage and a big output capacitor is used to mitigate ripple voltage on the virtual ground.

Users Design:

- Front panel: frequency, duty/skew, amplitude and level controls. Range selection. BNC connector for crocodile clips.
- Rear: power banana sockets.

Software measurements:

- Custom made windows application that reads mic-in at 48kHz, This calibrates with 1kHz V_{peak} signal and can report the Amplitude, duty cycle, skew ,frequency and THD which can be compared with actual oscilloscope results to verify.

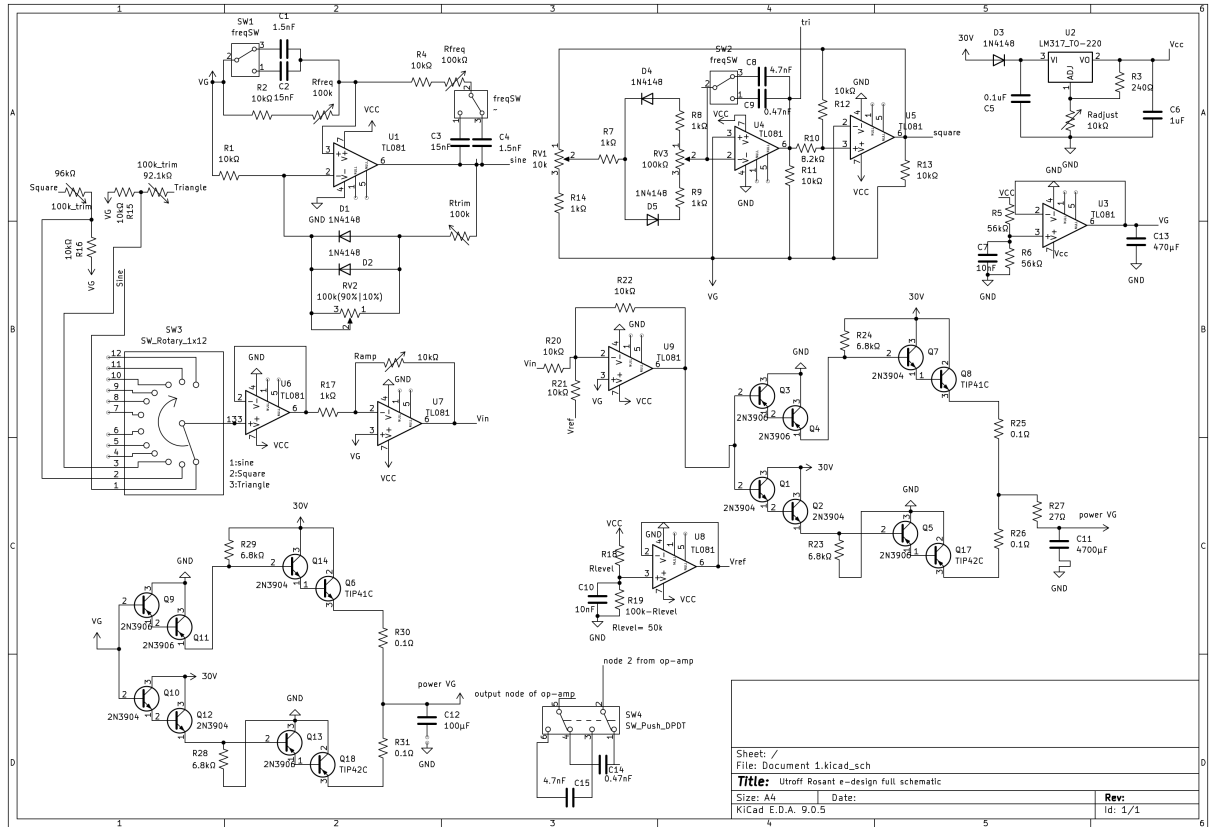


Figure 2 Full system schematic

The full system design is depicted in figure 2. Each sub system is described in detail in section 3. The Wien-bridge is the upper most left design, and its RC feedback system contains 2 switches to switch out the capacitor, effectively changing the frequency ranges. Both switch's combine to a single DPDT switch used in the actual system development. The same switch that is depicted on the left of the Title block in the schematic.

3 Hardware design and implementation

Designing a signal generator to create 3 distinct waveforms. A Sin, square and triangle waveform. That can change amplitude, duty cycle, skew, frequency and level adjustable.

3.1 Hardware block diagram and description of interaction

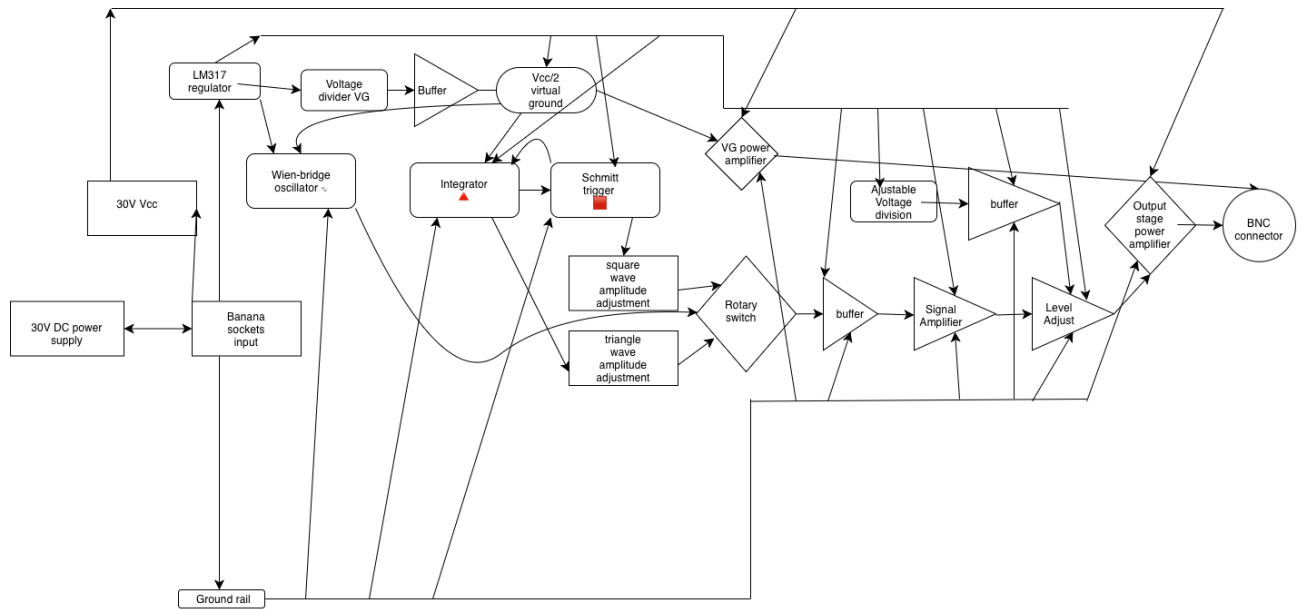


Figure 3 full system hardware block diagram

The system begins from the 30V bench power supply which feeds in through banana sockets into a LM317 voltage regulator, which is an undisturbed rail from any transients. A voltage divider plus buffer generates a stable virtual ground at 13.5V used as the mid-rail to bias all op-amps. The power virtual ground provides a safe return path from the output after the 27Ω load. The three oscillators produce a sine, square and triangle wave using a Wien-bridge for sinewave and a comparator and integrator to make a square and triangle. The three waveforms are routed through an amplification matching stage then through a rotary switch. The stages that follow are a global inverting amplifier, sets the overall gain, then passed to a summing/level adjust stage that adds a DC offset relative to virtual ground. Separating these stages keeps the amplitude independent of level and frequency. The final stage is a Class AB amplifier that provides high current to drive low resistive loads via a BNC connector. The return current gets sunk into the power virtual ground for symmetrical headroom on a single supply.

Using this topology separates low power signal generation and high-power current driving stages providing low distortion at the output.

3.2 Square and triangle wave generator

3.2.1 Square & triangle wave generation

Using an op-amp configured as a non-inverting Schmitt Trigger which changes quickly between Vcc and ground. Due to the op-amps output stage the saturation is about -2V on either side of the rail less than the Vcc. This is due to the transistor voltage drops inside the IS on the output stage. In the Datasheet for the TL081 Op-amp the maximum output swing for low resistance is about ±13V which compares with my practical readings. Using the following online resource, I found for a triangle and square generator I configured my circuit as follows. The R12, R10, R11 and R13 resistor was referenced from this online resource and the POTs and resistors in the integrating path are justified below. [2](Triangle square generator)

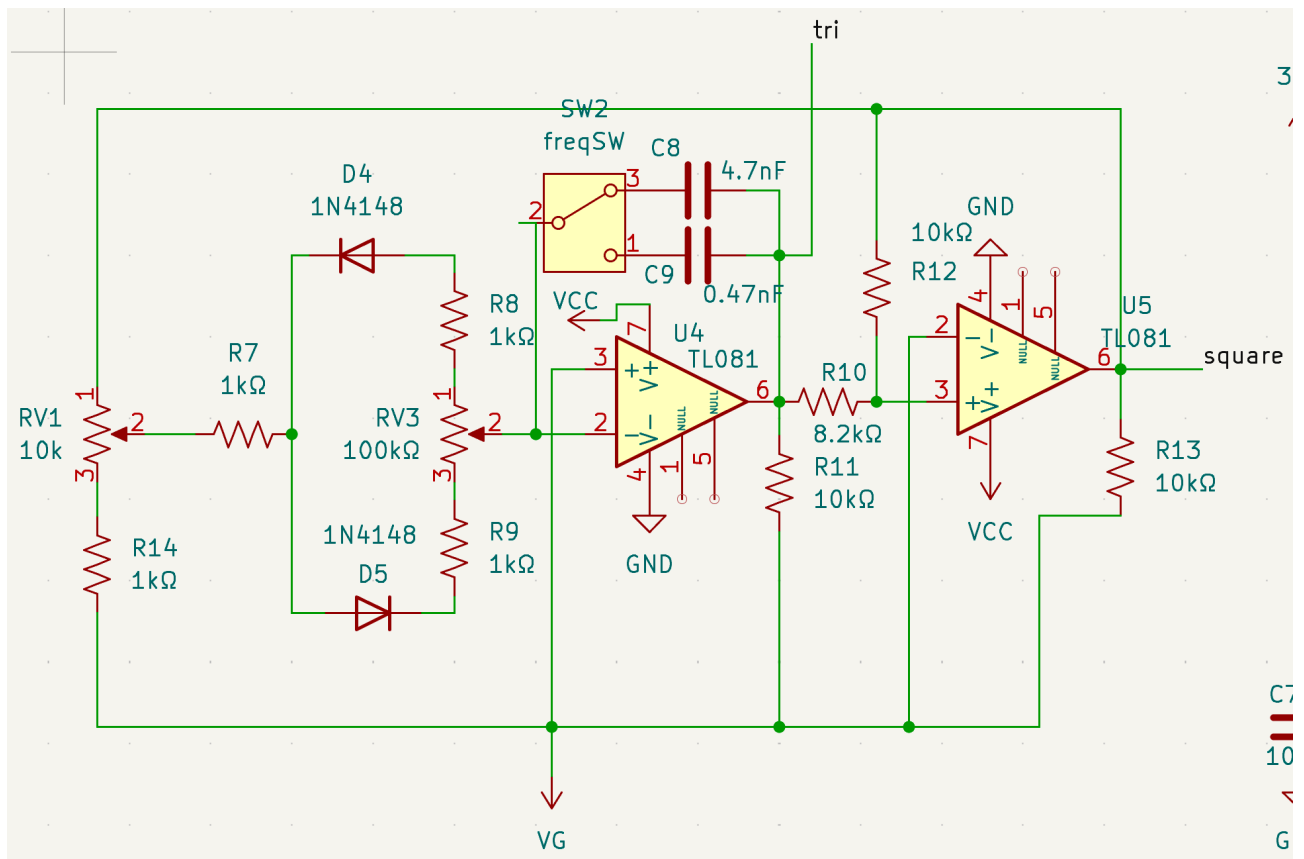


Figure 4 Triangle and Square Generator

As the input signal crosses defined threshold voltage level the comparator output switches between positive Vcc and ground producing a square wave. The threshold voltage is set by a voltage divider. This Voltage divider divides the triangle output by the ratio of R10 and R12 in the above figure 2 and compare it to virtual ground voltage. Every time the voltage of the triangle slope is larger than the virtual ground which is around 13.6V.

The triangle is generated through an Integrator made up of a feedback capacitor from the negative node of the op-amps input to the output of the op-amp. A capacitor will charge linearly if it is fed with a constant current source, the resistor network before it. The output of the square wave generator drives the triangle generator. When the square wave is high, the capacitor charges at a steady-rate which outputs a steady-ramp on the output. When the square wave transitions low the capacitor discharges and the ramp decrease in the negative direction. The triangle wave is driven to $V_{upperthres}$ down to $V_{lowerthres}$

3.2.1.1 Reason for resistors

The R14 resistor helps set a minimum resistance and limit the range of the adjustment for RV1, Improving stability and predictability of frequency. It is always a fixed resistance value. It also helps to control the charging of capacitor. The R7 resistor helps with the discharging working in unison with the 10kΩ POT and the D4 diode. R13 and R11 resistors help with setting up the amplitude of the oscillators. It also is always a low fixed resistance value. A 1kΩ resistor is chosen for the R13, R11 and R14 resistors.

3.2.2 frequency of generator

The frequency is determined by the time it takes to charge. If the resistance is increased the capacitor is charged at a slower rate thus the frequency decreases therefore reduce the resistance it will increase the charge rate and the frequency. These are a few formulas for the frequency that are derived below to get the theoretical frequency range by change a 10kΩ potentiometer. Using R10 and R12 the comparator threshold is derived.

Equation 1 comparator threshold

$$\beta = \frac{R10}{R12} = \frac{10k\Omega}{8.2k\Omega} = 1.2195$$

$$Vt = \beta \times Vcc$$

Where:

β : The Schmitt trigger threshold voltage

Vcc: Input voltage for the op-amp

Assume diode turn-on voltage is 0V.

The voltage in the integrator path is a fraction of the Vt. Using the following voltage division to get the voltage at node 2 of RV1.

Equation 2 voltage division fraction

$$k \uparrow (p) = k \downarrow (p) = k(p) = \frac{(1-p)Rp + R14}{R7 + Rp + R14} = \frac{(1-p) \times 10k + 1k}{12k} \quad p \in [0,1]$$

Rp = 10kΩ POT

Rs = 100kΩ. POT

C= 4.7nF or 0.47nF depending on the frequency range

To get the change in the potential from the lower threshold to the upper threshold the voltage is a indication of the potential difference from -Vsat→+Vsat

$$\Delta Vup = Vthres+ - Vthres- = (Vt+) - (-Vt) = 2Vt$$

Since time is defined as the distance over the speed. Speed in this case is the voltage over time. Voltage in this case is equivalent to distance charge must travel.

$$\left| \frac{dV}{dt} \right| = \frac{Ic}{C} = \frac{Vcc}{C(Rin \uparrow \downarrow)}$$

$$Vrv1 = k+(p) \times Vcc$$

For all signals assume duty cycle is at 50% therefore $k \uparrow = k \downarrow = k$

$$t = \frac{\Delta Vup}{\left| \frac{dV}{dt} \right|} = \frac{2VtCRin}{kVcc}$$

Total time = T = t↑+t↓ [10]

$$T = \frac{2VtC}{k(p)Vcc} (Rin \uparrow + Rin \downarrow)$$

$$Rin \uparrow + Rin \downarrow = 2R7 + 2pRp + R8 + R9 + Rs = p \times 20k + 104k$$

Equation 3 frequency for square and triangle wave

$$f(p) = \frac{1}{T} = \frac{k(p)}{2C} \times \beta \times \frac{1}{2R7 + 2pR_p + R8 + R9 + R_s} = \frac{k(p)}{2C} \times 1.2195 \times \frac{1}{p \times 20k + 104k}$$

If p is equal to 1 the resistance is at a maximum in the denominator, so the frequency is at its lowest. If $p = 0$. The denominator is at minimum, so the frequency is at its highest. Since all the resistance values stay constant as well as the potentiometer sweep range and that the equation is linear. To increase the frequency, range from 100Hz-1kHz to 1kHz-10kHz only the Capacitor needs to be changed by a factor of 10. Initially the Capacitor is 4.7nF for [100,1000]Hz and 0.47nF for [1000,10000]Hz.

3.2.2.1 Double pole double throw switch

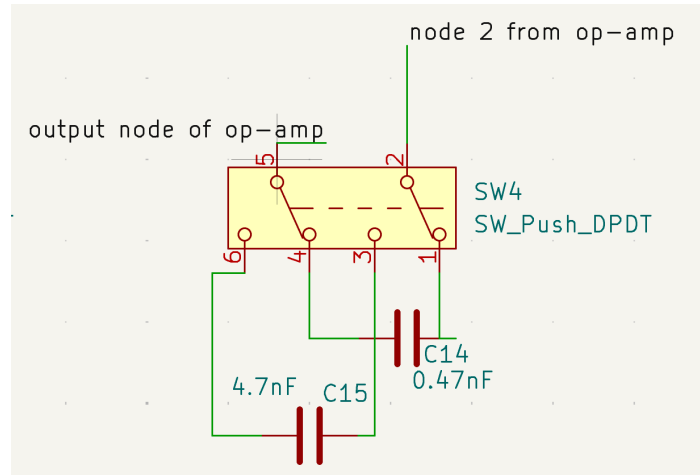


Figure 5 configuration of a DPDT switch in triangle/square circuit.

Using a double pole double throw switch with either Capacitor connect on either side frequency selection can be achieved with a flick of a switch when the switch actuator is all the way to the left then node 1 and 4 in figure 3 is connected. The input of the node is from the non-inverting side of the op-amp and the output is connected to node 5 in figure 3. If the switch actuator is flipped all the way right the 4.7nF capacitor is active, node 2 connects to 3 and node 5 connects to 6, and the frequency range switches.

$$f(0) = 1143\text{Hz} ; f(1) = 87.187\text{Hz for } C = 4.7\text{nF}$$

$$f(0) = 11430\text{Hz}; f(1) = 871.87\text{Hz} = \text{Hz for } C = 0.47\text{nF}$$

3.2.3 LT-spice simulations

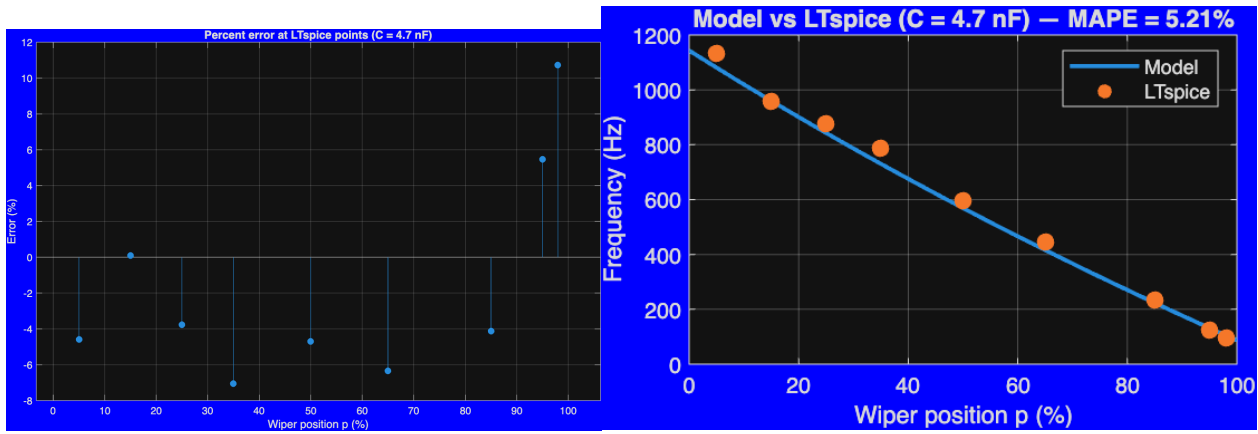


Figure 6 100-1kHz frequency comparison with frequency equation and spice

Table 1 100-1kHz frequency comparison with equation and spice

p	Frequency on LT-spice (C=4.7nF)Hz	Frequency equation(p/100 and C=4.7nF) Hz	Error(%) (avg=5.2%)
5	1133	1081.1	4.57
15	959	959.87	0.091
25	876	843.07	3.758
35	786	730.48	7.06
50	597	569.01	4.69
65	444	415.81	6.35
85	233	223.37	4.13
95	125	131.84	5.474
98	94.8	104.96	10.72

From table 2 and figure 3 it is seen that the equation derived for the triangle and square oscillator circuit is a good approximation. The diode voltage drop was not considered and was assumed to be ideal diodes with 0 turn-on voltage. This could add to the error that is seen in the equation; however the approximation is quite accurate as the average error is approximately 5% compared to the spice graphs. At middle frequencies. When the wiper is 50% the spice follows the equation closely. This percentage indicates that the theoretical model is robust for predicating circuit performance under the simulated condition and validates the analytical approach to select appropriate components for the circuit. All the calculations are done with 50% duty cycle and 1skew

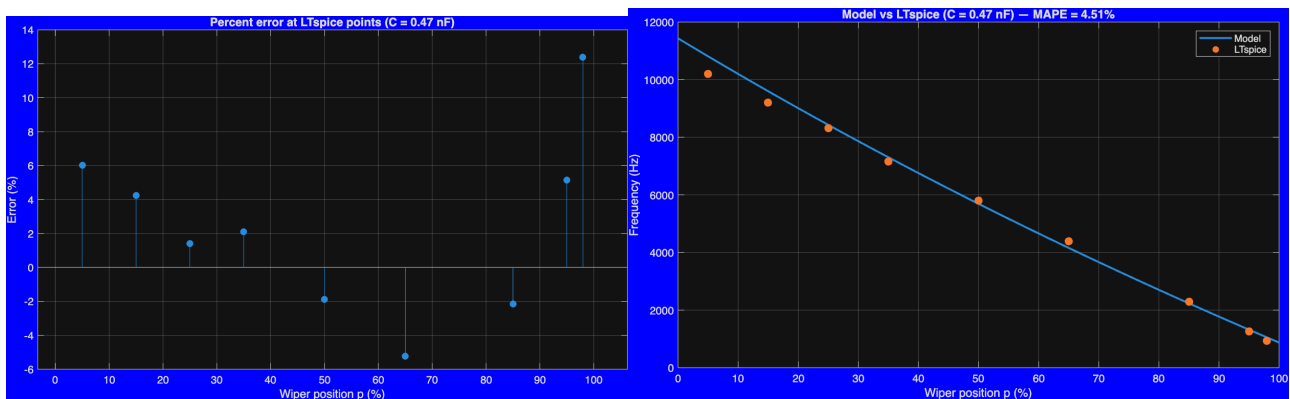


Figure 7 1kHz-10kHz frequency comparison with frequency equation and spice

Table 2 1kHz-10kHz frequency comparison with equation and spice

p	Frequency on Lt-spice (C=0.47nF)	Frequency equation(p/100 and C=0.47nF) Hz	Error(%)
5	10198	10811	6.01
15	9209	9598.7	4.23
25	8315	8430.7	1.39
35	7154	7304.8	2.11
50	5800	5690.1	1.895
65	4388	4158.1	5.24
85	2283	2233.7	2.16
95	1254	1318.4	5.14
98	934	1049.6	12.38

The same can be said for table 3 and figure 4. The equation approximates the spice waveform very accurately with a lower average error of 4.5%. This indicates that this equation is more accurate for higher frequencies.

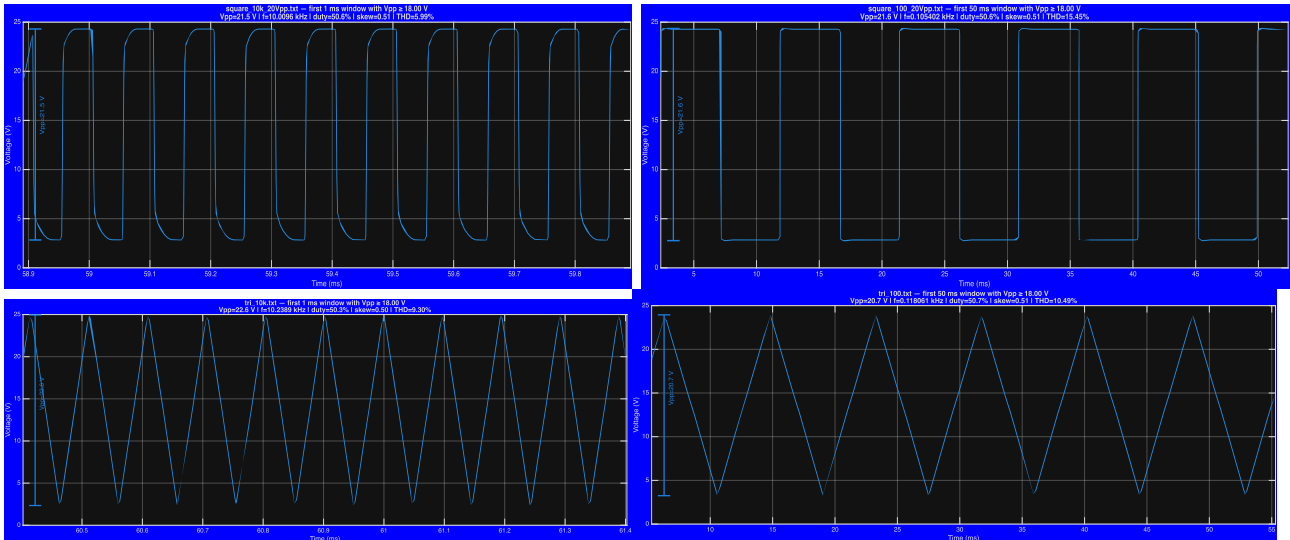


Figure 8 triangle and Square at 100Hz and 10kHz

In figure 4 the triangle waveforms display high linearity for both the rising and falling edge, this indicates a good integrator performance and minimal distortion. The charge and, discharge, RC and diode circuits are all well balanced which results in a symmetrical triangle. There are no rounding or curves at the peak of the triangle therefore there is no SR limitation.

The square waveforms show fast transitions between +Vsat and -Vsat. This is ideal for a good Schmitt trigger. The sharp vertical edges confirm that the op-amp slew rate is sufficient for the operating frequency except at high frequencies that above 10kHz. In both signals there is no server distortion, and all the signals are based around the same mean as well as a constant amplitude.

3.2.4 duty cycle and skew of generator

The duty cycle is determined by using opposite polarity diodes in the resistive path to change the high and low charge time of the capacitor. This creates an asymmetry between charge and discharge current paths and this adjusts the duty cycle of the output square wave. The capacitor charges through a resistor and diode path when the output of the square wave is high. As soon as the output of the square wave transitions low, the capacitor discharges through a different resistor and diode path. If resistance path

of the charge and discharge are biased unequally by using a potentiometer. The charging time can change which increases the corresponding half of the duty cycle. The Derivations for the Duty cycle equation is in the appendix but below is the formula to calculate the duty cycle of the square wave as well as the skew of the triangle wave.

Equation 4 duty cycle equation

$$D(p,s) = t \uparrow \frac{1}{t \uparrow + t \downarrow} = \frac{R7 + pRp + R8 + sRs}{2R7 + 2p + R7 + R9 + Rs} = \frac{p \times 10k + s \times 100k + 2k}{p \times 20k + 104k} \quad p, s \in [0,1]$$

For the skew the following formula will have the skew adjust from 0.1-10

Equation 5 skew equation

$$S(p,s) = \frac{t \uparrow}{t \downarrow} = \frac{R7 + pRp + R8 + sRs}{R7 + pRp + R9 + (1-s)Rs} = \frac{p10k + s100k + 2k}{p10k + (1-s)100k + 2k} \quad p, s \in [0,1]$$

CHOOSE A P VALUE 0 AND 1 THEN TEST MAX DUTY WITH EQUATION ONLY DO 5 AT 100 AT 10K

Rp: 10kΩ; Rs: 100kΩ

The reason why the setup for duty cycle adjustment works is because the network is places after the diode which directs the current flow. It adds resistance to one half-cycle while subtracting the same amount from the other side. Since the period is dependent on the Resistance seen on the charge and discharge cycle it remains the same because of the potentiometer. The frequency potentiometer changes the total current flowing into the path thus decreasing how fast the capacitor charges and discharges. therefore, a 10kΩ POT is used for the frequency as it can let through more current and provide higher frequencies that is needed. The 100kΩ duty cycle potentiometer will allow the extreme skew required by the triangle as it can limit the charge and discharge path a lot more than a 10kΩ POT.

In practice at high frequencies, >10kHz there is a coupling with the duty cycle and frequency. Over 90% skew at high frequency causes the triangle to saturate and this can be due to the slew rate of the op-amp

3.2.4.1 spice simulations

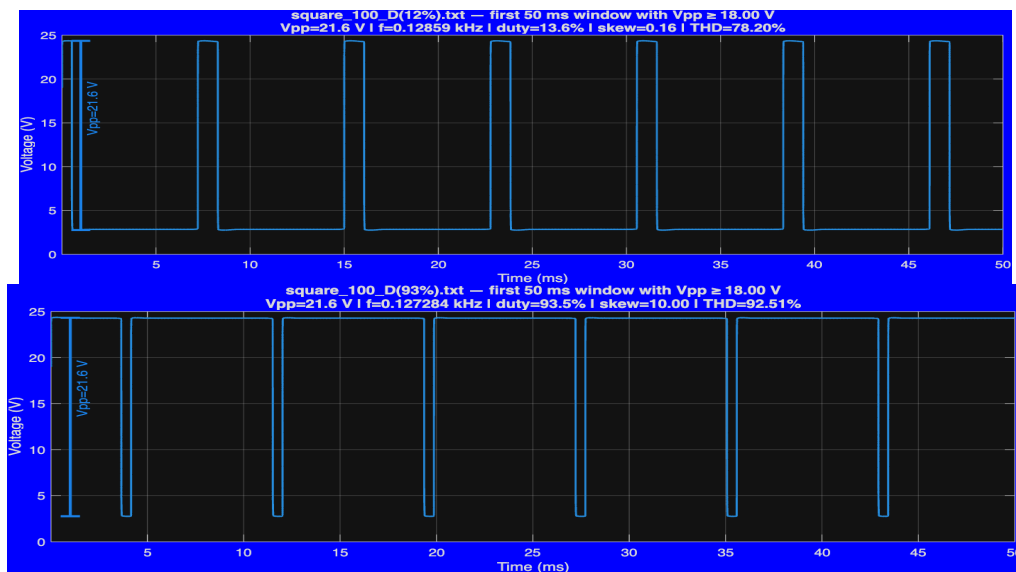


Figure 9 Duty cycle at 10% and 90% for 100Hz Square

In Figure 9 the duty cycle of a 100Hz square wave is tested. It demonstrates precise control over both the duty cycle with the amplitude remaining constant, with clean fast transitions from $-V_{sat}$ to $+V_{sat}$. This confirms that the op-amp is not limited by the SR at 100Hz.

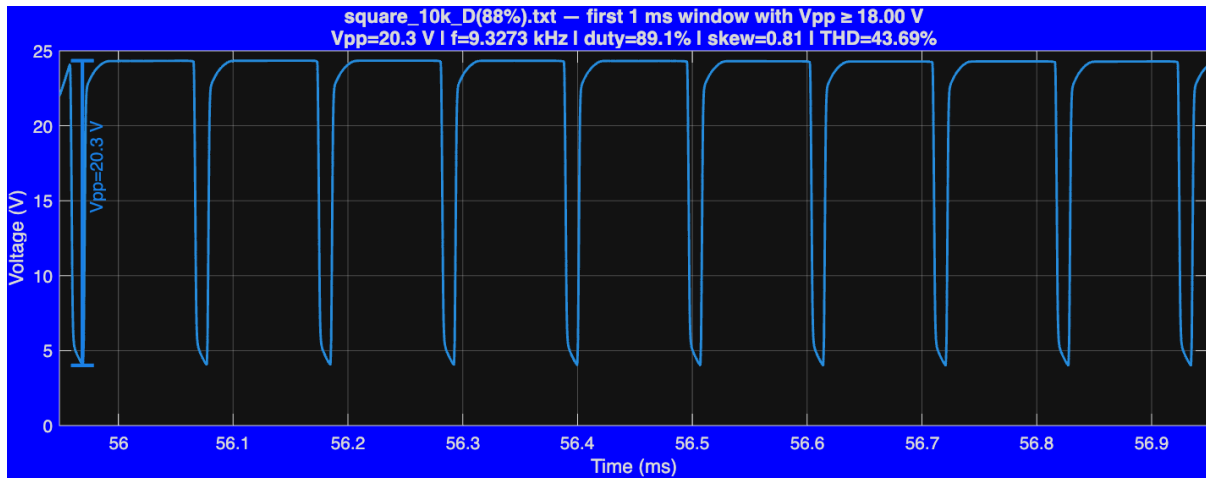


Figure 10 square wave at 10kHz with 90% duty cycle

In Figure 10 the slew rate of the op-amp starts to limit the fast switching from $-V_{sat}$ to $+V_{sat}$. This demonstrates the limitations of the op-amp. The amplitude does decrease from 21.6Vpp from figure 6 to 20.3Vpp. The frequency also drifts from 10kHz. Showing the limitations of the design at high frequencies to precisely control the duty cycle and keep it uncoupled from the frequency.

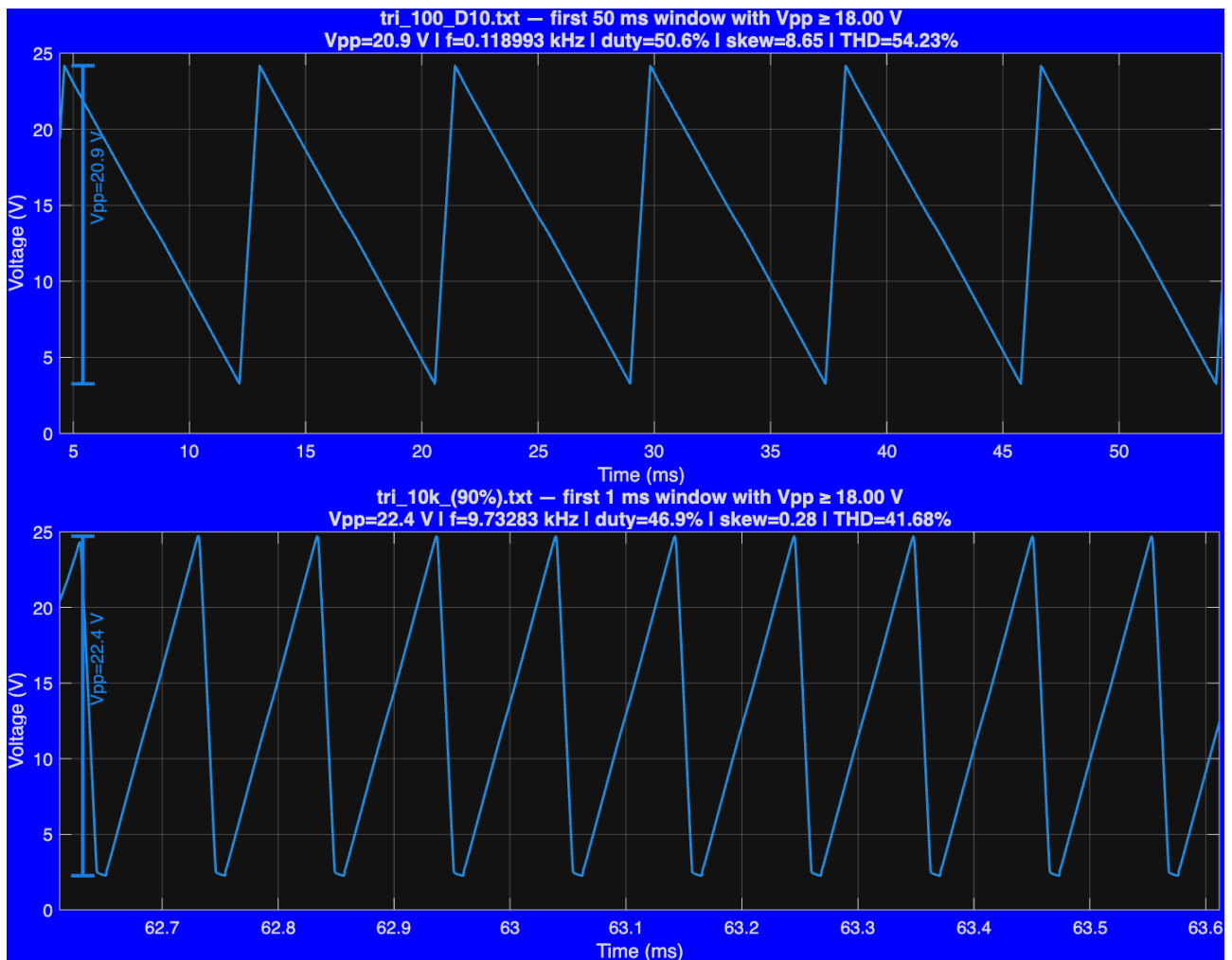


Figure 11 triangle wave at 10kHz and 100Hz with 0.1 skew and 10skew

The triangle wave can achieve about 0.2 skew at 10kHz before it starts to clip at the rails. Amplitude does stay constant compared to Figure 11 for the 10kHz waveform, however there is non-linear distortion at the peaks and the troughs. One reason for the non-linear distortion at the bottom of the wave is due to both diodes turning off simultaneously causing an interrupt in the current flow into the capacitor. When the control voltage is set in such a manner that the transition point is very close to the diode turn-on voltage and since it is a high frequency. Any interruption on the integrator path can cause non-linearity. That's why that non-linearity is not seen in the 100Hz waveform as the charge discharge interval is a lot longer and the capacitor is not as sensitive.

Mathematical derivations for frequency, duty cycle, skew and gain are validated against spice to to validate low errors of equation as well as validate that the system would work. When a practical design is made.

3.3 Sine wave generator

3.3.1 What are oscillators

The way a operational amplifier in feedback works as the output signal is feedback to the input signal on the inverting or non-inverting input causing and the difference of the tow signals is amplified. For oscillation to occur the output is feedback and summed inphase with the input allowings the signal to self sustain. The Barhausen Criterion is required for this. It states that the loop gain, open loop gain x frequency factor, must be greater than one. The total phase shift around the loop is an integer multiple of 360° .

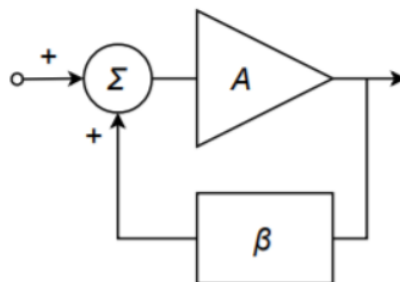


Figure 9.2.1: Positive feedback.

Figure 12 Feedback system [3]

When designing oscillators, we do not need to provide a “start-up” signal. There should be enough energy in the start-up transient analysis. Usually, the oscillation frequency is stored within the turn-on transient and noise signal. The oscillation signal will start to increase as time progresses but will reach a saturation point due to amplifier clipping. To control this, we can design a gain roll-off before it starts clipping. Selecting a resonant frequency and then designing the filters to attenuate signal frequencies above and below the resonant frequency is advised.

3.3.2 The Wien Bridge Oscillator

To Wien-bridge Oscillator. A simple lead/lag circuit is used with resistors and capacitors, creating a frequency-sensitive voltage divider. At this point the resistors and capacitor values are matched. At low frequencies, the capacitive reactance is infinite; therefore, the series Capacitor appears as an open circuit, with a zero-output voltage. At high frequencies, the capacitive reactance approaches zero, and the parallel capacitor is seen as a short circuit to ground.

In addition to amplitude response the Wien-bridge is configured so that at one unique frequency, oscillation frequency. The output from the network has zero phase shift relative to the input. This comes down to the fact that the feedback signal is perfectly in-phase with the amplifier input. The circuit requires positive feedback at just the right phase to sustain the sine wave. At any other frequency, the phase shifts introduced by the reactive elements of the circuit prevent sustained oscillation. This ensures that stable undistorted output is generated only at the selected resonant frequency.

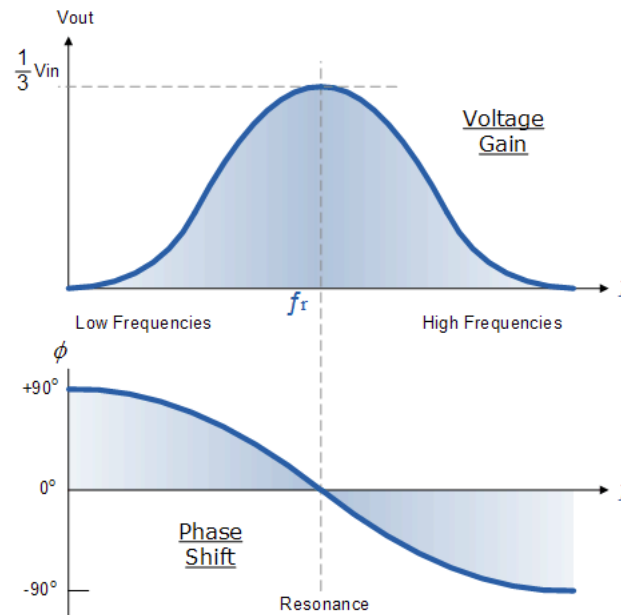


Figure 13 oscillation frequency requirements and phase shift.

At very low frequencies it is seen that the phase shift is positive and for high frequencies the phase shift is negative. In the middle there is a resonant frequency where it is exactly in phase.[4]

The feedback loop functions as a lead-lag network. At low frequencies, the reactance of the series capacitor is high, acting as an open circuit. It acts as a high-pass filter. The Shunt capacitor acts as a low-pass filter because the reactance becomes negligible, causing the capacitor to be seen as a short circuit. There lies a resonant frequency between these ranges, and this point sets the oscillation frequency. [3]

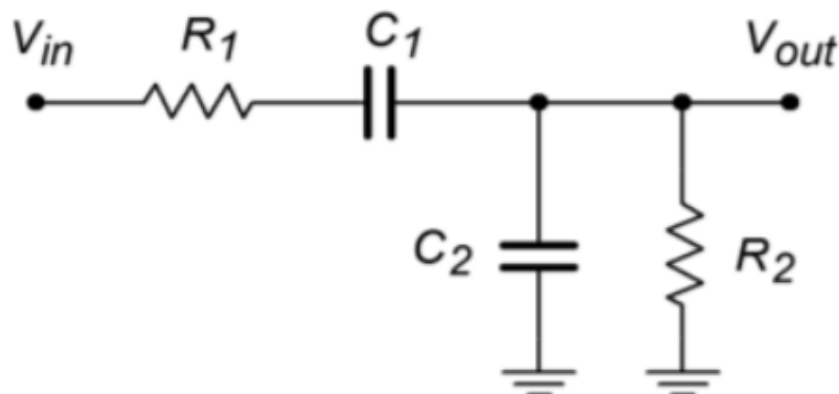


Figure 14 lead lag circuit[3]

$$\beta = \frac{Z_2}{Z_1 + Z_2}$$

Where $Z_1 = R_1 - jX_{C1}$, $Z_2 = R_2 || -jX_{C2}$ and $X_C = 1/\omega C$

$$Z_2 = \frac{R_2}{1 + \frac{R_2}{-jX_{C2}}} = \frac{R_2}{1 + j\omega R_2 C_2}$$

$$Z_1 = R_1 - \frac{j}{\omega C_1}$$

Substitute into β

$$\beta = \frac{R_2}{R_2 \left(1 + \frac{C_2}{C_1}\right) + R_1 + j(\omega R_1 R_2 C_2 - \frac{1}{\omega C_1})}$$

Using the imaginary part of the above equation and where $R_1=R_2=R$ and $C_1=C_2=C$. We get the following

$$f = \frac{1}{2\pi RC} \text{ and } \omega = \frac{1}{RC}$$

Using the Real part of β we determined that the forward gains needs to be exactly 3 to maintain oscillation. Thus, the frequency of the oscillation can be set using a dual potentiometer.

$$\beta = \frac{R_2}{R_2 \left(1 + \frac{C_2}{C_1}\right) + R_1} = \frac{R}{3R} = \frac{1}{3}$$

The sine wave Generator used was a Wien-bridge oscillator. The Wien-bridge oscillator is configured as follows.

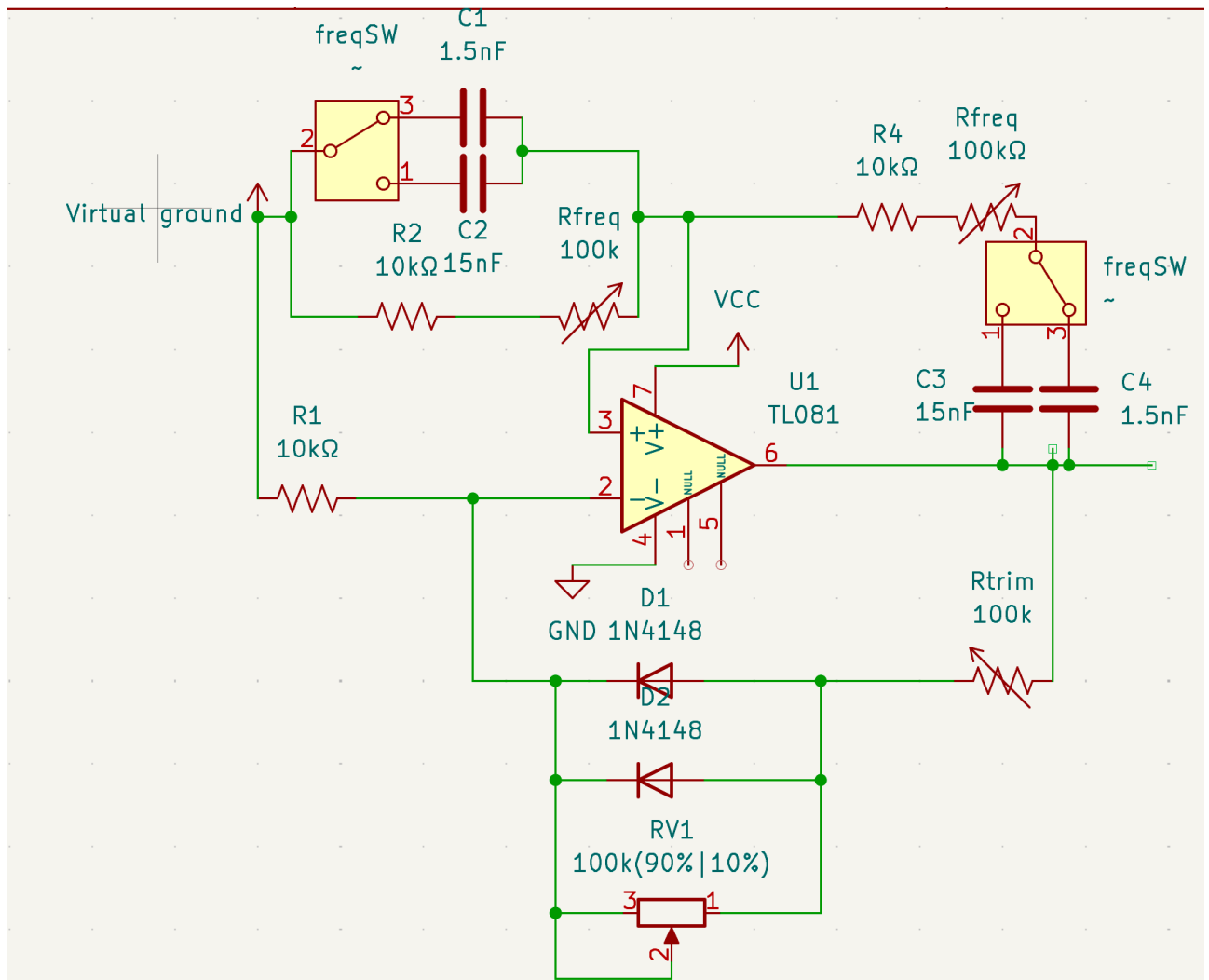


Figure 15 Wien-bridge oscillator, design inspired from [11]

3.3.2.1 Use of Diodes

The reason for the diodes is as an amplitude stabilization mechanism for the output sine wave. The function of the diodes is to provide a nonlinear feedback path. This path automatically limits the gain of the amplifier stage as the output amplitude is increased during oscillation.

The gain of the amplifiers is set slightly above the minimum that is needed for the oscillation, therefore the sinewave rapidly grows in amplitude. As the output approaches the diodes forward voltage of 0.65V. The diodes turn on and effectively reduce the feedback resistance which in turn reduces the amplifier's gain. This dynamic adjustment of the gain stabilizes the amplitude at a relatively constant level, keeping the circuit oscillating without the output being distorted or clipping.

Diodes are also cost effective and also have a fast response time since they react immediately to the signal amplitude changes, helping to control the output even for high frequencies, however diodes are still non-linear and its transition from off to on is relatively abrupt. This can introduce harmonic distortion, especially if the gain is set to high or the output amplitude is allowed to approach the saturation point.

Equation 6 frequency equation for Sinewave [9]

$$f = \frac{1}{2\pi RC}$$

3.3.2.2 Capacitor and resistor choices

To tune the Wien-bridge in order to not have a short circuit in the lead-lag circuit. Resistors in series with the potentiometer to limit the lower bound of the frequency range. Since a 1.5nF for C1,C4 and 15nF for C2,C3 are chosen. The series resistor has to be designed around this. The dual pot provided is 100kΩ. Since the frequency has an inverse relationship to the change in the resistance and the capacitance remains constant. This means that the lower the resistance the higher the frequency. Since a 0Ω resistance will cause a short circuit and the oscillator will stop oscillating. A series resistor has to be designed to mitigate this. Choosing the frequency at its highest required value, 10kHz, a series resistor is calculated as follows.

$$f = \frac{1}{2\pi RC}$$

$$\text{Rearrange the equation to } R = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 10k \times 1.5n} = 10.6k\Omega$$

Using 1.5nF capacitor value and the desired frequency of 10kHz.

As seen above the required resistance is 10.6kΩ, however this is not a E12 resistor value. The ATP allows for the frequency to go 20% below the required frequency bound. The Practical frequency bound is [800-12kHz]. The total resistance seen in the lead-lag network is the sum of the series resistor and the potentiometer with the following equation.

$$R = R_s + R_{pot}$$

The table below describes the different frequency for potentiometer extremes. The resistor value varies from 10kΩ-110kΩ. The bottom resistance is attained when the potentiometer is at 0Ω. This will change the total resistance for the above equation to be 10kΩ. Plugging that into the equation 1 will result in the frequency of , If the potentiometer is at maximum the total resistance seen is 100kΩ which results in a frequency of x. Changing the magnitude of the capacitor by a factor of 10 will decrease the frequency range by a factor of 10 as well.

Where:

$R_s = 10k\Omega$

$R_{pot} = [0-100k\Omega]$ dual potentiometer

Table 3 Verifying sinewave frequency formula

Configuration of Capacitance and Resistance	Frequency (theoretical)	Frequency (Hz)	Frequency (Hz) (LTspice simulation)	Error (%)
C=1.5nF R= 10kΩ	10610		10472	1.3
C=1.5nF R= 110kΩ	964.57		974	9.6%
C= 15nF R =10kΩ	1061.03		1085	2.21%
C=15nF R=110kΩ	96.45		98.23	1.81

As seen the frequency at the extremes for the potentiometer match's the frequency in LTspice simulation, this shows us that equation 1 works as well that the 10kΩ series resistor used will keep the frequency in the desired range for each extreme of the pot.

3.3.2.3 better amplifier gain

Using a trim potentiometer in feedback path of the wien-bridge oscillator allows for precise adjustment of the amplifier gain- This is a critical factor for minimizing THD. This allows for the gain to be tuned for perfect oscillation while the circuit is turned on. With careful adjustment the output amplitude can be fined tuned to exactly 3 to produce a cleaner sinewave.

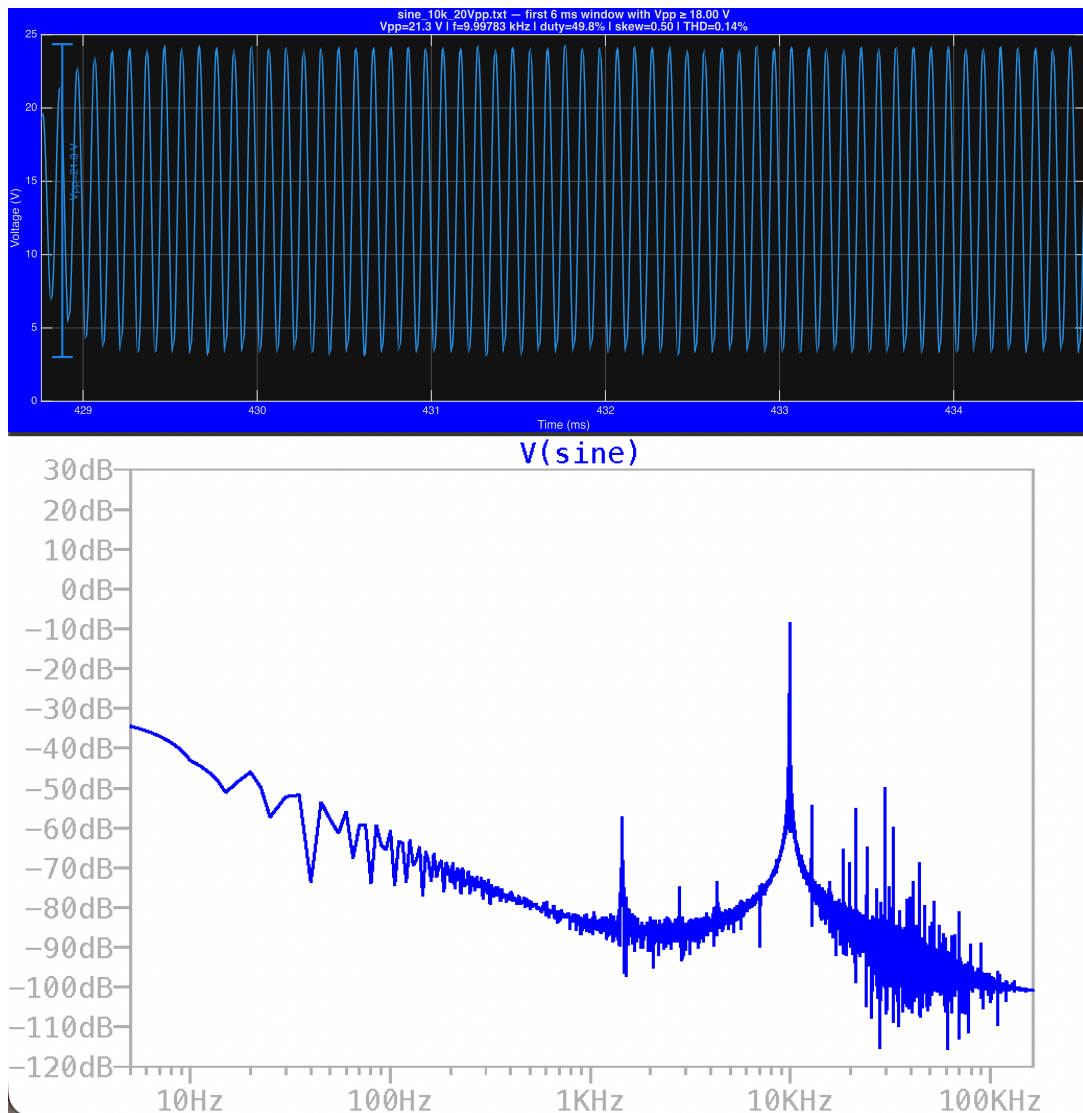


Figure 16 LT-spice 10kHz sinewave

Above is the FFT and sinewave at 10kHz. There is a lot of noise at the higher frequency spectral components. There is also consistent amplitude throughout the waveform which confirms stable oscillating operation. The strong isolated peak at 10kHz. Indicates that the signal energy is concentrated at the fundamental frequency. The lack of major secondary spectral components and the steep drop off into the noise floor after the fundamental harmonic confirms that the THD is low.

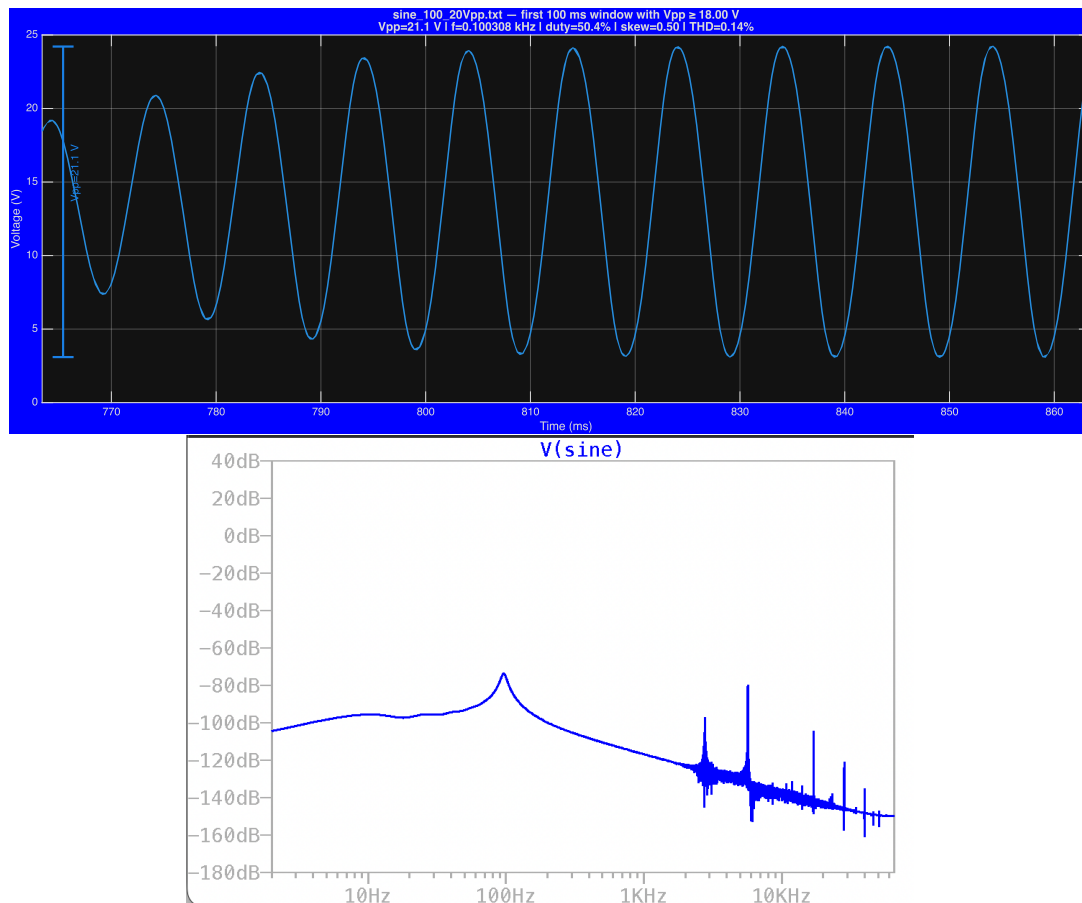


Figure 17 LT-spice sinewave and FFT at 100Hz

In figure 9, the FFT and sinewave from LTspice is presented. At the fundamental harmonic ,100Hz, there is almost not distortion between subsequent harmonics thereafter. This indicates that majority of the output signals power is transmitted in the fundamental harmonic. The steep amplitude drop as the frequency increases above 100Hz shows that the output is nearly perfect sinusoid. There are no significant harmonics at 200 or 300 Hz. The high frequency Harmonics seen could be due to non-linearity in the circuit. This is a very clear indication of a low THD.

Using the spice Fourier analysis. Four parameters in LTspice `.four 100 V(sine)` command. The actual THD can be calculated from the sinewave's FFT. Below is the figure results of the different THD's in LTspice.

```

Fourier components of V(sine)
DC component:13.6788

```

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	9.700e+1	8.404e-1	1.000e+0	59.33°	0.00°
2	1.940e+2	2.544e-2	3.027e-2	-43.50°	-102.83°
3	2.910e+2	8.594e-3	1.023e-2	51.25°	-8.08°
4	3.880e+2	3.670e-3	4.367e-3	-72.01°	-131.34°
5	4.850e+2	3.405e-3	4.052e-3	136.52°	77.19°
6	5.820e+2	1.589e-3	1.891e-3	-105.43°	-164.76°
7	6.790e+2	2.127e-3	2.531e-3	138.85°	79.52°
8	7.760e+2	1.375e-3	1.636e-3	-97.00°	-156.33°
9	8.730e+2	1.291e-3	1.536e-3	106.79°	47.46°

```

Partial Harmonic Distortion: 3.273457%
Total Harmonic Distortion: 3.305520%

```

Figure 18 THD of sinewave at 100Hz measured in LTspice

As seen in figure 9 each harmonic is shown with its magnitude. The fundamental harmonic is at approximately 97Hz and all the other harmonics are expressed as a fractions of this amplitude. The second harmonic is about 3% of the fundamental one and the rest are less than 1%. The small nonzero

values of the higher harmonics indicate minor non-linearities in the oscillator design. This THD is typical of a wien-bridge oscillator and can be improved amplitude control by using a lamp or filament. Another aspect to note is that the fundamental harmonic exhibits a 0° phase shift while the higher harmonics exhibit various phase-shift. This is important because the Wien-bridge oscillator only oscillates at the frequency where the lead-lag network produces a zero phase shift which is a defining condition for the Wien-bridge.

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	1.002e+4	8.786e-1	1.000e+0	-0.39°	0.00°
2	2.005e+4	9.542e-3	1.086e-2	-9.92°	-9.53°
3	3.008e+4	9.234e-3	1.051e-2	-118.89°	-118.50°
4	4.010e+4	3.098e-3	3.526e-3	-22.74°	-22.35°
5	5.012e+4	1.729e-3	1.967e-3	16.21°	16.60°
6	6.015e+4	2.968e-3	3.378e-3	13.33°	13.73°
7	7.018e+4	2.697e-3	3.070e-3	2.34°	2.73°
8	8.020e+4	1.401e-3	1.594e-3	28.49°	28.88°
9	9.022e+4	2.412e-3	2.745e-3	20.99°	21.39°
Partial Harmonic Distortion:		1.660175%			
Total Harmonic Distortion:		1.688460%			

Figure 19 THD of sinewave at 10kHz measured in LTspice

Doing the same analysis as before figure 10 shows the harmonic information of the sinewave at 10kHz. The fundamental harmonic being 10025kHz and the rest of the harmonics a fraction of that harmonic's amplitude. The THD at 10kHz is approximately 1.688% yielding better THD than at 100Hz.

The reason for this is that at lower frequencies. The diode-based amplitude stabilization tends to be more nonlinear. Additionally, the op-amp imperfections are more prevalent as the signal tends to be at a higher amplitude for a longer period, however at higher frequencies the diode conduction interval per cycle is shorter, which in turn decreases the non-linearity. The effect of input bias currents is reduced due to the faster signal transition; however, the SR and bandwidth of the op-amp does still play a role but fortunately the frequency is still low enough for it to not be that much of a nuance

3.4 Summing and signal selection

3.4.1 Signal selection

Using the 12-position rotary switch, Figure 20, enables the ability to select between the 3 signals with ease and then feed the selected signal into a buffer and an Inverting gain op amp circuit. The Wien-bridge oscillator's sinewave output was chosen as a nominal reference due to the adjustable amplitude by trimming the feedback resistors. This not only reduces the THD but also fine tuning loop gain to 3. Then matched the triangle and square wave signals output to match the sinusoid gain using a voltage divider with a trim POT and a 10k Ω resistor.

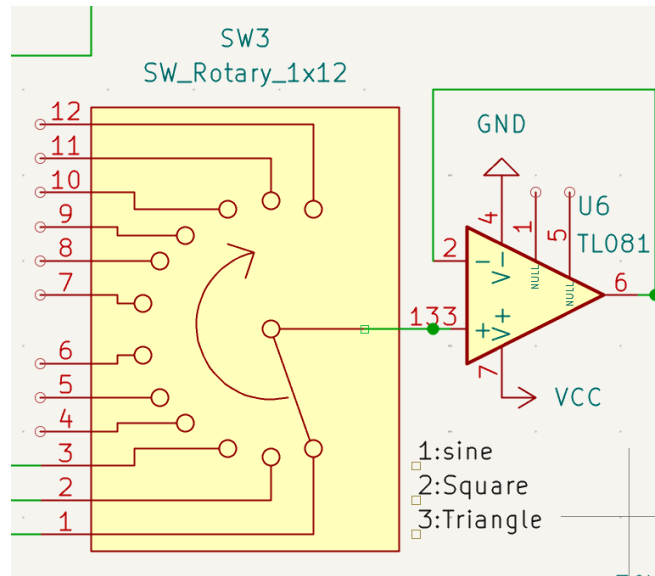


Figure 20 12 position rotary switch for signal selection

Once a pure sinewave output was established. The amplitudes of the triangle and square waves were adjusted to match the sine reference through the voltage divider circuit. This enables the high initial peak-to-peak of the triangle (23.5Vpp) and square 24Vpp to be scaled down to match the sine output of (2.3Vpp).

$$G = \frac{\text{sine } V_{pp}}{\text{triangle } V_{pp} \text{ or square } V_{pp}}$$

The new measured gain adjustment factors are 0.098V/V for the triangle and 0.093V/V for the square wave using the above equation. This adjustment ensures that both the square and triangle waveform outputs are directly comparable to the sinewave downs stream. Figure 21 below shows the voltage divider circuit for the triangle waveform, the square waveform has a similar circuit design.

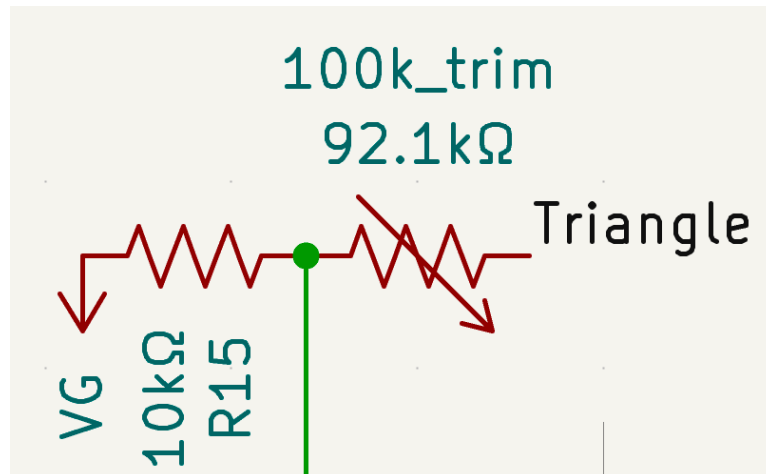


Figure 21 voltage divider circuit for square and triangle

Using Voltage division and choosing a bottom resistor of 10kΩ, we have:

$$\text{Triangle: } \frac{V_{out}}{V_{in}} = \frac{R_{bottom}}{R_{top} + R_{bottom}} \quad R_{top} \approx 92.1k\Omega$$

$$\text{Square: } \frac{V_{out}}{V_{in}} = \frac{R_{bottom}}{R_{top} + R_{bottom}} \quad R_{top} \approx 96k\Omega$$

Due to component tolerance, temperature and all non-idealized use a 100kΩ trim potentiometer for the gain adjustment in the real circuit.

3.5 Level adjustment

3.5.1 Adjustable virtual ground

A virtual ground sets a reference voltage point that acts as a ground for the circuit. It can be anywhere from 0-Vcc in a single supply circuit. By using a 100kΩ POT as the resistance divider, any point between 0 and Vcc can be set as a reference point. Adding a capacitor, C10 at the wiper will filter out any unwanted noise and transients which ensures for a stable output. To further stabilize the output an op-amp is configured as a buffer at the wiper output. This allows the output of the op-amp to track the input voltage regardless of load changes, it also mitigates loading effects from other parts of the circuit.

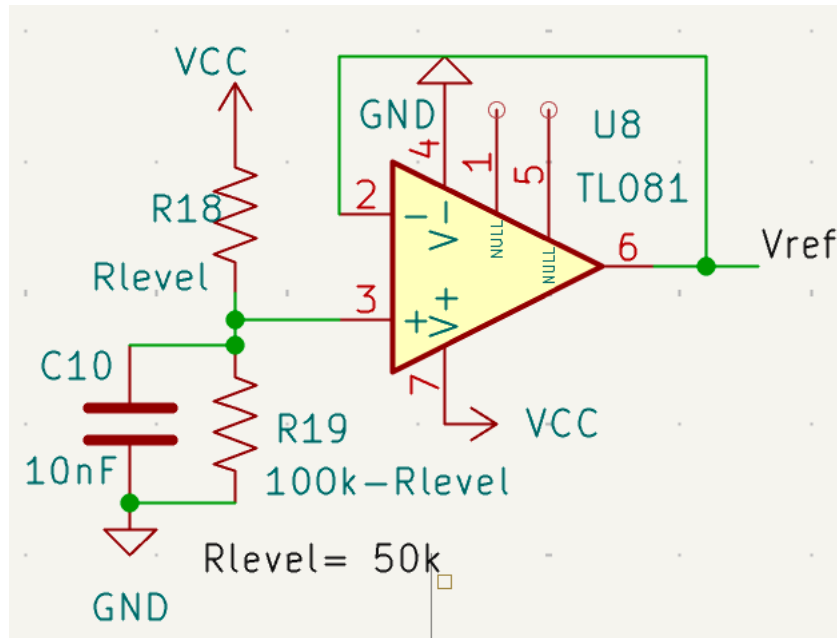


Figure 22 adjustable virtual ground circuit

3.5.2 Summing Amplifier

In Figure 23, A summing amplifier can combine multiple input voltages into a single output signal. The summing signal follows from the amplification stage and mixes the input wave with a reference voltage from the adjustable virtual ground. This achieves a vertical shift. The inverting node serves as the summing node. It is held at virtual ground. Each signal reaches the node via a 10kΩ resistor, and the feedback resistor is also 10kΩ. Which allows for unity gain.

Using a summing amplifier the input signal after the amplification stage is mixed with a reference voltage from the adjustable virtual ground in 3.6.1. The mixing of the input signal and reference voltage cause a vertical shift of the signal at the output of the op-amp. Since both input paths have a 10kΩ resistor in series with the outputs and the feedback resistor is also 10kΩ. We end up getting unity gain

and the equation of the output is. The full derivation of the output voltage of the op-amp is in Appendix below.

Doing a KCL at node 2 of the op-amp the following equation for the output is derived.:

Assume:

$V_n = V_p = V_{\text{ground}} = 13.5$: treat virtual ground as 0.

$I_n = I_p = 0$.

$$\frac{V_g - V_{in}}{10k\Omega} + I_n + \frac{V_g - V_{ref}}{10k\Omega} + \frac{V_g - V_{out}}{10k\Omega} = 0$$

Adding in assumptions:

$$3V_g - \frac{1}{10k\Omega}(V_{in} + V_{ref}) = \frac{V_{out}}{10k\Omega}$$

Equation 7 Level adjust equation

$$V_{out} = -(V_{ref} \pm V_{in}) + 40.5$$

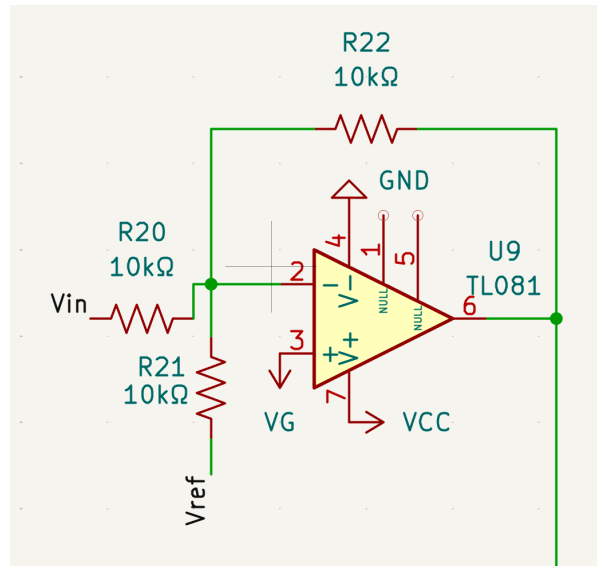


Figure 23 summing amplifier with nominal gain

3.5.3 spice simulation of level shifter

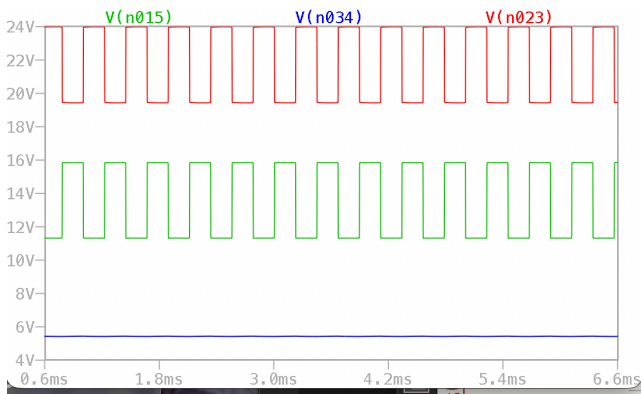


Figure 24 level shifting capabilities positive shift

Each signal in Figure 24 shows us different points before level shift and after. The Top trace indicates

The level shifted signal. The bottom traces indicate the DC shift added and the middle trace indicates the signal before shifting. The summing amplifier combines their input signals, so adding a DC voltage to the original signal will cause the entire signal to shift up by that amount. Summing simply adds DC offset to each point of the input waveform. The amplitude remains constant at 4.52Vpp because it only moves the signal up or down.

- Top trace,(red trace): Level shifted signal
- Middle trace(Green trace):Signal before shifting
- Bottom trace (blue trace) : DC shift added.

If the wiper is set to 80:20 ratio. The voltage division is $V_{ref} = \frac{20}{80+20} \times 28V$. Then Vref becomes 5.6V. Using the equation for the level shifter. If the middle signal is cantered at 13.5 and the DC level added is 5.6V. The new wave is cantered at

$$V_{out} = -(13.5 + 5.6) + 40.5 = 21.4V$$

This proves that for lower DC voltage added the signal gets shifted upward. Since there is no resistors that stop the adjustable virtual ground wiper from turning to the maximum. The signal can be shifted up until Vcc and down to ground.

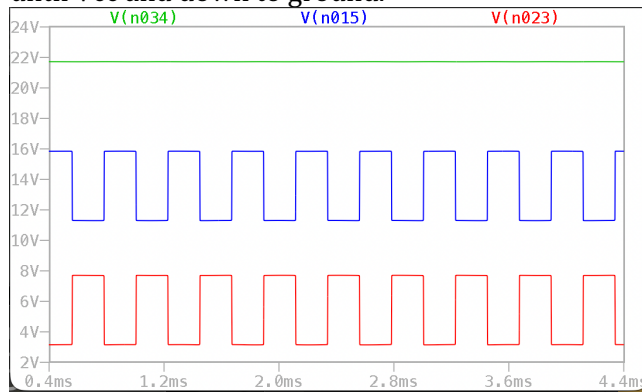


Figure 25 level shifting capabilities negative shift

- Top trace,(green trace): DC shift added
- Middle trace(blue trace):Signal before shifting
- Bottom trace (red trace) : Level shifted signal.

In figure 22 The reference voltage is a lot higher than in figure 21, about 21.67V. This causes the original signal to move downwards using the same level shifting equation

$$V_{out} = (-13.5 + 21.67) + 40.5 = 5.33V$$

The signal's new cantered value is a lot lower.

3.6 Amplification

3.6.1 Inverting Amplifier design

Using the following circuit design. After the signals pass through a buffer All the signals current is limited through a nominal 1kΩ resistor.

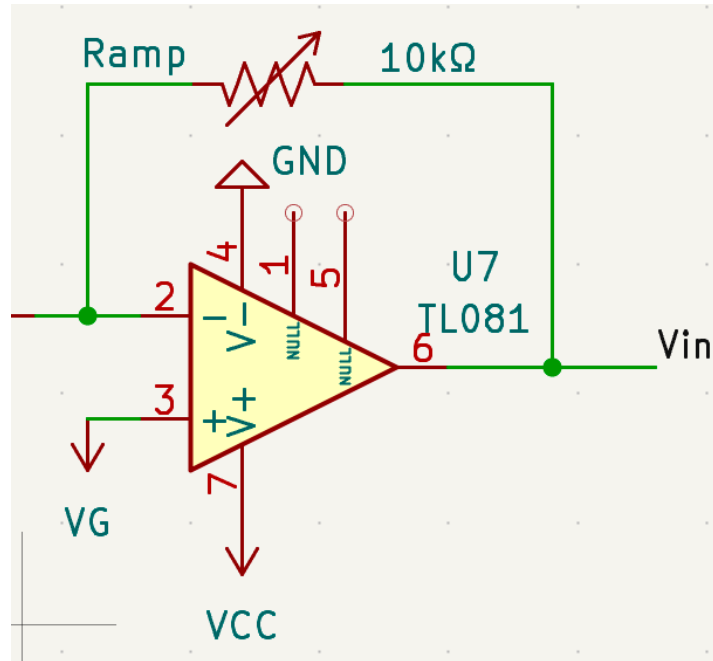


Figure 26 Signal Amplifier

Since this is a normal inverting amplifier that is referenced at virtual ground. The equation for a standard non-inverting op-amp with variable gain is

Equation 8 Amplifier equation

$$V_{out} = -\frac{R_f}{R_{in}} V_{in} = -\frac{R_{ramp}}{1k} V_{in} .$$

Ramp is defined as 10kΩ potentiometer. The maximum attainable gain on the input signal is 10. The Sine, triangle and square wave all have about 2.3V_{pk-pk} when it comes out of the oscillators into the resistor divider and out of the signal selection switch. Since all the signals have the same base amplitude. The maximum amplitude of all the wave forms is 23V_{pk-pk}. This allows for variable gain of all the signals.

Splitting up the Amplification stage and level adjust stage as well as the signal selection stage. All the signals are independent of previous stages and this mitigates loading effects going into the next stage.

3.7 Voltage regulation, virtual ground and output drive capability

Describe the design of any voltage regulation circuitry, the virtual ground and the driver design to meet the output source and sink current requirements. Include circuit schematics.

3.7.1 Voltage regulator

Regulating the input voltage from the banana connectors and the 30V from the bench power supply. A 28V supply is derived that can sink up to 1.5A of current using the LM317 adjustable voltage regulator. The following circuit which is provided in the LM317 datasheet was used as the designing circuit [5]

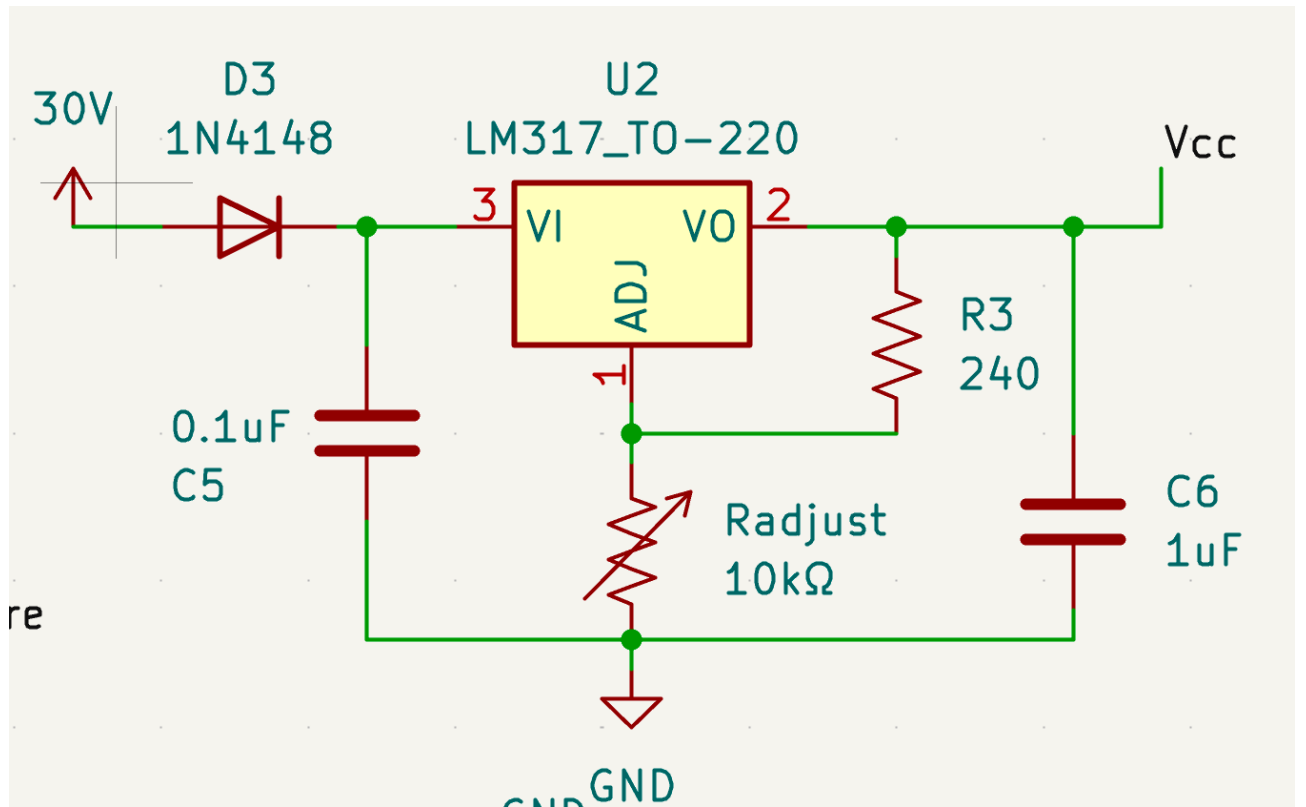


Figure 27 voltage regulator design

3.7.1.1 Reverse polarity protection and regulator properties

The use of the input Diode is to protect the circuit from incorrect polarity if the system is incorrectly connected. The turn on voltage for the diode used 1N4148, is about 1V/10mA. This means that the regulator only sees about 29V after the voltage drop of the diode[6]. The voltage regulator also has a drop-out voltage of 2-3V depending on the load its driving [5]. After the regulator there is approximately 27.1V defined as the Vcc node. The equation for the voltage measured as Vcc is.

$$V_{out} = V_{in} - V_{diode} - V_{dropout} = 30 - 1 - 2 = 27V$$

That is the expected Vcc which all the circuits will receive. The reason the regulator is used is so that if the line from the power supply is too long. The regulator can mitigate the transients and oscillations on the line.

The C5 capacitor is used to decrease noise on the input as well as decrease oscillation from source if the source is far from the regulator. The C6 capacitor ensures that the regulator remains constant with its output as there is a change in load on the line. It improves transient responses.

The equation below is used to derive the output voltage dependent on the Resistor you use at R2.

$$V_{out} = 1.25 \times \left(1 + \frac{R_2}{R_1}\right) + I_{Adj} \times R_2 \quad \text{Equation 9 Regulator equation} \quad \text{Eq.1}$$

Since I_{adj} is controlled to less than 100μA the second term can be neglected in Eq.1. Using a R2 of >5.1kΩ and the R1 of 240Ω. A 27.1V output can be achieved on the output

3.7.1.2 spice and matlab simulation of regulator

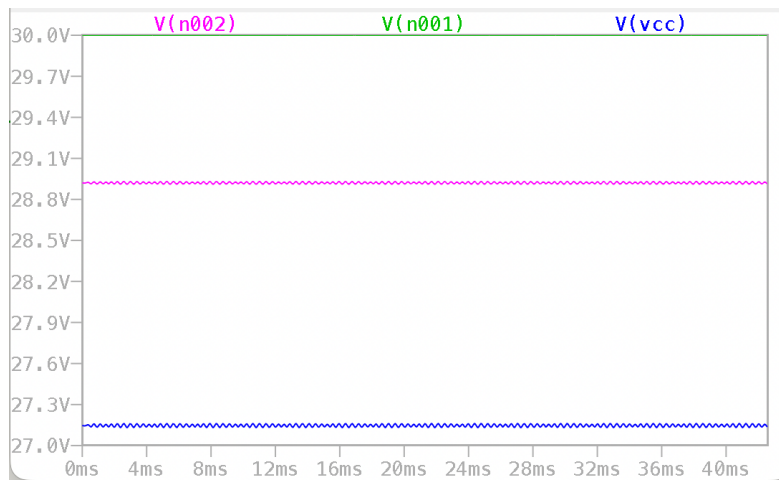


Figure 28 spice simulation of voltage at different stages on the regulator

- Top trace (green trace): Input voltage
- Middle trace (pink trace): Voltage after diode drop
- Bottom trace (blue trace): voltage after regulator drop-out voltage.

Table 4 voltage regulator comparison

Radj(k Ω)	Voltage output LT-spice(V)	Voltage with equation(V)
10	27.1	53.3
7	27.155	37.7
5	27.1	27.3
3	17.65	16.8
1	6.75	6.45
0.5	4	3.85
0.24	2.89	2.5
0.1	2.91	1.77

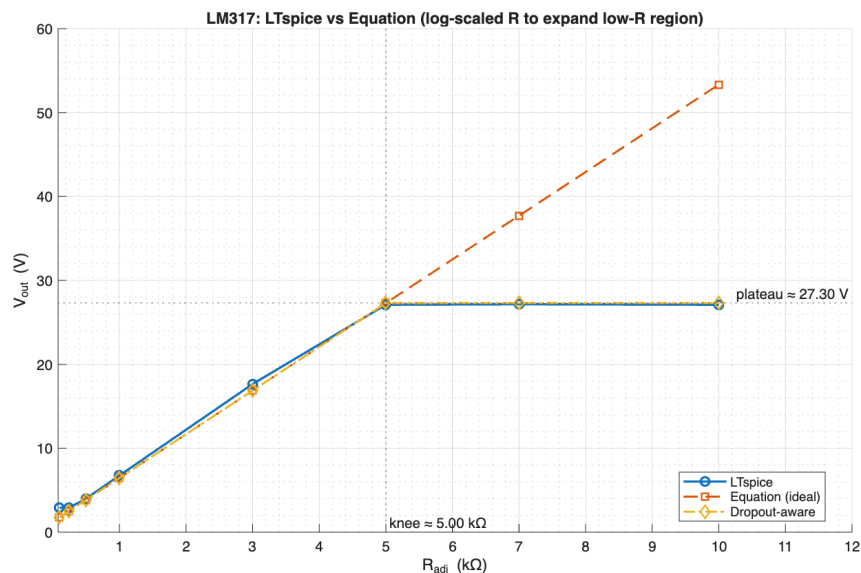


Figure 29 regulator equation compared to spice output

In figure 26 the blue graph depicts the LTspice simulation of the regulator as R_{adj}/R_2 is changed. At lower resistance values the analytical equation closely follows the spice output. However at about 5k Ω the spice simulation starts to saturate which represents the regulator dropout region. This means that the regulator cannot further increase the voltage as the supply voltage is at maximum. The Dropout

aware curve which is the spice circuit excluding the voltage constraints. It closely follows the ideal equation showing that it is a good approximation of the voltage at the output of the regulator.

3.7.2 virtual Ground

The virtual ground used two 56kΩ resistors as well as an op-amp configured as a buffer. Then, a Class AB amplifier on the output the virtual ground can sink/source more current.

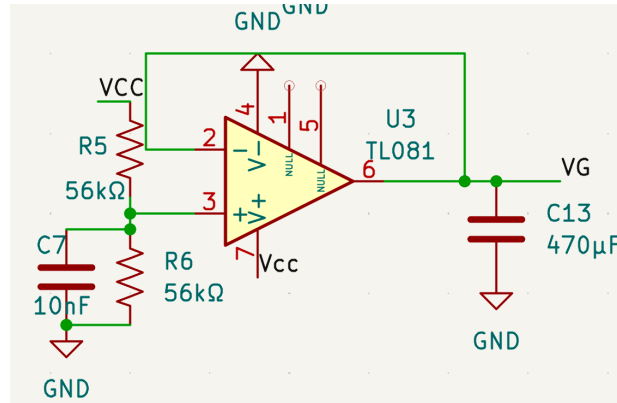


Figure 30 virtual ground design

The C7 capacitor is used to remove any transients and stabilizes the virtual ground due to noise from extra load on the virtual ground. The reason for using a buffer is that the two resistors will equally divide the voltage however, due to Temperature over time, the voltage can vary as the resistance changes with the change in temperature over time. The Op-amp on the output is used as a voltage follower and outputs the same voltage as the input. The 470μF capacitor is used to filter out any high frequency oscillation due to feedback from other components. The virtual ground is used to bias all op-amps around the same oscillation point. Since the virtual ground does not have to provide high current to bias the other op-amps. It does not need extra current supplying network/power amp network. The op-amp can only source/sink up to 10mA before saturating. Since it is only used to bias the other op-amps and the input resistance of an op-amp is very high. It does not draw a lot of current to bias for example the non-inverting node, therefore no power amplifier needed to keep for the virtual ground into the oscillator circuit

3.7.2.1 virtual ground power amp

At the output stage of the signal generator, it needs to drive a 27Ω load with a maximum of 10Vpk. Using ohms law, it will require the following current.

$$I_{out} = \frac{V_{pk}}{R} = \frac{10}{27} \approx 370mA$$

The op-amp will not be able to source or sink that current and will saturate therefore a power amplifier is needed.

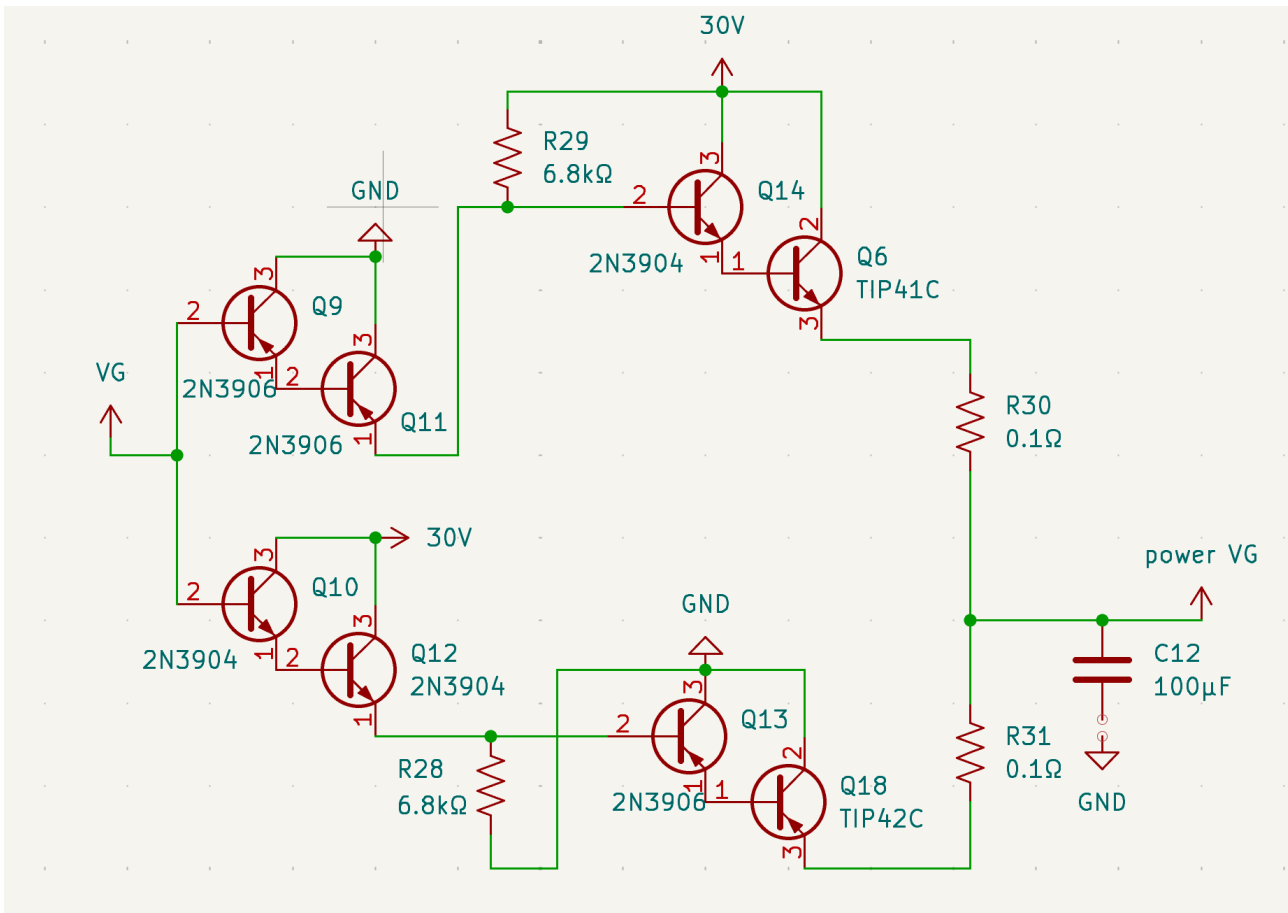


Figure 31 virtual ground power amplifier

A Class AB power amplifier which is designed to combine the efficiency of a Class B amplifier with low distortion of a Class A amplifier during operation. It is achieved by biasing the output transistors just above cut-off. For small signals both conduct like a Class A amplifier. Each transistor will take over more than half of the waveform for large signals like a Class B amplifier, thus eliminating Cross-over distortion, which is seen in a class B amplifier.

Using a Darlington Pair on the Input Buffer. It provides the correct voltage drop over the Darlington Pair for the Push-pull stage transistor pairs. It also has a high Input Impedance drawing very little current from the previous stage as well as loading effects from the previous stage. The choice of a Darlington Pair on the Output stage in a Push-pull configuration. The reason for use of a Darlington pair instead of a single transistor, is so that enough current is provided for low-impedance devices without putting significant load on the biasing stage. A tiny input current can control a much larger output current.

3.7.2.2 current source design

For the power amplifier design. The current gain, β , for the 2N3904 and 2N3906 50 was chosen and for the Tip41C and Tip42C a current gain of 100 was chosen. The reason for the choice was to not under-drive the output circuit while not dumping unnecessary current into the transistors increasing the heat. To get the current entering node 2 of Q14, the following equation for Darlington pairs was used.

$$I_B = \frac{I_{load}}{(\beta_{tip} + 1)(\beta_{2n3904} + 1)} = \frac{370m}{(101)(51)} = 72\mu A$$

The I_{bias} current was then chosen as $I_{bias} = 5mA + 72\mu A = 5.072\mu A$.

The design decision was to make the bias current 5.5mA to increase head room so that the power amp could be used to drive bigger loads. For Instance, in the demo, A 8Ω ,8W load was driven without changing power amp values. Since on $72\mu A$ is required at the base of node 2 of Q14. The additional current is dissipated as heat in the resistor as well as sunk by the input buffer transistors as heat. This

can cause issues if the left-over current is too much as can overheat the input buffer causing thermal runaway.

Doing a KVL from the Vcc node down until the output node of the 27Ω load. The following equation is derived to calculate the resistor required to provide the bias current.

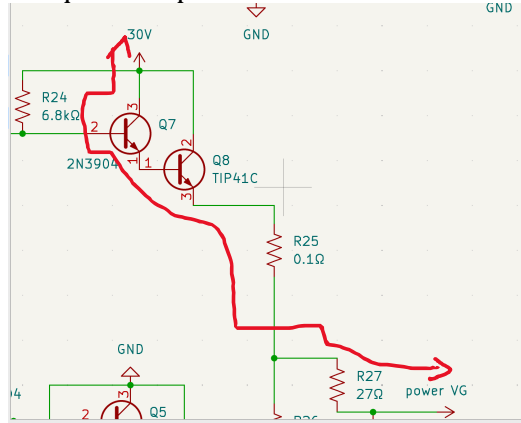


Figure 32 KVL loop for resistor calculations

$$R_{bias} = \frac{30 - 1.3 - 370m \times 0.1 - 10 + 13.5}{5.5m} = 5756\Omega$$

6.8kΩ resistor is chosen as in the lab this is the closest E12 resistor to 5756Ω. The optimal resistor choice would be a 5.6kΩ resistor however to minimize the extra current flow through input buffer transistors a higher resistor value was chosen.

$$I_{bias} = \frac{30 - 1.3 - 370m \times 0.1 - 10 + 13.5}{6.8k} = 4.65mA$$

The resistor sets a stable bias current through the Darlington Pair, defining the quiescent (idle) current of the output transistors. There is no need for a BJT current source as the signal produced from our signal generator is not used in audio equipment. The resistor will set a stable operating current without active current source.

The main output is formed using complementary Darlington pairs arranged in a push-pull fashion. This allows for high current delivery to low impedance loads and maintains system robustness. The use of emitter resistors are low power resistors, 0.1Ω, resistors used to ensure that the current is shared more evenly on the output. It helps with Thermal stability which helps with thermal runaway. The transistors will then self-limit as the temperature increases.

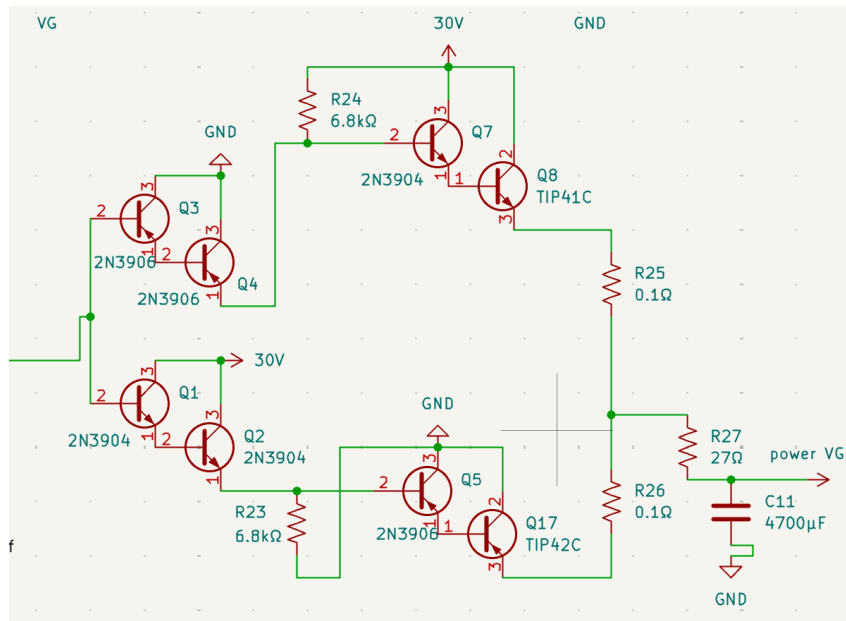


Figure 33 power amplifier for the output stage into 27Ω load

The 4700μF capacitor is used to filter out transients and noise at the higher frequencies seen on the load.

3.7.2.3 Thermal calculations

Calculating the thermal capabilities of the 2N3904. The thermal calculations can be done. When the upper tip is turned off and the input buffer transistors are kept on for both transistors. Using the .OP command in LTspice to get the transistor Vce at full amplitude and duty cycle the the current that needs to be sunk into those transistors are as follows:

Name:	q17	q1	q2	q5	q6
Model:	2n3904	2n3904	2n3904	2n3906	2n3906
Ib:	1.60e-07	1.57e-08	5.39e-06	-3.53e-08	-8.06e-06
Ic:	5.40e-05	5.37e-06	1.84e-03	-8.02e-06	-1.83e-03
Vbe:	5.77e-01	5.17e-01	6.68e-01	-5.27e-01	-6.68e-01
Vbc:	-1.25e+01	-1.37e+01	-1.42e+01	1.37e+01	1.43e+01
Vce:	1.31e+01	1.42e+01	1.49e+01	-1.43e+01	-1.49e+01
BetaDC:	3.38e+02	3.41e+02	3.41e+02	2.28e+02	2.28e+02
Gm:	2.09e-03	2.08e-04	7.08e-02	3.10e-04	7.06e-02
Rpi:	1.62e+05	1.64e+06	4.80e+03	7.33e+05	3.21e+03
Rx:	2.00e+01	2.00e+01	2.00e+01	2.00e+01	2.00e+01
Ro:	2.08e+06	2.12e+07	6.21e+04	1.42e+07	6.23e+04
Cbe:	1.26e-11	1.14e-11	3.74e-11	1.44e-11	4.05e-11
Cbc:	1.55e-12	1.51e-12	1.49e-12	1.69e-12	1.67e-12
Cjs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
BetaAC:	3.37e+02	3.41e+02	3.40e+02	2.27e+02	2.27e+02
Cbx:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Ft:	2.35e+07	2.56e+06	2.89e+08	3.07e+06	2.66e+08

Figure 34 Vce and Ic for Q6 2N306 transistor

$$I_{bias} \approx 1.83mA \text{ and } V_{ce} = 14.9V$$

$$P_{Q11} = V_{ce} \times I_{bias} = 14.9 \times 1.83m = 27mW$$

This is validated with the following power plot of the transistor.

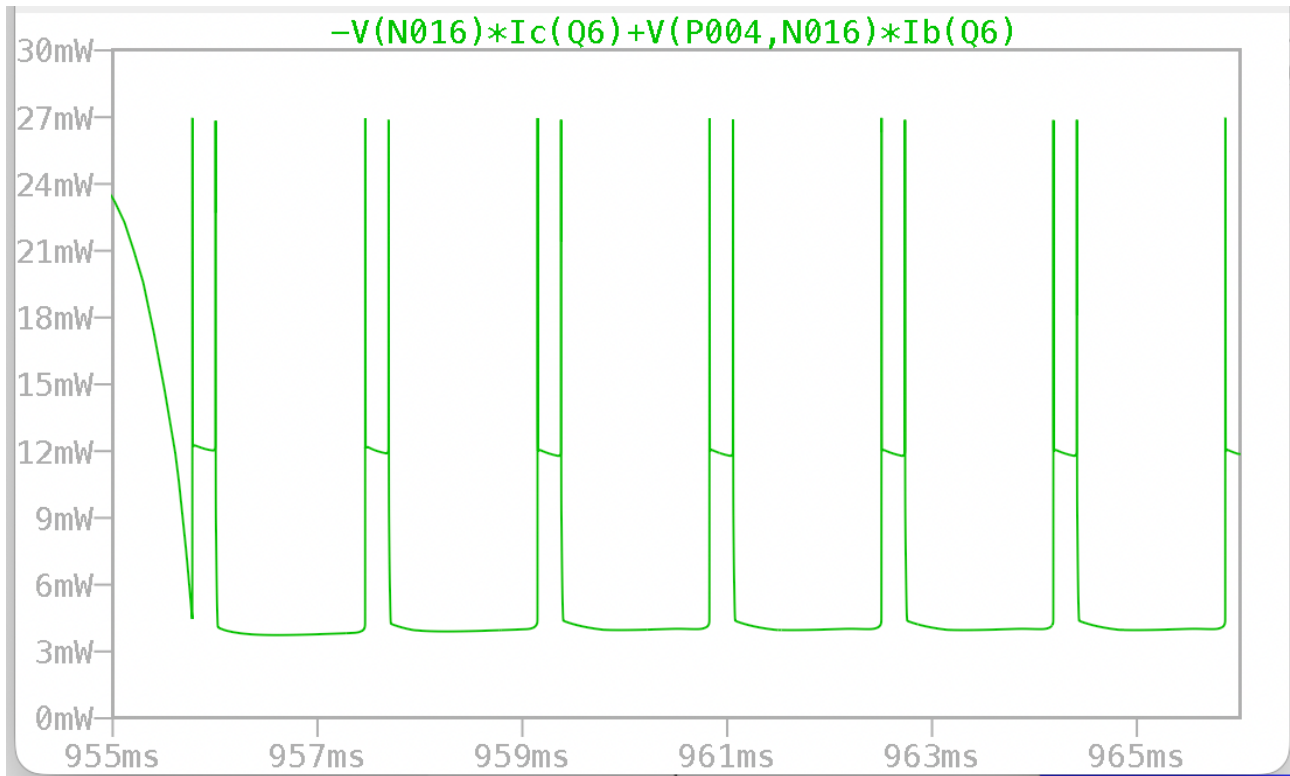


Figure 35 power of input buffer transistors

Since this is the maximum power dissipated in the input buffer transistor to calculate the Temperature of the transistor is as follows:

$$T_j = T_{amb} + P_{Q11} \times R_{\theta} = 25 + 27m \times 83.3 = 27.2^{\circ}C$$

The maximum operating temperature without a heat sink is $150^{\circ}C$ and this is far below that so no heat sink is required. The R24 bias resistor maximum current supplied was designed for

$$Pr_{29} = 28.7 \times 4.22m = 0.12W$$

Need to use $\frac{1}{4}$ watt resistor that can handle the current flow. The resistors in the lab can handle up to 0.25W of power so that is more than enough.

For the TIP transistors, they can dissipate 65W at $25^{\circ}C$. The datasheet does not provide the thermal resistance. But using the maximum power dissipated and the maximum allowable junction to case temperature. The heatsink calculation can be done. The Thermal resistance of the junction to ambient is $62.5^{\circ}C/W$ [7].

Equation 10 Thermal resistance for TIP

$$T_{max_J/C} = \frac{150 - 25}{65} = 1.9^{\circ} \frac{C}{W}$$

Running a .OP command in LTspice and getting the Vce of the TIP41C. The TIP I am referencing is Q10 with a Vce of 13.7V which in Figure 33 Q8 transistor.

	Name: q16	q10	q18
Model:	tip42c	tip41c	tip41c
Ib:	-4.47e-05	5.44e-05	5.42e-05
Ic:	-4.43e-03	4.42e-03	4.40e-03
Vbe:	-6.42e-01	5.89e-01	5.89e-01
Vbc:	1.31e+01	-1.31e+01	-1.31e+01
Vce:	-1.38e+01	1.37e+01	1.37e+01
BetaDC:	9.90e+01	8.12e+01	8.12e+01
Gm:	1.71e-01	1.71e-01	1.70e-01
Rpi:	6.56e+02	5.59e+02	5.61e+02
Rx:	1.00e-01	1.00e-01	1.00e-01
Ro:	2.56e+04	1.43e+04	1.43e+04
Cbe:	4.65e-09	7.16e-09	7.13e-09
Cbc:	1.31e-10	9.07e-11	9.07e-11
Cjs:	0.00e+00	0.00e+00	0.00e+00
BetaAC:	1.12e+02	9.54e+01	9.53e+01
Cbx:	0.00e+00	0.00e+00	0.00e+00
Ft:	5.68e+06	3.74e+06	3.74e+06

ch

Figure 36 OP command in LTspice to get Vce

The junction temperature at maximum current flow, when at fully duty cycle for square or maximum peak for sinewave is:

$$P_{tran} = V_{ce} \times I_e \times \frac{\beta}{\beta + 1} = 13.7 \times 0.370 \times \left(\frac{100}{101}\right) = 5.018W$$

This LTSpice plot to verify the power is below:

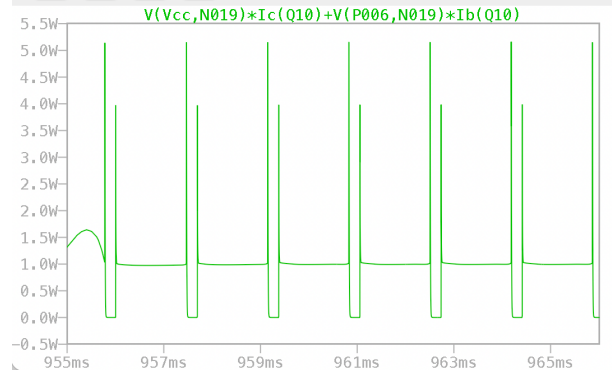


Figure 37 power of the TIP41C transistor for a full duty cycle, full amplitude square wave.

This verifies that the tip will draw up to 5W of power at maximum current draw.

Then to calculate the junction Temperature:

$$T_j = T_a + (\theta_{cj} + \theta_{ja})P_{tran} = 25 + (1.91 + 62.5) \times 5.018 = 348.21^\circ C$$

This is far above the rated temperature of $150^\circ C$ therefore a heatsink is required. The heat sink provided is a flat-plate aluminium. With a heat sink the maximum allowable power is 65W. The maximum power for a TIP41C transistor without a heat sink is:

$$P_{max} = \frac{T_j - T_a}{\theta_{ja}} = \frac{150 - 25}{62.5} = 2W$$

Since the transistor has to dissipate more than 2W this further verity's that a heatsink is required.

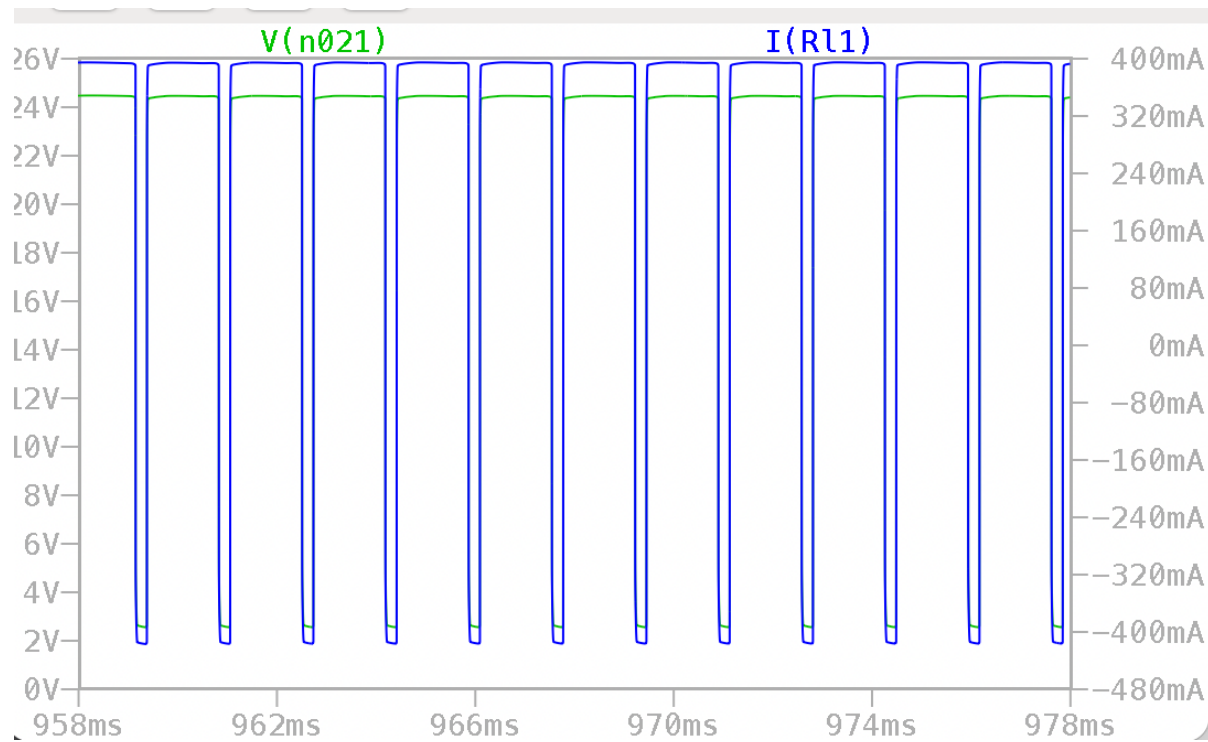


Figure 38 current sink\sourcing capabilities of the power amp

The power amp can source and sink up to 400mA of current in spice showing the driving capabilities of the power amp is sufficient for a 27Ω load. The virtual ground power amp and regular output stage power amp have the same design.

In the design there are 2 virtual grounds one for the biasing of the oscillator circuit around virtual ground and one for sink/sourcing current to the load. Below is the spice simulation of each plot at full load and duty cycle.

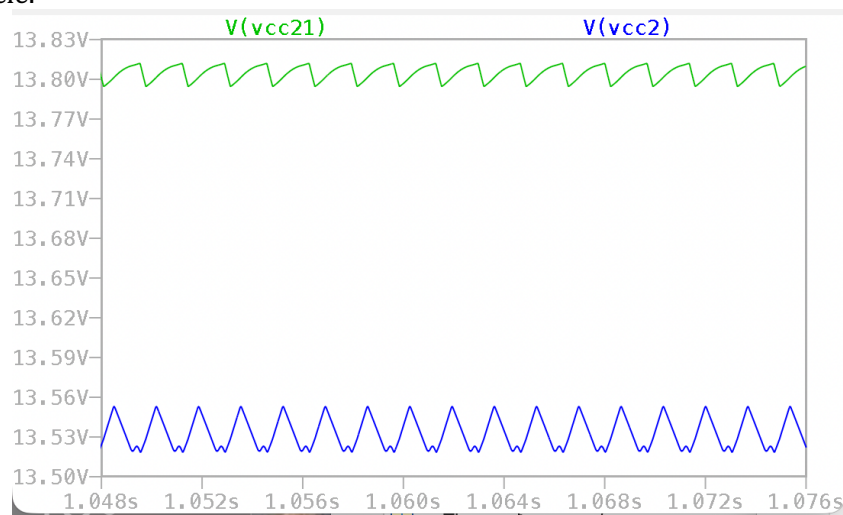


Figure 39 virtual ground spice simulations

The bottom trace, blue trace, this is the virtual ground traces which was design in section 3.7.2. It establishes a stable virtual ground for the low-current circuits, such as oscillators and preamp stages. The op-amp ensures that the virtual ground voltage remains steady and closely tracks the designed setpoint. The voltage stays within a narrow band of 13.53V to 13.56V, very small oscillation.

The power stage virtual ground, which is the top trace, green graph, this circuit acts as the sink/source for the high-current branch of the output of the amplifier with the same small ripple at 13.80V-13.83V about 300mV higher than the normal virtual ground. This is due to the circuit having to sink/source greater current and respond to dynamic load variations which results in the larger ripple.

The power stage virtual ground has a larger capacitor at its output to help with these transients. It has a 4700 μ F Capacitor and a 100 μ F capacitor on the normal virtual ground.

The main reason for both virtual ground is to separate the power stage and the lower current stages. A op-amp based virtual ground is good for referencing voltages in lower-power side and provides precise bias and lower noise.

The power stage virtual ground is required when significant current must flow through the load and back into the virtual ground.

3.8 Switch design

Using the PC-16 POT with a built switch serves a dual purpose. It serves as a mechanical variable resistor to change the amplitude of the output signals and is the POT used in the inverting amplifier design. Built into the end of its rotation, if it is turned fully counterclockwise, is a switch. This switch is used to connect and disconnect the power supply while simultaneously starting the input current low as when the switch is on the output amplitude is at its lowest.

The switches maximum current it can allow through without any damage to the component is 1A at 250VAC [8]. This is far above the electrical specifications for the signal generator.

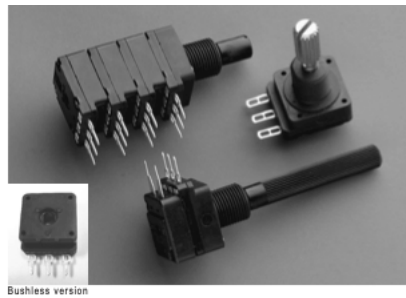


Figure 40 POT with integrated switch [8]

3.9 Construction

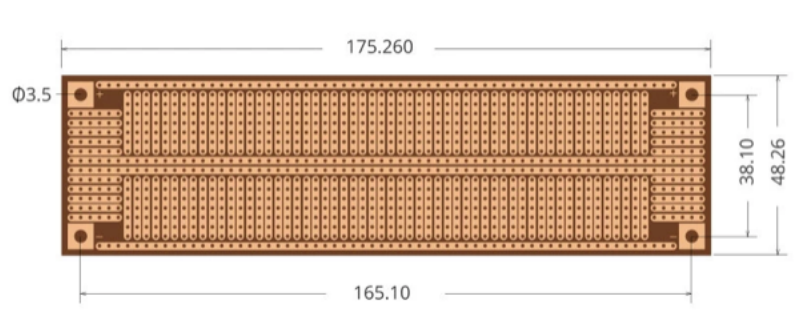


Figure 41 protoboard used for soldering

The system is constructed with 3 standard protoboards [9]. Each with a integrated vcc and ground rail. These rails are used to distribute the Vcc voltage throughout each board as well as ground return paths from all the circuits. All 3 oscillators and regulator are assembled onto a single board to optimize space. The Wien-bridge oscillator is designed on a separate 16-pin board for ease of use when adding trim POTs.

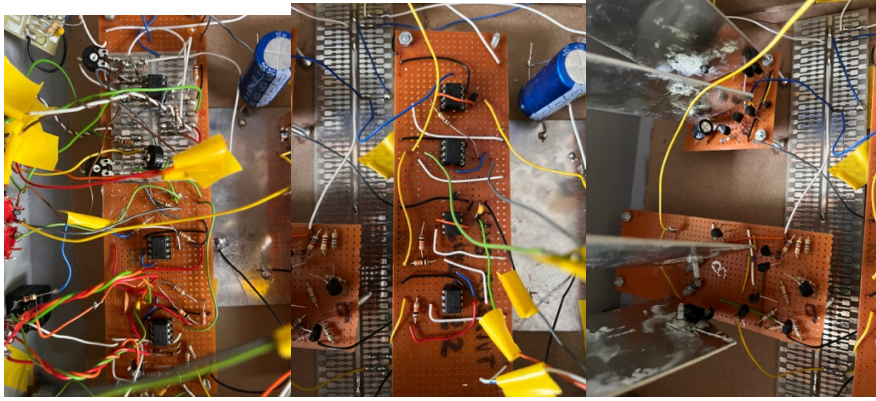


Figure 42 oscillator circuits (left) , gain/amp/level stage (middle), power amplifier(right)

The signal selection, gain, amplifier and level adjust stages are integrated on a single board for separation between stages. Both power amplifiers are designed on one board to save space, but the board is broken in half to separate the amplifiers for easy prototyping. A single 16-pin soldering board is used for the op-amp virtual ground module, and it place on the ground.

For power and ground interconnections a soldering strip is used to distribute the V_{cc} , virtual ground and system ground between all the different stages. All the ground connections are then tied to a larger ground plane, which is the common return path of the power supply. The large surface area of the ground plane helps to reduce ground loops and provides a low impedance return path for currents to minimize noise.

The system is power by a single 30V supply input connection, The main voltage is split internally, as one branch goes to the power amplifiers and another branch goes to the regulator as depicted in figure 24.

The enclosure is a plastic container, figure 35, with holes on the front and back. All potentiometers and switches are mounted on the front panel while the input power Bananna connector is mounted on the back panel. This gives a nice ergonomic and sleek design.

The circuit includes polarity protection and current limiting resistor to ensure not component damage. The use of a LM317 regulator for stable V_{cc} at low power stage and minimal component usage throughout the system design show case environmental knowledge and easy end of life replacement. Components used are through hole thus mitigating the use of excessive wires and for reusability.

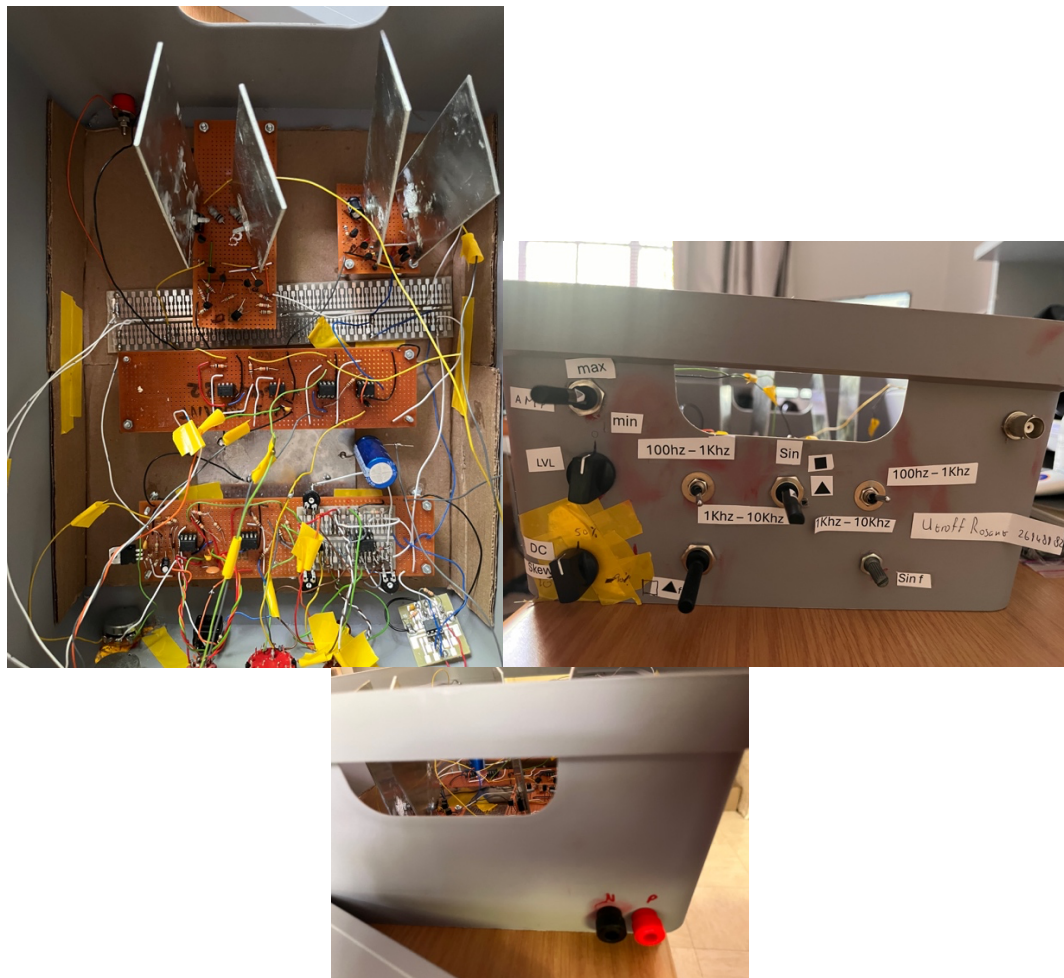


Figure 43 full system inside packaging including front panel and back panel

4 Software design and implementation

4.1 Programming language

There are multiple different languages to choose from to develop a oscilloscope MVP. The choices were between Rust, C++ and python. The reason C++ was not chosen was the time it would've taken to develop the same GUI and Numpy/Scipy in python already could run single channel analysis at native speeds. The reason rust was not chosen is the fact that there are better libraries on python to do the exact task at hand. Therefore, python was chosen due to its scientific stack, Numpy and scipy. These libraries provide vectorised array operations back by optimized C, allowing the use of heavy operations like a Hann-windowed FFT while staying at native speed and keeping the code compact and readable.

The core algorithms used all had a $\log(N)$ or $O(N)$ time complexity meaning they didn't use python loops which are slow. The GUI uses libraries like PyQt5 and pyqtgraph, which can draw tens and thousands of points a second to have the feeling of a real oscilloscope. Using pyInstaller the software could be bundled with the interpreter and bytecode and be turned into a .exe file.

Even though there are a lot of upsides to using python there are also downsides. The downsides of using python is the audio latency or driver mismatch. The sound device library relies on the OS drivers. This can introduce latency. There is also dependency issues where some libraries lag the newest python

versions and this can break installations. Also, some code can behave differently when installed on different OS audio stacks. For instance, a .exe file will not run on a linux machine like mac. Python is also more memory intensive than C++, so having a simple UI will help mitigate the RAM usage. Other than those issues python offers a wide variety of libraries and is a very user-friendly language.

4.2 Software block diagram and description of interaction

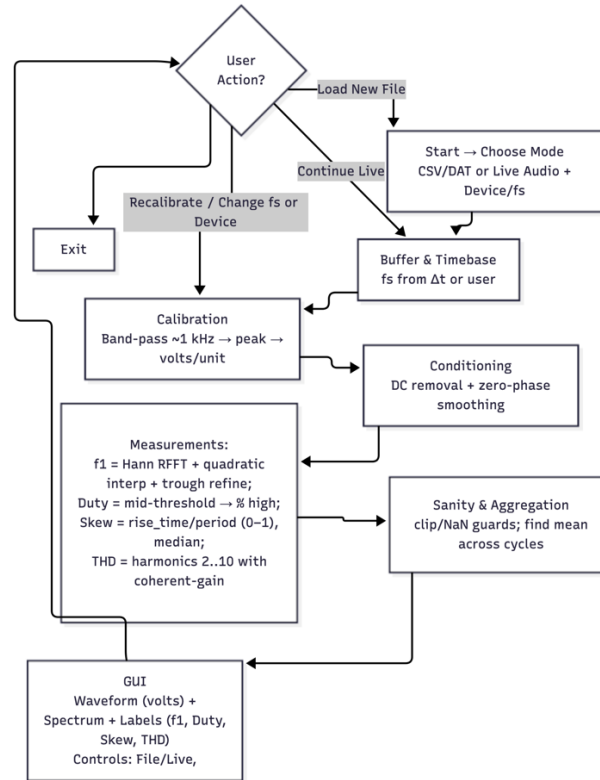


Figure 44 program functionality flow diagram

As the Application opens the user can choose to read the data in via CSV or by doing a live input through the microphone jack. The sample rate, fs , is set to the soundcard capabilities configured in settings. It is default set to 48kHz. If it is using a CSV and there is a column in the CSV then it can calculate the sampling rate by the following formula:

$$fs = \frac{1}{average(\Delta t)}$$

4.3 Use of AI tools

The two LLM's used in this design was chat GPT-5 thinking and vscode copilot. GPT-5 was the main LLM as I prompted it to give me an MVP for an oscilloscope application. Using the specifications which the oscilloscope had to do. It then gave me a boiler plate design of a oscilloscope and told me to install all the required libraries and create a new virtual environment. AI reduced the design time so that the focus was on the algorithm choice and parameter tuning.

Prompts as follows were used to refine the product that GPT-5 provided. The procedure in which the csv must be fixed. By providing the LLM with the CSV format, which was used to read in sample values, It could fix the csv loading function. Originally the code was not perfect so there was a lot of iterations and back and forth on specific specification that had to be there. One prompt that had to be formulated

was due to the mis-reported frequency of a triangle waveform due to the extra local minima it was measuring. The LLM then corrected this error by using another zero-phase 2nd order filter at 5% of the Nyquist before finding peaks (-x).

To verify if the code was correct the test CSV's on stem learn were used to see if it displayed the correct output, The CSV files from stem learn had the frequency and duty cycle values embedded in the name of the file. This allowed for the software to be tested with these files to verify if it was displaying the correct results e.g the THD was lower than 1% the frequency was correctly interpreted. The duty cycle and skew displayed correctly. Using the live input. Audio settings on windows had to be adjusted, turning off any filters from window for pure waves to be filter out, that setting had to be turned off. Since my machine had a dedicated microphone input only a tip-ring-sleeve connector was required. Testing a 2Vpk-pk signal and checking if the calibration button worked as intended.

4.4 Data read-in and calibration

For the live audio input it is captured from the laptop's Mic/Line-in via sounddevice library. The default sample rate is 48kHz at 16-bit. The file input is a secondary input with CSV and in the file if a time column exists the app auto calculates the sample frequency. The device should be formatted as follows. If the user selects live-audio in and sets the sample rate. They should ensure the windows settings for mic boost is off, enhancements are off and mic input level is initially 5%-15%. The signal generator can exceed sound card limits so the input signal should be attenuated with a 100kΩ series resistor.

"Sounddevice.InputStream()" calls an interrupt that pushes the audio block into a circular buffer. The GUI reads the latest 5000-10000 samples for plotting; this keeps the UI responsive and in real time. Calibrating a live 1kHz 1Vpeak signal a short signal window is put through a bandpass filter and the raw maximum value is measured and the scale is set to:

$$\frac{V}{unit} = \frac{V_{refpeak}}{\max|sample|}$$

Now the signal is calibrated, and all signals now are scaled to this new calibration factor. Before measurements. Before doing any calculation on the sample values. The DC is removed from the signals if there is any DC offset and put through a like zero-phase low pass filter to smoothen out the waveform from higher harmonics.

The Frequency is measured using a Hann-windowed RFFT with a quadratic peak interpolation specifically for the triangle and square wave. The duty cycle of the square wave is calculated by finding the average value of the square wave and then calculating the number of samples over the threshold over one period. The skew is as the rise_time/period of the triangle. This is incorrect as it should be the rise_time/fall_time which is defined in equation 5. The THD is calculated by using the following formula for 10 harmonics.

Equation 11 THD equation for software

$$THD = 100 \sqrt{\sum_{k=2}^{10} \frac{A_k^2}{A_1^2}}$$

The GUI uses PyQt5 library and uses the pyqtgraph to render the waveforms in volts as well as the spectrum in dB.

4.5 Software functions

Describe and show results for functions such as amplitude measurement, duty cycle/skew determination, frequency analysis (FFT), total harmonic distortion calculation, etc.

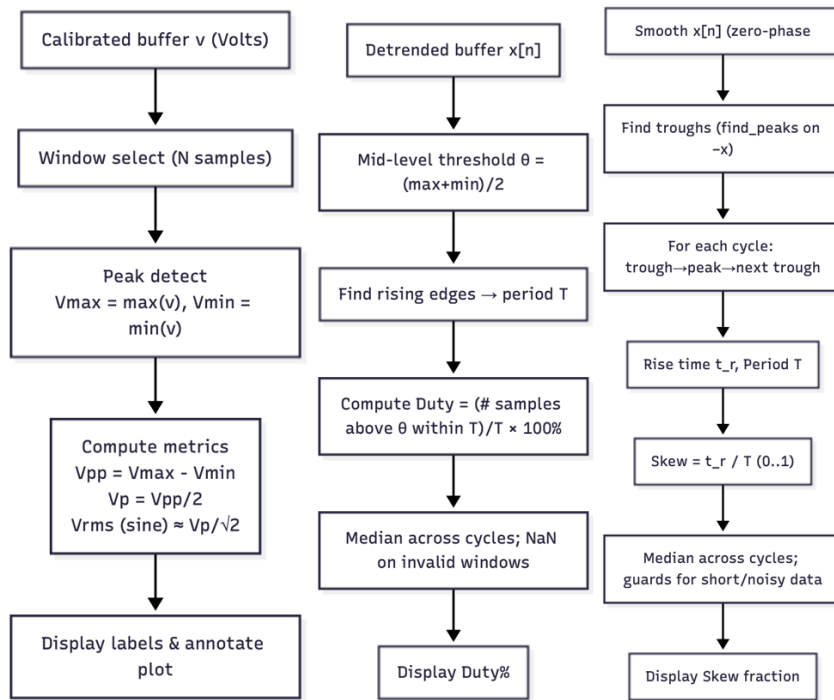


Figure 45 Amplitude adjustment flow diagram, left; `calc_duty_cycle()` function, middle; `calc_triangle_skew()`, right;

This function routinely checks to see if the waveform is outputting the correct peak-peak voltage. This links to the amplitude results seen in section 5.2.2 where the y-axis is the amplitude of the signal should be around 19-20Vpk-pk.

The threshold level is the mean voltage across all cycles. The software compares to the measurements taken in section 5.2.1 in Figure 54. Confirming the accurate 10%-90% duty cycle control and showing the same minor amplitude drop at the extreme duty ratios observed in the hardware section.

For the triangle skew the `_smooth` function is called first which filters out any local extrema which can alter the frequency and skew calculation. The skew-calculations matches the software skew fractions seen in Figure 58 and Figure 59. This shows good agreement with oscilloscope results at 100Hz and the expected deviation due to SR at 10kHz is shown.

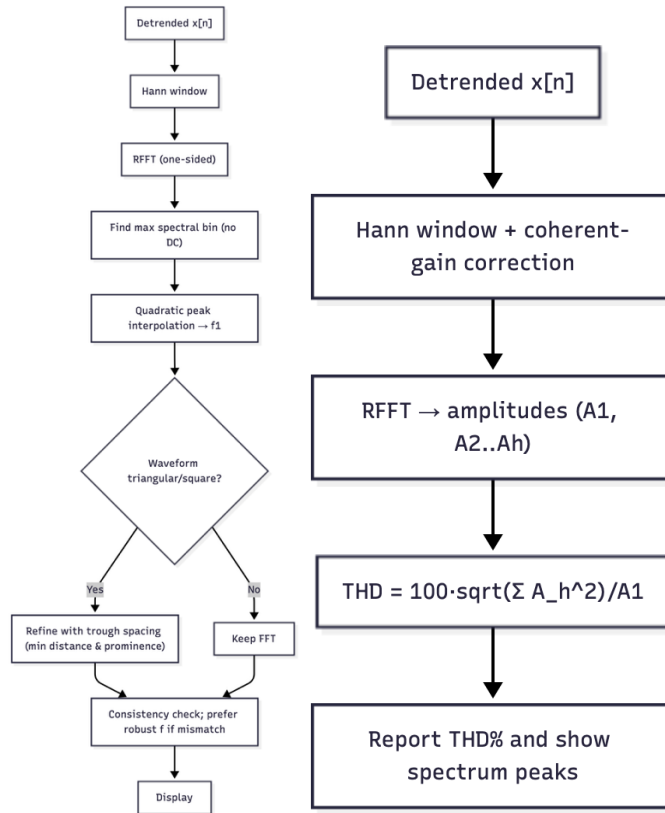


Figure 46 freq_fft_estimate(),left; calc_THD, right

The FFT-based frequency detection function calculates and plots the FFT correctly as depicted in Figure 61. This remains constant for high and low frequencies. The Hann window is a smoothen taper applied to a finite window just before the FFT, this is used to reduce spectral leakage. The RFFT which is a single side FFT that only returns a one-sided spectrum.

The calc_THD function produces the THD percentages shown in Figure 62 and uses Equation 11 to calculate the THD from the FFT window and only calculates the THD up until the 10th harmonic.

5 Measurements and Results

5.1 Square, triangular and sine waves

5.1.1 Triangle, Square and Sine at 10kHz,1kHz and 100Hz

The below measurements are from the actual oscillator circuit at 20 V_{peak-peak} over a 27Ω load resistor. Using Matlab script, it reads the csv file from the oscilloscope in the laboratory and then plots the signals overtime. Each signal is not in phase as the csv file sampled each signal at different rates, however the frequency, duty cycle and skew are displayed for each graph. Using Ohms law the maximum current drawn from the power-amplifier is as follows:

$$I = \frac{V_{max_{pk}}}{R} = \frac{10.2}{27} = 0.38A$$

This maximum current is only sustained when the duty cycle of the square wave is greater than 95%. Only then the average voltage over the load is 10.2V_{peak}. Due to the nature of an oscillator the voltage varies from 0-30V. Although the voltage is positive due to the virtual ground point. All op-amps are biased around this virtual point created. The square wave output closely matches that of the spice plots output.

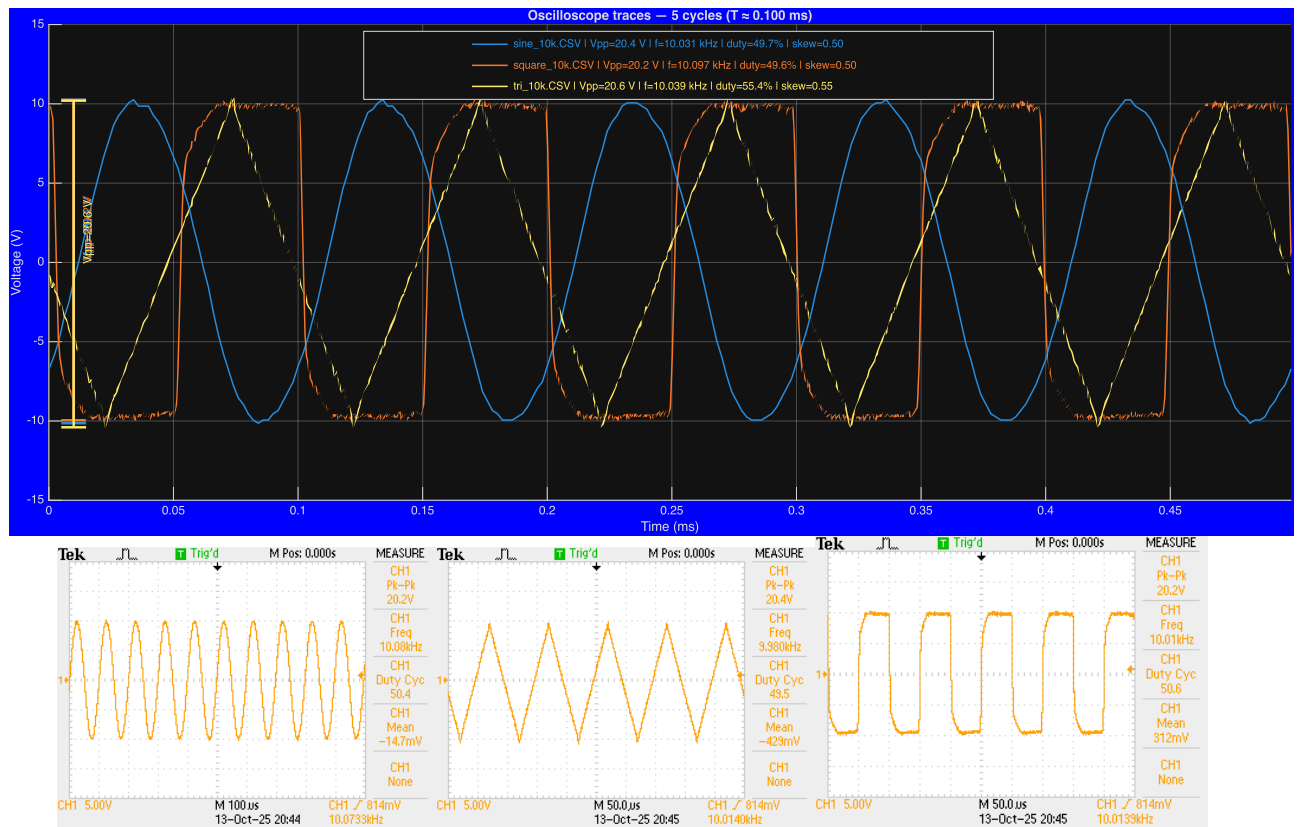


Figure 47 10kHz waveforms on MATLAB and oscilloscope measurements

At high frequencies and high peak-to-peak voltages. The square wave has rounded transitions between $+V_{sat}$ and $-V_{sat}$. This is due to the slew rate of the op-amp. The slew rate is the maximum rate at which the output can change voltage over time. The Slew rate of the TL081 op-amp is $13V/\mu s$. At a frequency of 10kHz and large amplitudes the voltage change required per unit time exceeds the limitations of the op-amp. It matches the spice outputs in Figure 8 since a non-idealized op-amp was used in LTspice for simulations.

Due to the instantaneous need for a square wave to transition from $-V_{sat}$ to $+V_{sat}$ at low frequencies the period of the square wave is much larger than $1\mu s$, thus the slew rate is not as noticeable. However, as the frequency increases and the period of the square wave decreases. The slew rate limitation of the op-amp becomes more noticeable. Better practical implementation would be to use a op-amp with a higher SR or reducing the amplitude of the output signal.

At higher frequencies the triangle waveform starts to exhibit slight nonlinearity, noticeable as a “bubble” or rounding at the peak of the waveform. This imperfection is also caused by the SR limitation of the op-amp. Additional factors like non-ideal capacitor behaviour also contributes to this curvature making the top of the triangle wave deviate from expected sharp edge as seen in the lower frequency diagram, Figure 8

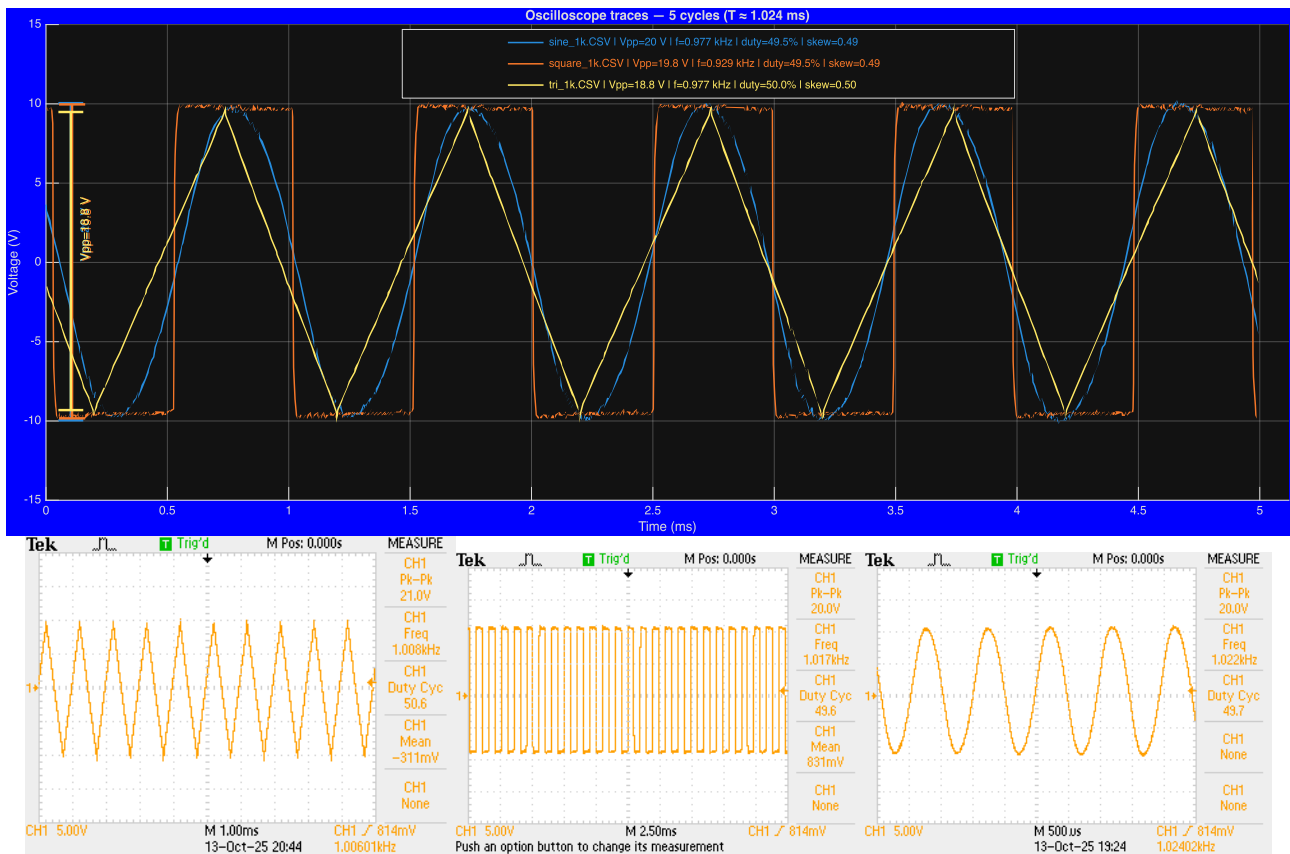


Figure 48 1kHz waveforms on MATLAB and oscilloscope measurements

As seen in the Figure 48above the square, triangle and sine wave looks near perfect. It matches the spice design in section 3 perfectly. All the harmonic information is maintained in the square wave therefore steep transitions between +Vsat and -Vsat is seen.

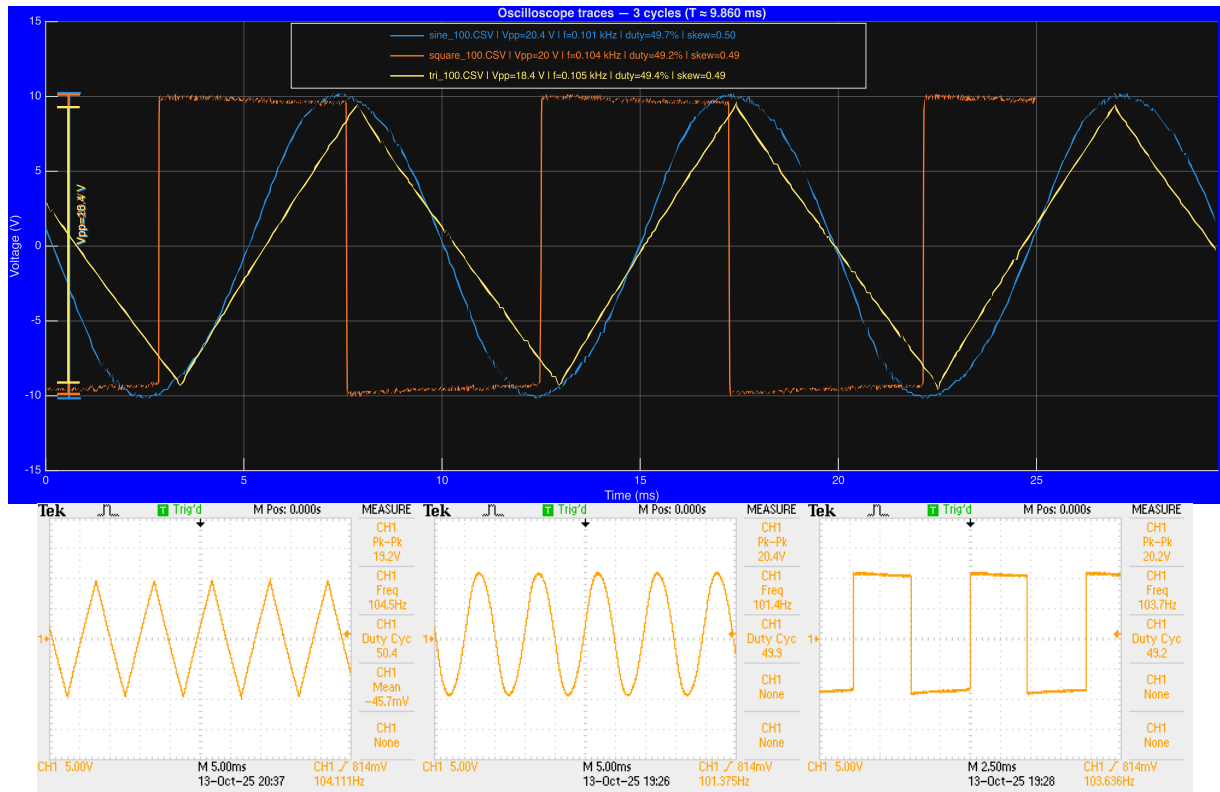


Figure 49 100Hz waveforms on MATLAB and oscilloscope measurements

In figure 49 the 100Hz, waveforms follow the desired response as a low frequency waveform does not exhibit SR limitations. Thus the sharp square edges and high linearity seen in the triangle waveform.

Overall, the measured waveforms at 10kHz, 1kHz and 100Hz closely replicate the results obtained in LTSPICE in section 3.2. This validates the theoretical oscillator model. There is no massive deviation from virtual ground as different signals are selected at the same frequency verifying the virtual ground biasing design.

However, at high frequencies and amplitude using a op-amp with better SR will result in a cleaner square and triangle waveform

5.1.1.1 Power driving capabilities

Table 5 Power and current capabilities of waveforms

Waveform	Average Vpk(V)	Ipeak over 27Ω(mA)	Average Vrms(V)	Irms(mA)	Peak power (W)	Rms power (W)
Sinewave	10.1	374	7.14	264	3.78	1.89
Square wave	10.1	374	10.067	373	3.78	3.78
Triangle wave	10.1	374	5.83	216	3.78	1.26

Using the RMS and peak voltage of the waveform the average power dissipation and maximum dissipation the circuit can be calculated. The RMS voltage of a symmetrical triangle is $V_{rms} = \frac{V_p}{\sqrt{3}}$, for a sinewave, $V_{rms} = \frac{V_p}{\sqrt{2}}$ and the Rms voltage for a 50% duty cycle square wave, $V_{rms} = \frac{V_{pk-pk}}{2}$. The Rms and peak power is calculated using $P = \frac{V^2}{R}$

Table 5 shows that the power rating of the load. Since a 5W 27 Ω load was used only 3.8W at maximum was dissipated into the load. This also verify that the signal generator was able to supply the current and power needed to drive a low resistive load.

5.1.2 power amplifier

how a single supply voltage system operates. With only one sided 30V supply and no negative rail a virtual ground needs to setup. Signals now need to remain between 0V and Vcc. In this section the virtual ground will be test to see if there are any oscillation or loading effects on it. In the system design there are two virtual ground points. One for biasing signals and only used for the oscillatory stage, low power stage. The other is for the power stage, sink/sourcing current coming from the load.

5.1.2.1 virtual ground measurement of biasing circuit

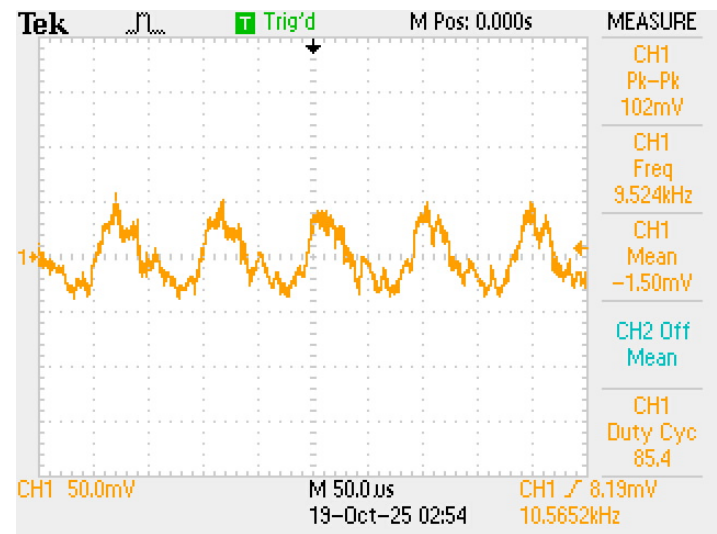


Figure 50 virtual ground measurement at full load

In Figure 50 virtual ground measurement at full load the virtual ground is measured at full load and compared to the LTspice plot inFigure 39. Both plots show a periodic ripple voltage related to loading effects from different circuit. The mean voltage does not drift over time showing a constant reference point under dynamic loads. The AC coupling on the Oscilloscope is on head why there is no DC-offset. The signal is measured from the virtual ground node to actual ground. In Figure 50 there is slightly higher noise content compared to the LTspice plot. This is a accurate representation of what is to be expected in a real-world scenario.

The use of small 100nF decoupling capacitors near supply pins and op-amps show case a reduction high frequency noise from components. Using a ground plane to mitigate ground return loops also show case better stability.

5.1.2.2 power virtual ground measurement

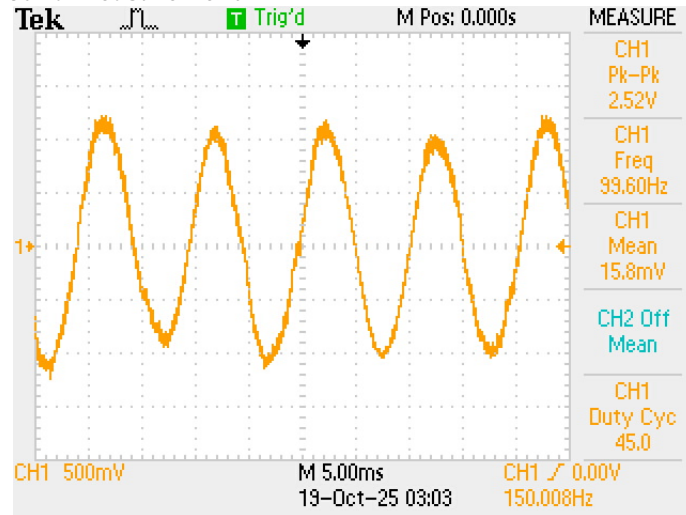


Figure 51 power virtual ground at full load

The ripple voltage for the power virtual ground after the load is a lot higher than that of the normal virtual ground. The peak-to-peak voltage is about 2.52V. The high amplitude ripple is due to limitations of the shunt capacitor on the power VG node. Any impedance seen on the VG node results in voltage fluctuation. The capacitor acts as a battery under big transient requirements from the load after fast switching. To decrease the ripple voltage on the output a larger capacitor would need to be used or increasing the frequency. That is due to the ripple voltage equation.

$$V_r = \frac{I_{load}}{fC} = \frac{400m}{100 \times 4.7m} = 0.426mV$$

This does not compare to the LT-spice measurements. The measured ripple voltage on the oscilloscope. The Design seems to have a short coming with the power VG measurement. Causing a large sinusoidal return voltage. To fix this problem daisy chaining multiple shunt capacitors on the power virtual ground to effectively increase the total capacitance.

5.1.2.3 maximum current supplied by circuit

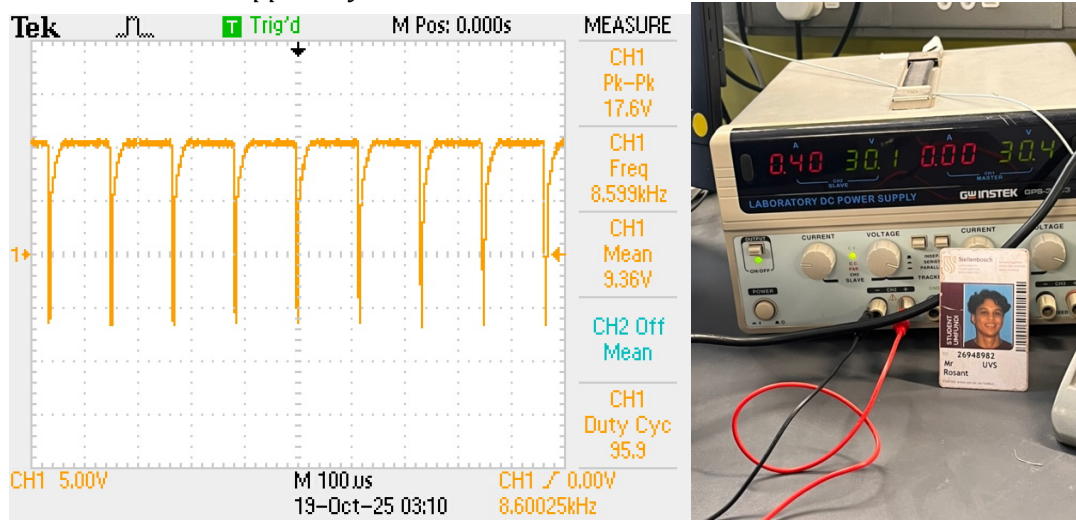


Figure 52 maximum current supplied at full duty cycle square wave

The maximum supplied current at a full duty cycle full amplitude square wave can be seen as 400mA from the DC power supply. Of the 400mA about 50mA is used for the rest of the circuit requirements and lower power stage and 350mA is used for the power amplifiers to drive the load. This

shows that the power amp was designed correctly to source and sink the required current to drive a 27Ω load at 10V amplitude.

5.2 Duty cycle and skew

5.2.1 duty cycle measurements

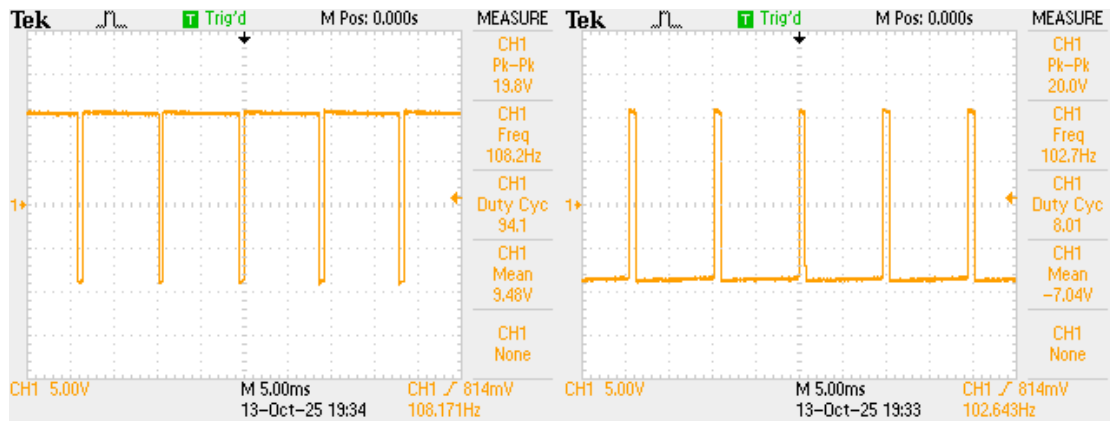


Figure 53 Duty cycle of square wave for 90% and 10% at 100Hz

At 100Hz. The oscilloscope measurement shows a clean, sharp transition with minimal rounding. Confirming that the slew rate limitation are negligible at low frequencies. The duty cycle is accurately maintained at **approximately 9.1%**, which are in the design specifications. At 90% duty cycle the circuit still produces a well-defined edge with a constant amplitude of 19.8V peak-peak. This show cases the stability of the system.



Figure 54 Duty cycle of of square wave for 90% and 10% at 100Hz using software

Using the CSV measurements in the software developed it was able to confirm the measurement and accurately calculate the Duty cycle. The software accurately calculated the above 90% duty cycle and compared to the software there was only a 0.2% difference.

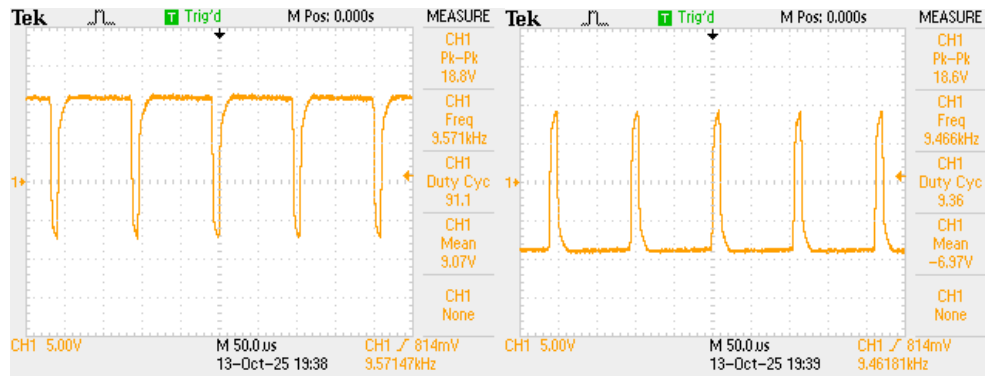


Figure 55 Duty cycle of square wave for 90% and 10% at 10kHz

For the 10kHz waveform. The 10% duty cycle have slight rounding at the edges. This is due to the op-amp slew rate limitation. Despite this a <10% duty cycle is still achievable. The slew rate limitation can be seen in the >90% duty cycle. The amplitude decreases from 20Vpp to 18.8Vpp as the duty cycle is increased. This is due to the diodes turning off momentarily which was noted in section 3.2.4.1, where the nonlinearity is seen in the triangle waveform clipping as the skew is adjust to maximum.

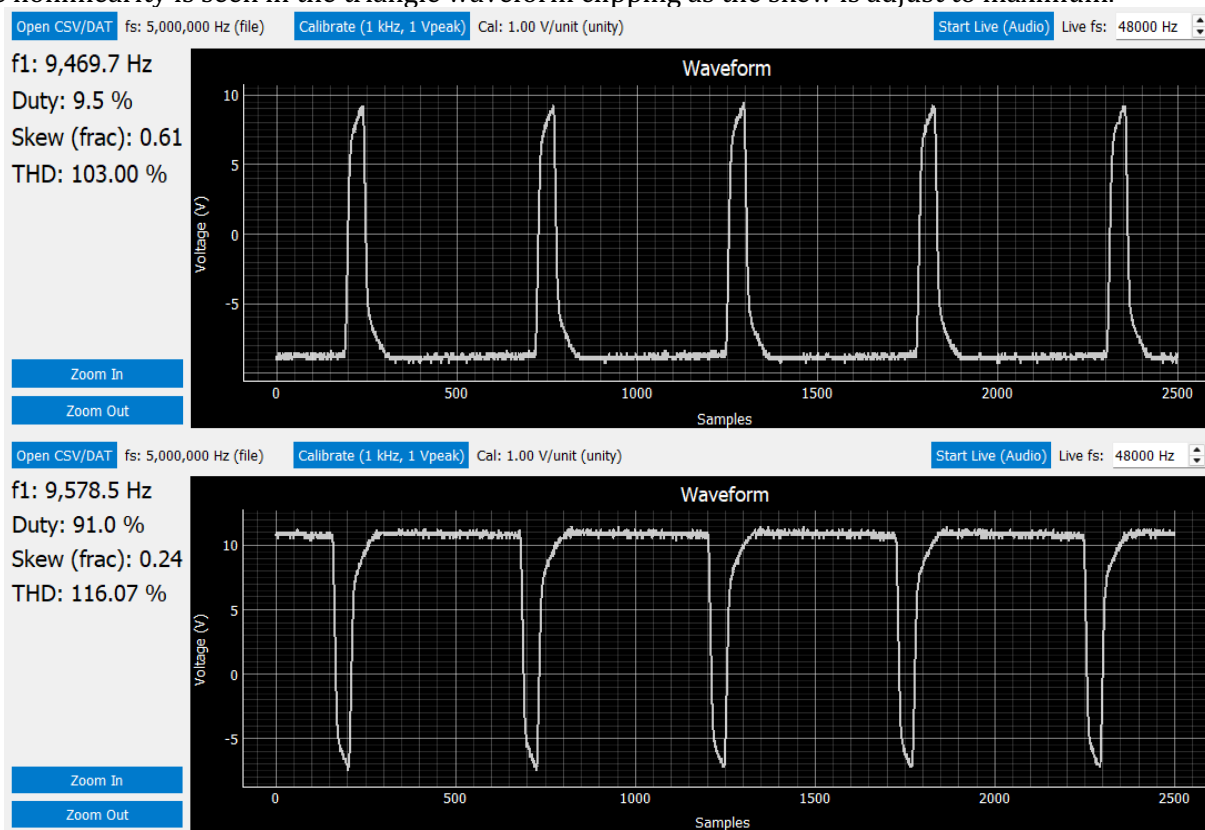


Figure 56 Duty cycle of square wave for 90% and 10% at 10kHz using software

The same discrepancies can be seen in the software measurement of the 90% and 10% duty cycle. The peak-to-peak amplitude decreases as well as the frequency that drifts from 10kHz. Down to 9.5kHz. the duty cycle is still accurately calculated and printed.

The frequency at 100Hz has no coupling on the duty cycle and has less than 2% deviation from 100Hz. Which show case the design choice for low frequency. At 10kHz there is a higher **error (5%)** due to slew rate limitations and high-frequency parasitic effects. The op-amp 13V/ μ s slew rate becomes a limiting factor at 10kHz.

Table 6 Duty cycle comparison with software and oscilloscope

Frequency (Hz)	Duty Cycle (%)	Oscilloscope (%)	Software (%)	Error(%)
100	10	8.01	8	0.124
100	90	94.1	93.9	0.212
10k	10	9.96	9.5	4.62
10k	90	91.1	91	0.11

The diode-based duty cycle adjustment successfully provides the control from 10% to 90% meeting design specifications across the frequency range, but as noted in section 3.2.4.1 the coupling between duty cycle and frequency occurs at high frequencies with extreme duty cycles >90% causing some performance degradation.

5.2.2 Skew measurements

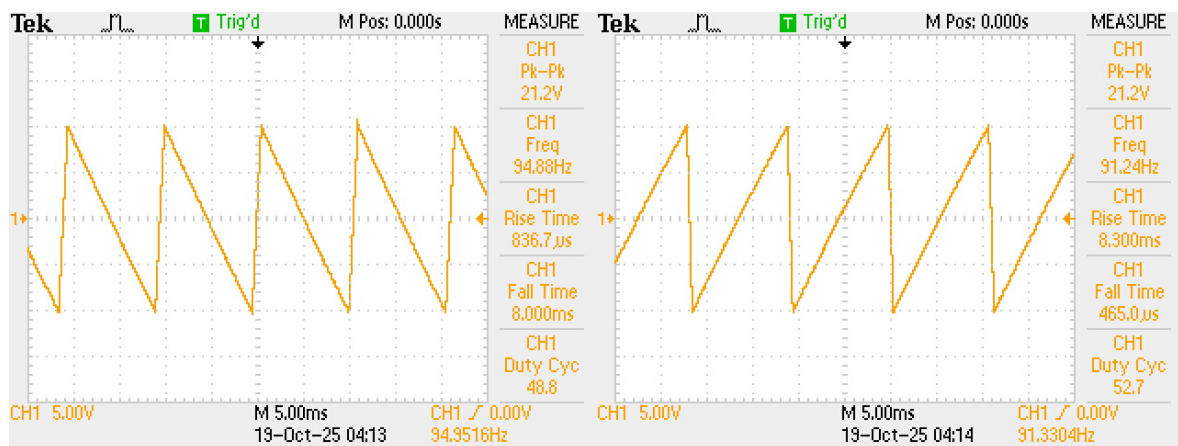


Figure 57 skew of triangle at 0.1 and 1 for 100Hz



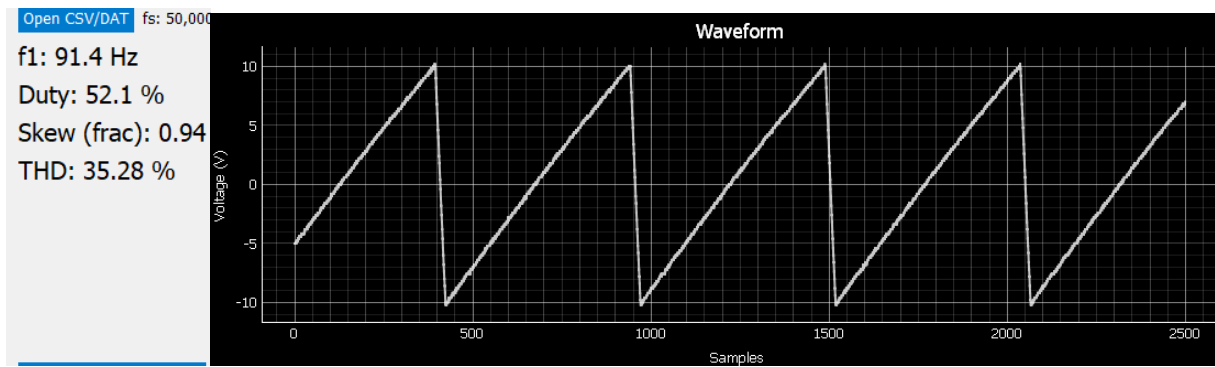


Figure 58 skew of triangle at 0.1 and 1 for 100Hz using software

The triangle wave form depicts both extremes of skew both in the software and on the oscilloscope. At 100Hz the waveform is sharp and responsive and does not show any slew rate limitation. Depicting highly linear waveforms

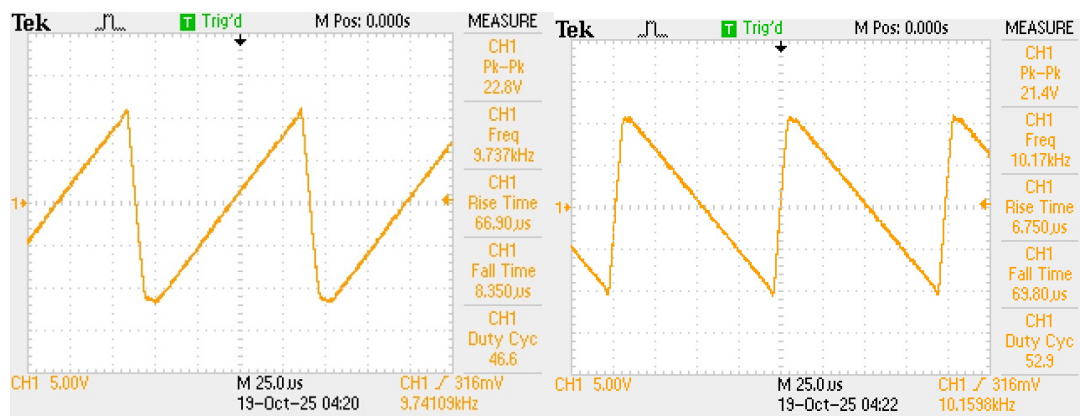


Figure 59 skew of triangle at 0.1 and 1 for 10kHz

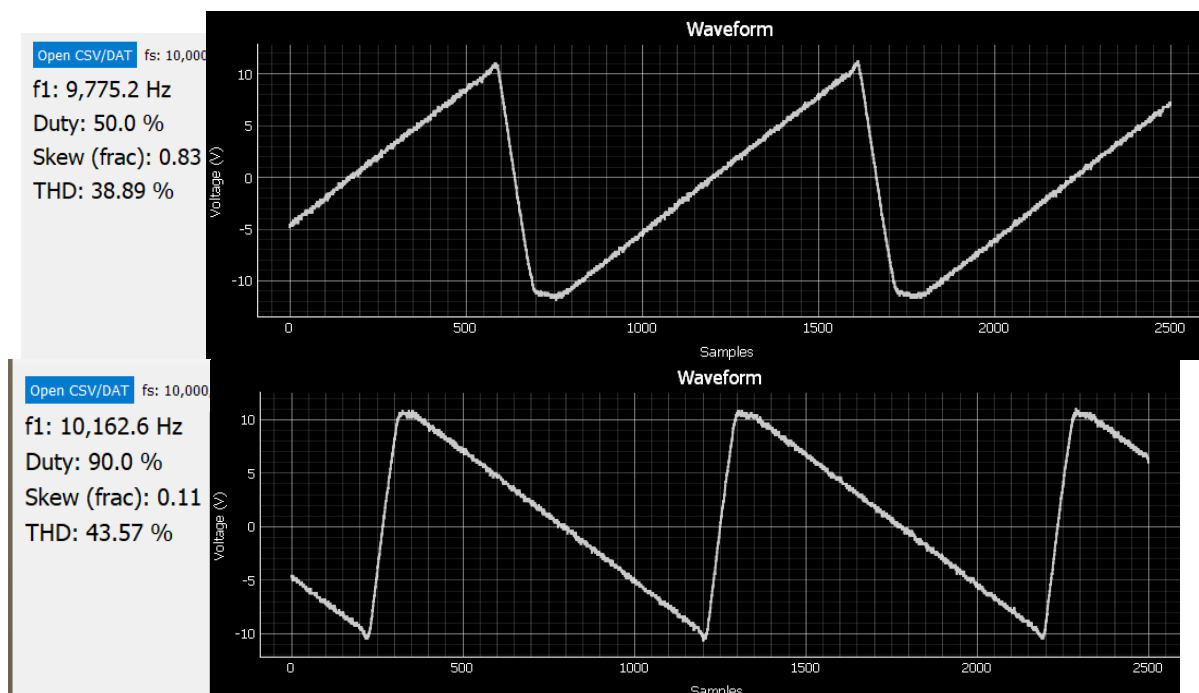


Figure 60 skew of triangle at 0.1 and 1 for 10kHz using software

However, the 10kHz waveform has slight rounding at the peak of the waveform. Which is due to the SR describe in section 5.1.1. The skew is effective but struggles to get above 0.9 or below 0.1. The amplitude is also no longer constant and grows by 500mV on either side.

To calculate the skew from the oscilloscope. The following equation is used:

$$skew = \frac{rise_{time}}{fall_{time}}$$

The skew calculation used in the software works only from 0.1-1, where 0.1 being skew left and 1 being skewed all the way right. This is incorrect and a shortcoming of my software. Hence Only skew at 0.1 can be validated. Skew at >0.5 needs to be multiplied by 10.

Table 7 skew of triangle from oscilloscope compared to software

Frequency (Hz)	Skew setting	Software skew	Oscilloscope skew	Error between software and oscilloscope(%)
100	0.1	0.1	0.095	5
100	1	9.4	9.47	0.74
10k	0.1	0.11	0.09	18
10k	1	8.3	8.0	3.6

At low frequencies the triangle waveform skew can be adjusted very accurately, and the waveform is linear. The software and oscilloscope measurements closely match each other but at high frequencies the practical limitations such as op-amp slew rate. This leads to rounding of the sharp edges smaller skew values and slight amplitude amplification since the slower ramp results in a larger voltage swing for a given period.

5.3 Sine wave total harmonic distortion

5.3.1 Measuring THD using FFT

Using a Fast Fourier transform on the oscilloscope the following plots were obtained at 10kHz and 100Hz.

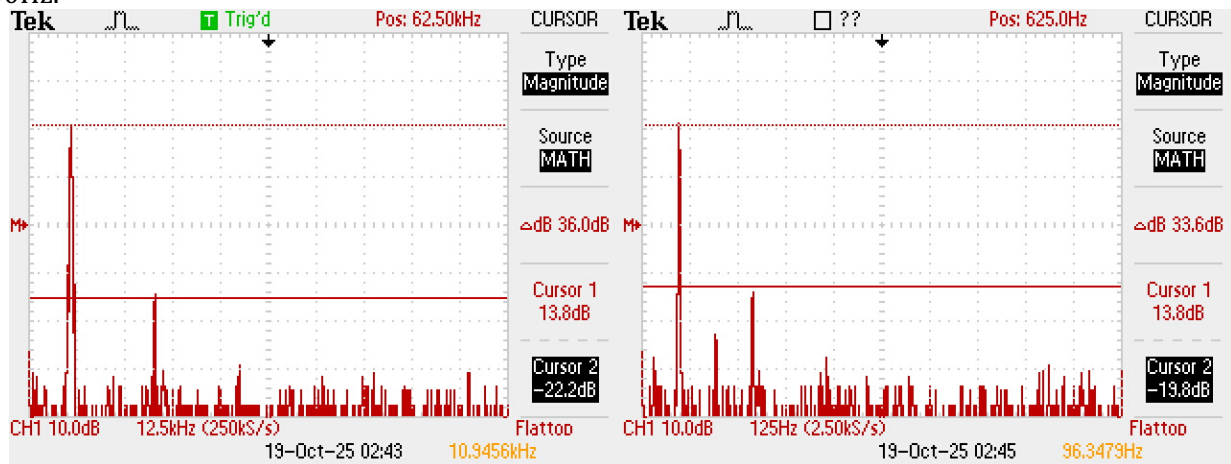


Figure 61 FFT of Sinewave at 100Hz and 10kHz

The specification for the sinewave was to have a THD of lower than 1% at the frequency bounds, 100Hz and 10kHz. Each spectral peak shows the strength of the frequency in decibels (dB). Extracting the total harmonic distortion from the dB values, they first must be converted to linear form. The fundamental peak is the main signal any harmonics after that will cause the sinewave to distort. Converting the spectral components to a linear amplitude uses the following equation.

$$A = 10^{\left(\frac{L}{20}\right)} \text{ eq. 2}$$

$$THD = \sqrt{\frac{x_1^2 + x_2^2 \dots + x_n^2}{x_1^2}}$$

For 10KHz

Fundamental Harmonic: $A_1 = 10^{\frac{13.8}{20}} = 4.897$

Second Harmonic: $A_2 = 10^{\frac{-22.2}{20}} = 0.077624$

$$THD = \sqrt{\frac{A_2^2}{A_1^2}} = \frac{A_2}{A_1} = 0.01585 \approx 1.59\%$$

For 100Hz:

$A_1 = 4.8977$

$A_2 = 0.10232$

$$THD = \frac{A_2}{A_1} = 0.02089 \approx 2.1\%$$

The THD at 10kHz and at 100Hz is not up to specification concluding that the wien-bridge is not a good choice for a sinewave oscillator. To decrease the THD of the sinewave a trim pot is used in the feedback from loop of the circuit in figure 6. This allows for the sinewave to be tuned while it is operating, however due to the tolerance difference of the dual POT trimming the feedback loop to decrease the THD causes the frequency and the Amplitude of the sinewave to be coupled. A trade off had to be made by slightly increasing the THD to keep the amplitude constant while shifting through the frequency range. For the THD to be lower than 1% the Gain of the feedback loop needs to be exactly 3.

5.3.2 FFT and THD using software

Using the software developed it automatically calculates the THD using FFT using the function in section 4.5 above.

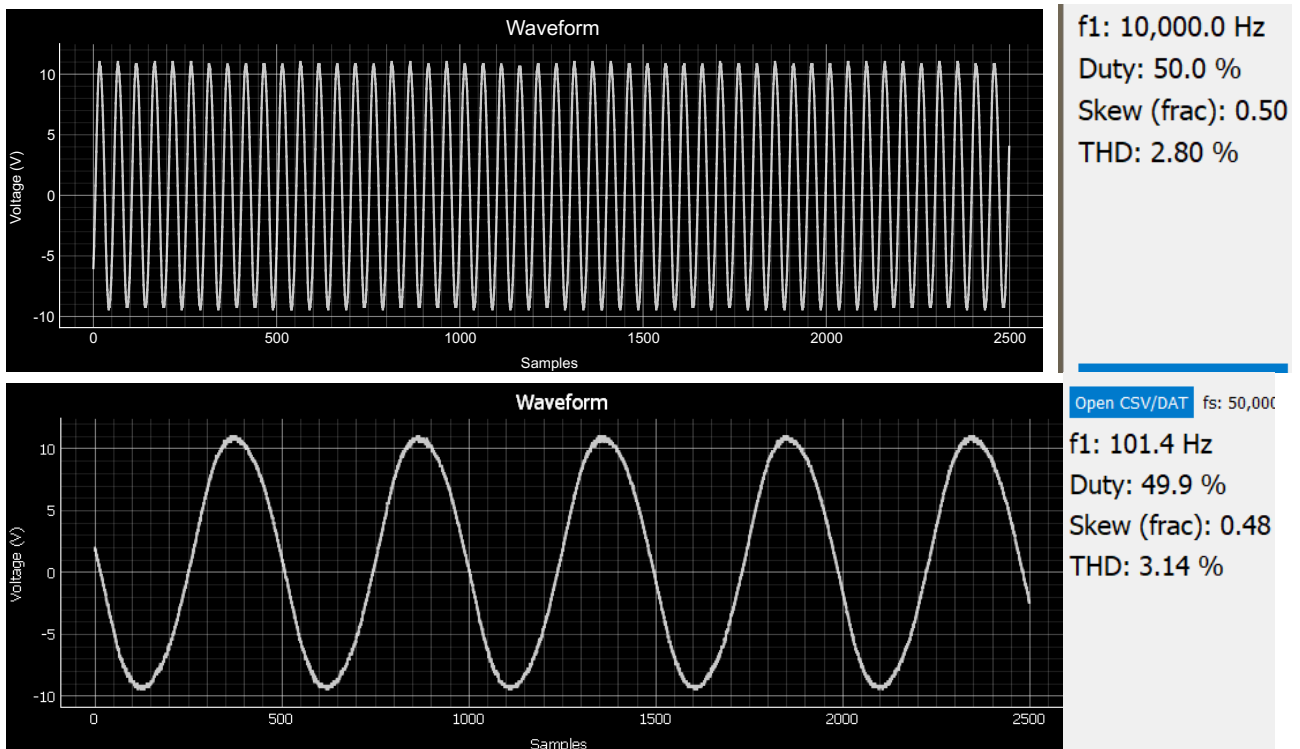


Figure 62 THD of sinewave at 10kHz and 100Hz

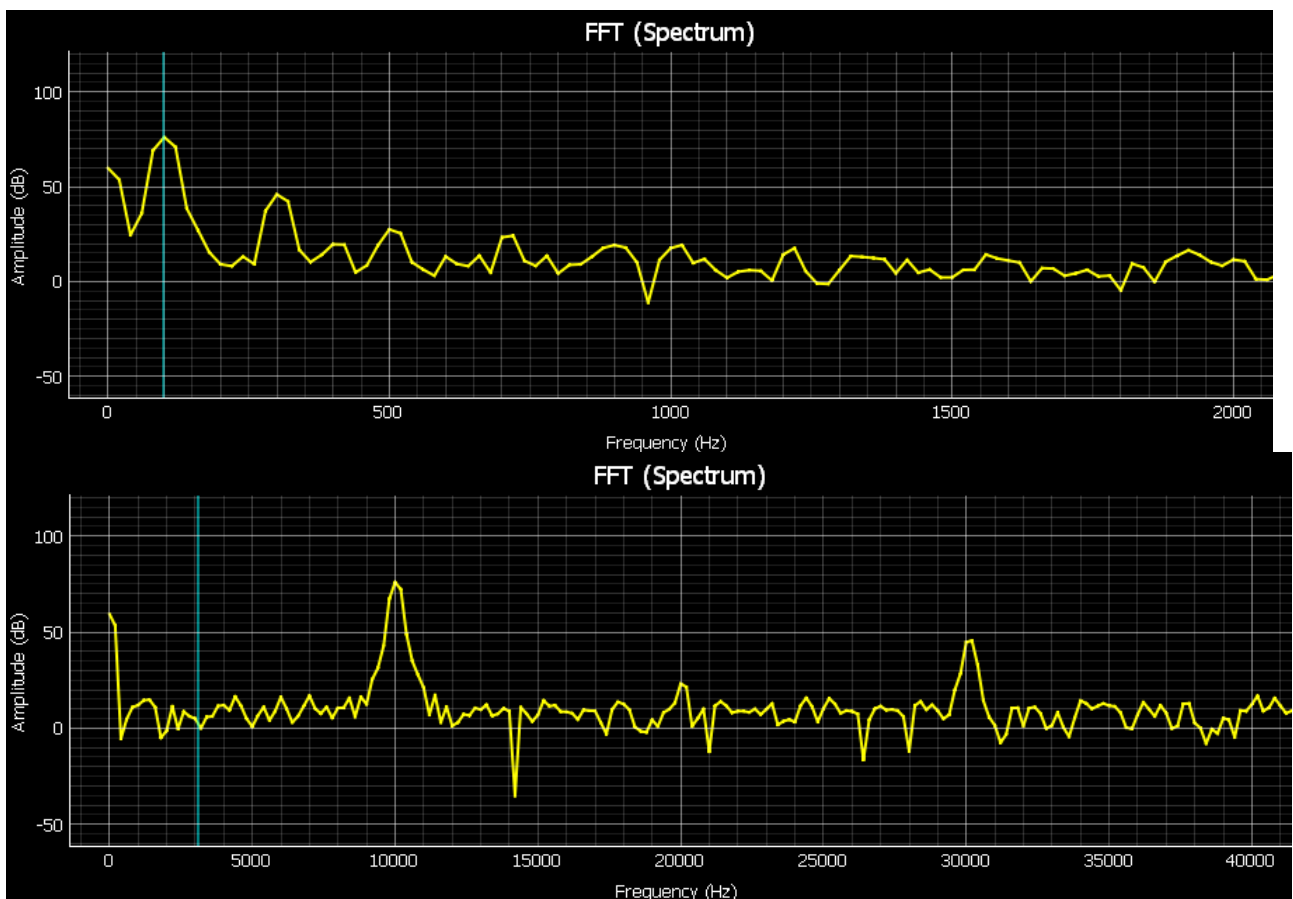


Figure 63 FFT using software

In Figure 63 the FFT from the software is seen. The top graph is the FFT of a sinewave at 100Hz. As seen there is a huge fundamental harmonic at the 100Hz, and any subsequent harmonic has a very small amplitude which strongly reflect the same results as the FFT from the oscilloscope. The bottom FFT plot is for the 10kHz sinewave, again the same fundamental harmonic at the 10kHz frequency point and then

a smaller harmonic at 30kHz. This again shows how accurate the software is to an actual oscilloscope measurement and verifies that the THD of the sinewave is low. Not under 1% but very close to 1%

As Seen the THD is not lower than 1% which is to be expected. The THD in the Wien-bridge rarely falls below 1% due to several practical sources and nonlinearity. Many factors influence the THD including the tolerance in the dual potentiometer provided. As well small discrepancies in the feedback path or the lead-lag network can deviate the from the precise gain needed from calculation resulting in the sinewave distorting and harmonics increasing. The diodes also play a major role as its non-linear switching can introduce major gain changes and add distortion.

The op-amp also contributes errors. One of the errors being the non-zero input bias current. These parasitic currents interact with the resistors in the lead-lag network, producing a unintended voltage drop. Even at low levels it can slightly shift the feedback conditions away from its ideal level which can cause amplitude errors or distortion on the sinewave.

5.3.3 How to better the THD

Here are a few solutions to better the THD of the sinewave. Using a stabilization element like a small incandescent light bulb to get automatic gain control will offer smooth and gradual resistance change with output amplitude, minimizing distortion far better than using simple diodes. Using precise dual potentiometers and then gain trimming the signal while in operation. The accurate gain adjustment alongside precision trimmers will prevent the amplitude from overshooting.

5.4 Level adjusts

The oscillator level shifting capabilities are from $-V_{sat}$ to V_{sat} . The Level-adjusting circuit is added after the amplification circuit. The reason for placing it after the amplification stage is to have separate control of the signal peak-peak voltage as which point the signal operates around. This allows the system to be more dynamic. The required level-adjust capability should be that the signal can be adjusted above $\pm 5V$ and the amplitude of the signal being independent of the signals level. Since an adjustable virtual ground is used as the reference voltage the output signal can swing to both rails.

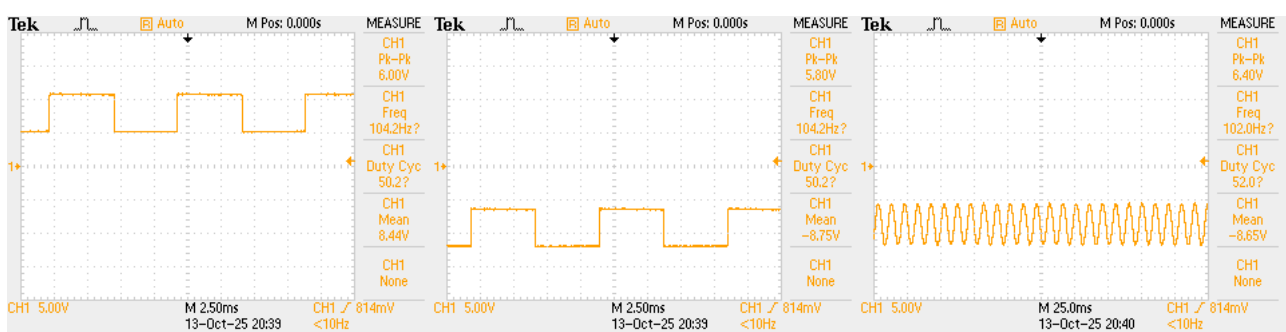


Figure 64 Level adjust of square and sinewave

As seen in figure 10 the level/mean stays constant for different selected signals, and the signals can also be adjusted above the required $\pm 5V$ specification. If a limit was required for the level shifter series resistors are to be implemented on each side of the wiper to limit the maximum threshold voltages that the reference voltage could swing to.

6 Conclusions

The analog signal generator was successfully designed hitting all the ATP defined in the project specifications. By producing a stable sine, square and triangle waveform over the frequency range. The waveforms frequency, duty cycle, skew and level adjust are all independent of each other. The theoretical oscillator equations derived for each oscillator stage closely matched both the LTspice simulations as well as the measurements taken in the lab. This validated the mathematical models and design assumptions made. The square and triangle wave exhibit precise duty-cycle and skew adjustment within 5% tolerance. The Wien-bridge oscillator achieved low THD from about 1.6%-2%. Although this is not below 1% for a Wien-bridge design this is very good.

The single supply architecture and virtual ground demonstrated a stable biasing with minimal ripple lower than 500mVpk-pk while the power virtual ground has more oscillation due to the huge capacitor on the return path. The Class AB power amplifier provided sufficient current to drive a 27Ω load without significant distortion. Limitations observed on the design were primarily due to the op-amp slew rate constraints as well as component tolerances. My design short coming lie in the non-linearity of the diode circuit in triangle/square circuit. Causing flat tops at high amplitude for my triangle circuit. The use of high tolerance resistors also cause deviations from the theoretical results of spice. To better the design. More decoupling capacitors needs to be implemented as well as more low tolerance POTs and resistors.

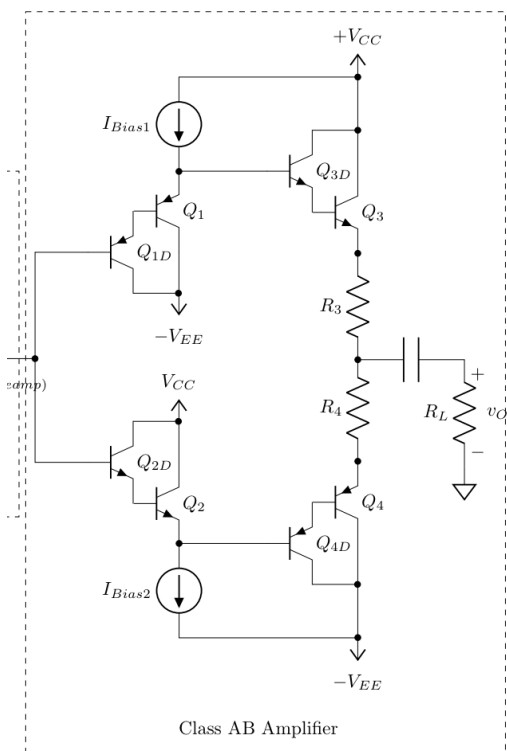
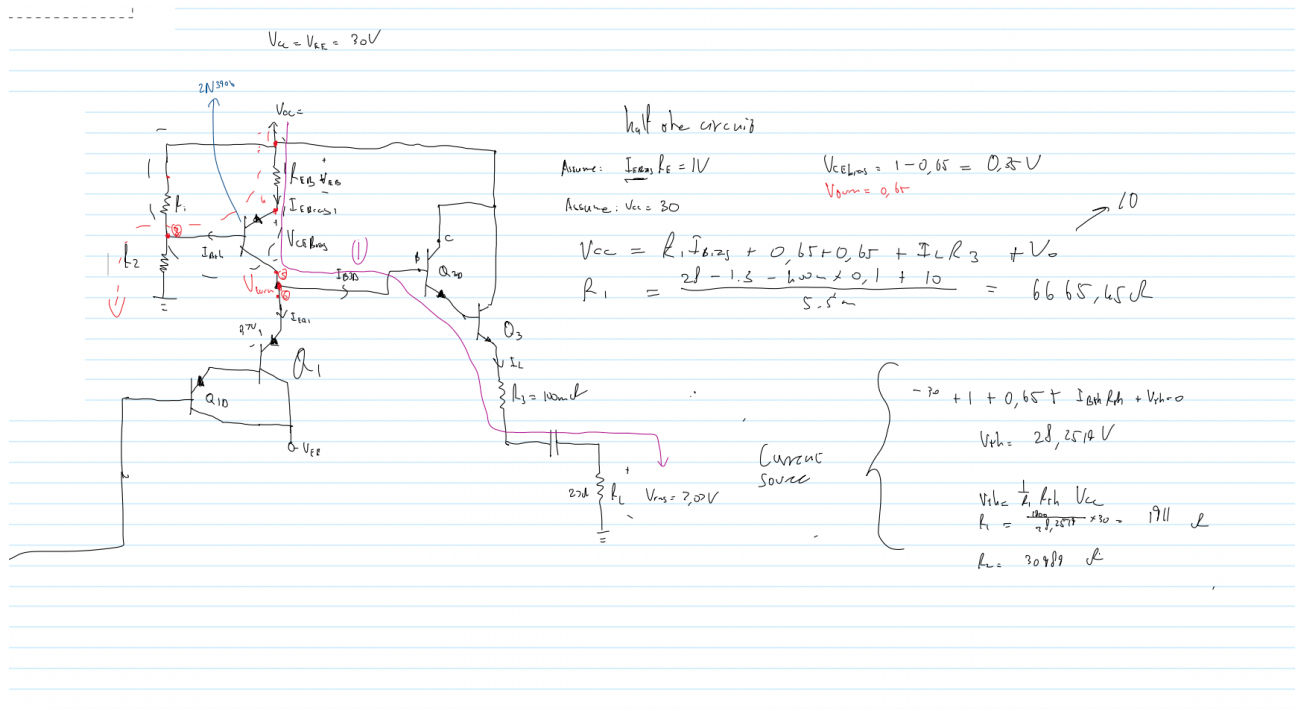
The custom python software validated all oscilloscope data perfectly and was able to correctly take in live signals from a signal generator. It confirmed excellent alignment with hardware reading which showcase the integration of analog and digital software measurement device.

Overall, the project reflects a robust use of analog design theory, practical design and prototyping. Future improvements would be to use a different sinusoidal oscillator like a phase shift oscillator to bring down the THD. Using better high slew rate op-amps or a more linear stabilization element like a lamp instead of diodes.

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7 Appendix



Full - Derivation:

Schmitt trigger configured as non-inverting.

Comparator threshold voltage.

$$\beta = \frac{R_2}{R_1 + R_2}$$

Frequency before display:

$$f_{low} = 1 \text{ kHz}$$

$$f_{high} = 1.31 \text{ kHz}$$

$$pH = 10 \text{ kHz}$$

Integrator input resistance

$$R_{in+} = R_2 + R_1 + R_3 + R_4$$

$$R_{in-} = R_2 + R_1 + R_3 + R_4$$

$$R_{in-} = R_2 + R_1 + R_3 + R_4 + (1 - \beta) R_2$$

Resistor before diode divides up Schmitt trigger voltage level

$$K_{CP} = \frac{R_2}{R_1 + R_2} = \frac{(1 - \beta) R_2 + R_2}{R_1 + R_2 + R_3 + R_4} = \frac{(1 - \beta) R_2 + 1.31 R_2}{12 \text{ kHz}}$$

only at duty cycle is the same

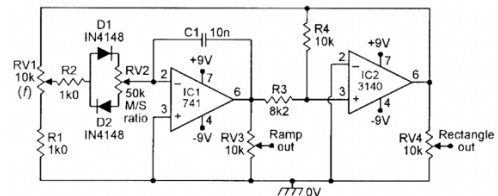


Figure 17. 100Hz-1kHz ramp/rectangle generator with variable slope-M/S ratio.

$$K_{CP} = \frac{R_2}{R_1 + R_2} = \frac{(1 - \beta) R_2 + R_2}{R_1 + R_2 + R_3 + R_4} = \frac{(1 - \beta) R_2 + 1.31 R_2}{12 \text{ kHz}}$$

only at duty cycle is the same

Geometry charge/discharge time

$$I = \frac{V}{R} \quad \frac{dV}{dt} = \frac{V}{RC}$$

To get the upper ramp is given f_{in} - V_{in} to V_{out}

$$\Delta V_{up} = V_{in+} - V_{in-} = (+V_1) - (-V_1) = 2V_1$$

Since time is defined as distance over speed. the slope of the angle is set by current source.

$$\left| \frac{dV}{dt} \right| = \frac{I}{C} = \frac{V_{CC}}{RC} \quad \text{Ramp/V}$$

Voltage at X_1 position

$$V_{X1} = k_1(p) V_{CC}$$

$$t_{up} = \frac{\Delta V}{\left| \frac{dV}{dt} \right|} = \frac{2V_1 C}{k_1(p) V_{CC}} \quad \text{Discrete Voltage period charge}$$

$V_1 = \text{threshold voltage}$

$$T_{total} = t_{up} + t_{down} = \frac{1}{f} = \frac{1}{f_{low}} + \frac{1}{f_{high}}$$

$$= \frac{2V_1 C}{k_1(p) V_{CC}} + \frac{2V_1 C}{k_2(p) V_{CC}}$$

for 50% duty cycle

$$k_1 = k_2 = k$$

$$\therefore T = \frac{2V_1 C}{k V_{CC}} (k_1 + k_2)$$

$$\therefore f = \frac{1}{T} = \frac{k V_{CC}}{2V_1 C (k_1 + k_2)}$$

$$f_{low} = 1 \text{ kHz} \quad V_{CC} = 9 \text{ V}$$

$$f_{high} = 1.31 \text{ kHz}$$

$$f_{CP} = \frac{k(p)}{2C} \cdot \frac{V_{CC}}{2V_1 + 2V_1 + V_1 + V_1 + V_1 + V_1}$$

$$= \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$f_{CP} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$f_{CP} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$k(1) = \frac{1}{1.2}$$

$$k(2) = \frac{1}{1.2}$$

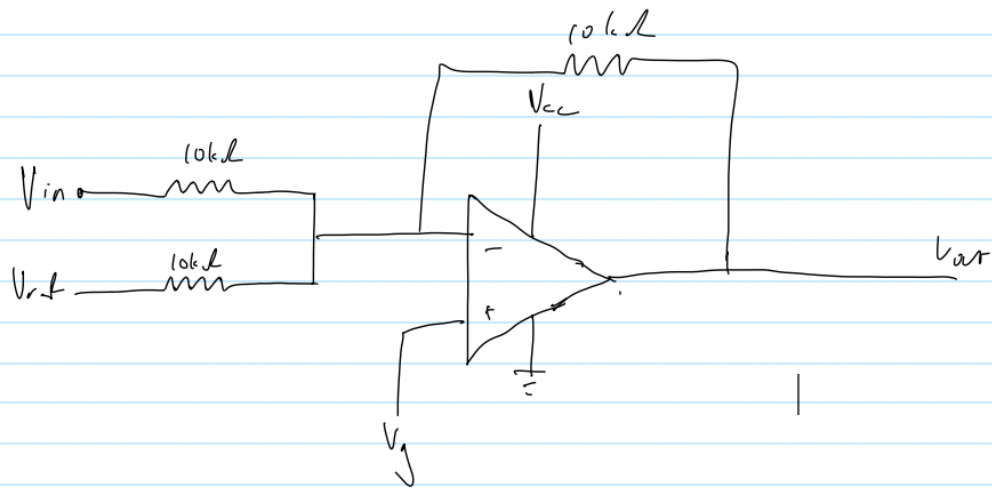
$$f(1) = \frac{1}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$f(2) = \frac{1}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$f_{CP} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

$$f_{CP} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}} = \frac{k(p)}{2 \cdot 1.31 \cdot 2} \cdot \frac{9}{2 \cdot 1.31 + 10 \text{ kHz}}$$

Figure 65 derivation of triangle



$$KCL @ V_n: \left(\frac{V_n - V_{in}}{10k\Omega} + \frac{V_n - V_{ref}}{10k\Omega} + \frac{V_n - V_{out}}{10k\Omega} \right) = 0$$

$$V_n \left(\frac{3}{10k\Omega} \right) - \frac{1}{10k\Omega} (V_{in} + V_{ref}) = \frac{V_{out}}{10k\Omega}$$

$$3V_n - (V_{in} + V_{ref}) = V_{out}$$

Since $V_n = V_{in} = V_{ref}$ ground is seen as 0

$$V_{out} = -(V_{in} + V_{ref})$$

Figure 66 level shift derivation

$$\beta = \frac{Z_2}{Z_1 + Z_2}$$

$$Z_1 = R_1 - jX_{C1} \quad \text{and} \quad Z_2 = R_2 \parallel -jX_{C2}$$

$$Z_2 = \frac{-jX_{C2} R_2}{-jX_{C2} + R_2}$$

$$Z_2 = \frac{-jX_{C2} R_2}{-jX_{C2} (1 + \frac{R_2}{-jX_{C2}})}$$

$$Z_2 = \frac{R_2}{1 + \frac{R_2}{-jX_{C2}}}$$

$$\beta = \frac{R_2}{R_2 (1 + \frac{C_2}{C_1}) + R_1 + j(\omega R_1 R_2 C_2 - \frac{1}{\omega C_1})}$$

$$\text{Now } X_{C1} = \frac{1}{\omega C_1} \quad \omega R_1 R_2 C_2 = \frac{1}{\omega C_1}$$

$$\therefore Z_1 = R_1 - \frac{j}{\omega C_1} \quad \omega^2 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$Z_2 = \frac{R_2}{1 + j\omega R_2 C_2} \quad \omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

If $C_1 = C_2 = C$ and $R_1 = R_2 = R$

$$\omega = \frac{1}{RC} \quad \text{or}$$

$$f = \frac{1}{2\pi RC}$$

To find magnitude of feedback factor

$$\beta = \frac{R_2}{R_2 (1 + \frac{C_2}{C_1}) + R_1}$$

$$\beta = \frac{1}{3}$$

$$\beta = \frac{1}{3}$$

Figure 67 full wien-bridge derivation

```
function ltspace_ascii_split_adaptive()
%% ===== USER CONFIG =====
files = { ...
    'tri_100_D90.txt', ...
    'tri_100_D10.txt', ...
    'tri_10k_D10.txt' ...
    'tri_10k_(90%).txt'...
};
max_figs    = 4;           % up to 4 figure windows
min_vpp_V   = 18.0;       % require at least this Vpp in the plotted window
hiF_thresh  = 0.9e3;      % Hz; above this, use 10 ms (else 50 ms)
save_png    = false;

%% ===== PROCESS EACH FILE =====
n = min(max_figs, numel(files));
```

```

for k = 1:n
    fn = files{k};
    if ~isfile(fn)
        warning("Skipping (not found): %s", fn);
        continue
    end

    % --- Read LTspice ASCII (tab-delimited) ---
    [t, y, yname] = read_ltspice_ascii(fn);
    t = t - t(1);

    % --- Estimate frequency from the full record (robust FFT)
    yd_full = y - mean(y);
    [f_est_full, ~] = estimate_freq_fft(t, yd_full);    % Hz

    % --- Choose window: 10 ms for >1 kHz, else 50 ms
    % --- Choose window length from estimated frequency (3 bands)
    f1 = 1e3;    % 1 kHz
    f2 = 5e3;    % 5 kHz

    if isfinite(f_est_full) && f_est_full > f2
        window_ms = 1;    % very high freq (e.g., 10 kHz)
    elseif isfinite(f_est_full) && f_est_full >= f1
        window_ms = 20;    % mid band: 1-5 kHz
    else
        window_ms = 50;    % low freq: < 1 kHz
    end
    w = window_ms/1000;
    % seconds

    % --- Find earliest window of chosen length with Vpp ≥ min_vpp_V ---
    [i0, i1] = first_window_over_vpp(t, y, w, min_vpp_V);
    if isempty(i0)
        fprintf('No %d ms window (%.0f Hz class) with Vpp ≥ %.2f V in %s -
skipping.\n', ...
            window_ms, hiF_thresh, min_vpp_V, fn);
        continue
    end

    % --- Extract that window and compute metrics INSIDE it ---
    tW = t(i0:i1);    yW = y(i0:i1);
    [vlo, vhi] = robust_bounds(yW, 0.5);    Vpp = vhi - vlo;

    yd = yW - mean(yW);

    % Frequency from Schmitt-style edge timing (robust for 10 kHz square)
    [f_est, Tper_edges] = estimate_freq_edges_schmitt(tW, yW);

    % Duty / skew as before (your function still returns T_mid if you need it)
    [duty, skew, ~] = duty_and_skew(tW, yd);

    % Fallback if edge timing failed for any reason
    if ~isfinite(f_est) || f_est <= 0
        [f_est, ~] = estimate_freq_fft(tW, yd);
    end
    thd_pct = thd_percent(tW, yW);

    % --- Plot one figure per file ---
    figure('Name', fn, 'NumberTitle','off', 'Color','b', ...
        'Position', [120+60*(k-1) 140+40*(k-1) 1120 360]);
    plot(tW*1e3, yW, 'LineWidth', 1.4); grid on; box on
    xlabel('Time (ms)'); ylabel('Voltage (V)');

```

```

    title(sprintf(['%s - first %d ms window with Vpp ≥ %.2f V\n' ...
                  'Vpp=%.3g V | f=%.6g kHz | duty=%.1f%% | skew=%.2f |
THD=%.2f%%'], ...
            fn, window_ms, min_vpp_V, Vpp, f_est/1e3, 100*duty, skew, thd_pct),
    ...
        'Interpreter','none');

% exact window limits
xlim([tW(1) tW(1)+w]*1e3);

% --- Vpp bracket on left ---
x0 = (tW(1) + 0.02*w)*1e3;    cap = 0.01*window_ms;    col = [0.1 0.5 0.9];
line([x0 x0], [vlo vhi], 'Color', col, 'LineWidth', 2);
line([x0-cap x0+cap], [vlo vlo], 'Color', col, 'LineWidth', 2);
line([x0-cap x0+cap], [vhi vhi], 'Color', col, 'LineWidth', 2);
text(x0 + 0.012*window_ms, (vlo+vhi)/2, sprintf('Vpp=%.3g V', Vpp), ...
    'Rotation',90,'HorizontalAlignment','left','VerticalAlignment','middle',
    ...
    'Color', col, 'FontSize',10);

if save_png
    out = sprintf('%s_%dms.png', erase(fn,{'.txt','.TXT'}), window_ms);
    exportgraphics(gcf, out, 'Resolution', 300);
    fprintf('Saved %s\n', out);
end
end

%% ===== HELPERS (same as before; no toolboxes)
=====

function [t, y, colname] = read_ltspice_ascii(fn)
    T = readtable(fn,'Delimiter','\t','MultipleDelimsAsOne',true, ...
        'ReadVariableNames',true,'TextType','string', ...
        'VariableNamingRule','preserve');
    if any(strcmpi(T.Properties.VariableNames,'Index')), T.Index = []; end
    t = double(T{:},1}); y = double(T{:},2});
    colname = string(T.Properties.VariableNames{2});
    [t, idx] = sort(t); y = y(idx);
end

function [i0, i1] = first_window_over_vpp(t, y, w, vpp_min)
% Robust "first good window" finder with re-anchoring to avoid long flat starts.
    n = numel(t); i0 = []; i1 = []; j = 1;
    for i = 1:n
        while j < n && (t(j) - t(i)) < w, j = j + 1; end
        if (t(j) - t(i)) < w, break; end
        yseg = y(i:j);
        lo = prctile(yseg, 1); hi = prctile(yseg, 99); vpp = hi - lo;
        mid = (lo + hi)/2;
        frac_hi = mean(yseg > mid + 0.1*vpp);
        frac_lo = mean(yseg < mid - 0.1*vpp);
        if vpp >= vpp_min && (frac_hi > 0.1) && (frac_lo > 0.1)
            % re-anchor to first strong excursion so window shows oscillation
            thr = mid + 0.25*vpp;
            k = find(yseg > thr, 1, 'first'); if isempty(k), k =
round(0.1*numel(yseg)); end
            istart = i + k - 1; j2 = istart;
            while j2 < n && (t(j2) - t(istart)) < w, j2 = j2 + 1; end
            if (t(j2) - t(istart)) >= w, i0 = istart; i1 = j2; else, i0 = i; i1
= j; end
        return
    end
end

```

```

        end
    end
end

function [f0, T0] = estimate_freq_fft(t, y)
    dt = median(diff(t));
    tu = t(1):dt:t(end);
    yu = interp1(t, y, tu, 'linear','extrap'); yu = yu - mean(yu);
    L = numel(yu); N = 2^nextpow2(L); wL = hann_win(L).';
    X = fft((yu(:).').*wL, N);
    mag = abs(X); mag(1)=0; kset = 2:floor(N/2);
    [~,k] = max(mag(kset)); k = kset(k);
    if k>1 && k<floor(N/2)
        a=mag(k-1); b=mag(k); c=mag(k+1);
        d = 0.5*(a - c)/(a - 2*b + c);
        f0 = (k + d - 1) * (1/(N*dt));
    else
        f0 = (k-1) * (1/(N*dt));
    end
    if ~isfinite(f0) || f0<=0, f0 = 1/(t(end)-t(1)); end
    T0 = 1/f0;
end

function [duty, skew, T] = duty_and_skew(t, y)
    % ----- DUTY from mid-level crossings (unchanged semantics) -----
    [lo,hi] = robust_bounds(y,10);
    thr = (lo+hi)/2;
    s = y - thr;

    % rising/falling crossings and linear interpolation
    ridx = find(s(1:end-1) < 0 & s(2:end) >= 0); % rising
    fidx = find(s(1:end-1) > 0 & s(2:end) <= 0); % falling
    if isempty(ridx) || isempty(fidx)
        duty = NaN; skew = NaN; T = NaN; return;
    end

    tr = t(ridx) - s(ridx).*(t(ridx+1)-t(ridx))./(s(ridx+1)-s(ridx)); % rising
times
    tf = t(fidx) - s(fidx).*(t(fidx+1)-t(fidx))./(s(fidx+1)-s(fidx)); %
falling times (fixed)

    % ensure each rising has a subsequent falling
    if tf(1) < tr(1), tf(1) = []; end
    n = min(numel(tr), numel(tf));
    tr = tr(1:n); tf = tf(1:n);
    if numel(tr) < 2
        duty = NaN; skew = NaN; T = NaN; return;
    end

    % period & high/low durations from mid-level crossings
    T_mid = mean(diff(tr), 'omitnan');
    Th = mean(tf - tr, 'omitnan');
    Tl = T_mid - Th;

    duty = max(min(Th / (Th + Tl), 1), 0); % unchanged semantics

    % ----- SKEW from extrema timing (ratio t_down / t_up) -----
    % find local minima/maxima from derivative sign changes (no toolboxes)
    % ----- SKEW from extrema timing (ratio t_down / t_up) -----
    % find local minima/maxima from derivative sign changes (no toolboxes)
    dy = diff(y) ./ diff(t); sgn = sign(dy);
    kmax = find(sgn(1:end-1) > 0 & sgn(2:end) <= 0) + 1; % + -> - (peaks)

```

```

kmin = find(sgn(1:end-1) < 0 & sgn(2:end) >= 0) + 1;    % - -> + (troughs)

if isempty(kmax) || isempty(kmin)
    skew = NaN; T = T_mid; return;
end

% Ensure the first extremum is a trough; if not, drop the first peak
if ~isempty(kmax) && ~isempty(kmin) && kmax(1) < kmin(1)
    kmax(1) = [];
end

% We need triplets: trough(i) -> peak(i) -> trough(i+1)
m = min(numel(kmin)-1, numel(kmax));
if m < 1
    skew = NaN; T = T_mid; return;
end

kmin1 = kmin(1:m);           % trough_i
kmax1 = kmax(1:m);           % peak_i
kmin2 = kmin(2:m+1);         % trough_{i+1}

% Durations using consecutive extrema
Tup = mean(t(kmax1) - t(kmin1), 'omitnan'); % trough -> peak
Tdn = mean(t(kmin2) - t(kmax1), 'omitnan'); % peak -> next trough
T_ext = mean(t(kmin2) - t(kmin1), 'omitnan'); % period

% Skew ratio and clamping to [0.1, 10]
S = Tdn / Tup;
if ~isfinite(S) || S <= 0, S = NaN; end
skew = min(max(S, 0.1), 10);

% Prefer extrema-based period if valid
if isfinite(T_ext) && T_ext > 0
    T = T_ext;
else
    T = T_mid;
end
end

function thd_pct = thd_percent(t, y)
    dt = median(diff(t)); tu = t(1):dt:t(end);
    yu = interp1(t, y, tu, 'linear', 'extrap'); yu = yu - mean(yu);
    L = numel(yu); N = 2^nextpow2(L); wL = hann_win(L).';
    X = fft((yu(:).').*wL, N);
    mag = abs(X); mag(1)=0; k1set = 2:floor(N/2);
    [~,k1] = max(mag(k1set)); k1 = k1set(k1); if k1<=1, thd_pct=NaN; return; end
    cg = sum(wL)/L; P = (abs(X)/N/cg).^2;
    nhmax = floor((N/2)/k1); P1 = P(k1); Ph = 0;
    for h=2:nhmax, kh = round(h*k1); if kh<=N/2, Ph = Ph + P(kh); end; end
    thd_pct = 100*sqrt(Ph/P1);
end

function [lo,hi] = robust_bounds(y, p)
    if nargin<2, p = 0.5; end
    ys = sort(y(:)); n = numel(ys);
    i1 = max(1, round((p/100)*n)); i2 = min(n, round((1-p/100)*n));
    lo = ys(i1); hi = ys(i2);
end

function w = hann_win(N)
    n = 0:N-1; w = 0.5 - 0.5*cos(2*pi*n/N); w = w(:);
end

```

```

function [f_est, Tmed] = estimate_freq_edges_schmitt(t, y)
% Robust frequency estimator using Schmitt edge timing.
% Returns f_est (Hz) and Tmed = median period from rising edges.

f_est = NaN; Tmed = NaN;
if numel(t) < 4, return; end

% Robust bounds & thresholds
[lo,hi] = robust_bounds(y, 1); % 1%/99% percentiles
Vpp = hi - lo;
if Vpp <= 0, return; end
thr_lo = lo + 0.30*Vpp;
thr_hi = lo + 0.70*Vpp;

% Debounced edge tracker (low/high state)
state = (y(1) > thr_hi); % 0=low, 1=high
rises = [];

for k = 1:numel(y)-1
    y0 = y(k); y1 = y(k+1);
    t0 = t(k); t1 = t(k+1);

    if ~state
        % LOW: look for crossing up through thr_hi
        if y0 < thr_hi && y1 >= thr_hi
            % linear interpolate crossing time
            tr = t0 + (thr_hi - y0) * (t1 - t0) / (y1 - y0);
            rises(end+1) = tr; %#ok<AGROW>
            state = 1;
        end
    else
        % HIGH: wait until we cross down through thr_lo before arming again
        if y0 > thr_lo && y1 <= thr_lo
            state = 0;
        end
    end
end

if numel(rises) < 2
    return;
end

dT = diff(rises);
% Remove extreme outliers (edge glitches)
q = quantile(dT,[0.1 0.9]);
dT = dT(dT>=q(1) & dT<=q(2));
if isempty(dT), return; end

Tmed = median(dT);
if Tmed > 0
    f_est = 1 / Tmed;
end
end

```

above matlab code used for LTspice measurements

```

%% === CONFIG ===
files = {'square_10k_D10.CSV'};
f_known = []; % Hz (set [] to auto-estimate)
f_hint = 10000; % Hz; [] if unknown (only narrows search)
Ncycles = 7; % show EXACTLY this many cycles

%% === FIGURE ===
figure('Color','b'); hold on; grid on
set(gcf,'Position',[100 100 1200 420]);

colors = lines(numel(files));

% collect handles/labels only for traces that actually plotted
hLines = gobjects(0);
labels = {};
metrics = struct('file','', 'Vpp',[], 'freq',[], 'duty',[], 'skew',[]);

%% === LOOP FILES ===
for k = 1:numel(files)
    assert(isfile(files{k}), "Can't find: " + files{k});

    % --- load (time = 2nd-last, volts = last) ---
    M = readmatrix(files{k});
    t = M(:,end-1); y = M(:,end);
    [t, idx] = sort(t); y = y(idx);
    t = t - t(1);
    y = y - mean(y); % remove DC

    % --- period / frequency ---
    if ~isempty(f_known)
        T = 1/f_known;
    else
        T = local_estimate_period(t, y, f_hint);
    end
    f_est = 1/T;

    % --- duty & skew ---
    [duty, skew] = duty_skew_robust(t, y);

    % --- Vpp (robust) ---
    Vpp = prctile(y, 99.5) - prctile(y, 0.5);

    % ----- PLOT EXACTLY Ncycles (x in cycles) -----
    cyc = (t - t(1)) * f_est; % cycles axis
    idxKeep = (cyc >= 0) & (cyc <= Ncycles); % first Ncycles only

    h = plot(cyc(idxKeep), y(idxKeep), ...
        'LineWidth', 1.4, 'Color', colors(k,:));
    xlabel('Cycles'); % cycles x-axis

    % Vpp bracket in cycles (left margin of the Ncycles window)
    y_lo = prctile(y(idxKeep), 0.5);
    y_hi = prctile(y(idxKeep), 99.5);
    x0 = 0.02*Ncycles; % 2% into the cycles window
    cap = 0.01*Ncycles;
    plot([x0 x0], [y_lo y_hi], '--', 'Color', colors(k,:), 'LineWidth', 2);
    plot([x0-cap x0+cap], [y_lo y_lo], '--', 'Color', colors(k,:), 'LineWidth',
2);
    plot([x0-cap x0+cap], [y_hi y_hi], '--', 'Color', colors(k,:), 'LineWidth',
2);
    text(x0 + 0.012*Ncycles, (y_lo+y_hi)/2, sprintf('Vpp=%.3g V', Vpp), ...

```

```

'Rotation',90,'VerticalAlignment','middle','FontSize',10,'Color',colors(k,:));

% accumulate legend + metrics only if a line actually plotted
if isgraphics(h,'line')
    hLines(end+1) = h; %#ok<AGROW>
    labels{end+1} = sprintf('%s | Vpp=%.1f V | f=%.3f kHz | duty=%.1f%% |
skew=%.2f', ...
                                files{k}, Vpp, f_est/1e3, 100*duty, skew);
%#ok<AGROW>
end
metrics(k).file = files{k};
metrics(k).Vpp = Vpp;
metrics(k).freq = f_est;
metrics(k).duty = duty;
metrics(k).skew = skew;
end

% lock the x-axis to exactly Ncycles (once, after all plots)
xlim([0 Ncycles]);
ylabel('Voltage (V)');

% ----- LEGEND (only valid handles) -----
valid = isgraphics(hLines,'line');
hLines = hLines(valid);
labels = labels(valid);
if ~isempty(hLines)
    legend(hLines, labels, 'Interpreter','none', 'Location','bestoutside');
else
    warning('No valid traces were plotted, legend suppressed.');
```

```

end

% ----- TITLE (show frequency, not period) -----
if ~isempty(f_known)
    title(sprintf('Oscilloscope traces – %d cycles at f = %.3f kHz', Ncycles,
f_known/1e3));
else
    % use the first non-NaN measured frequency
    f_first = NaN;
    for m = 1:numel(metrics)
        if isfield(metrics(m),'freq') && isfinite(metrics(m).freq) &&
metrics(m).freq>0
            f_first = metrics(m).freq; break
        end
    end
    if isfinite(f_first)
        title(sprintf('Oscilloscope traces – %d cycles at f ≈ %.3f kHz',
Ncycles, f_first/1e3));
    else
        title(sprintf('Oscilloscope traces – %d cycles', Ncycles));
    end
end

exportgraphics(gcf, sprintf('scope_overlay_%dcycles.png', Ncycles),
'Resolution', 300);

%% ===== LOCAL FUNCTIONS =====

function T = local_period_fft(t, y)
    dt = median(diff(t));
    y = y - mean(y);
    N = 2^nextpow2(numel(y));
```

```

Y = fft(y, N);
f = (0:N-1)/(N*dt);
[~,k] = max(abs(Y(2:floor(N/2)))); k = k+1;
f0 = f(k); if f0<=0, f0 = 1; end
T = 1/f0;
end

function [duty, skew] = duty_skew_robust(t, y)
% DUTY : fraction of time above mid-level per cycle (square should be ~0.10)
% SKEW : fraction of period spent on the RISING ramp (triangle/saw ~0.10 for
D=10%)
%
% Robust details:
% - Hysteresis around midlevel for duty (Schmitt trigger) to avoid chatter
% - Skew from derivative sign: Tup / (Tup + Tdown) per cycle

t = t(:); y = y(:) - mean(y(:), 'omitnan');
if numel(t) < 8, duty = NaN; skew = NaN; return; end

% ----- Duty via hysteretic crossings around midlevel -----
ylo = prctile(y,10); yhi = prctile(y,90);
mid = (ylo + yhi)/2;
hys = 0.05*(yhi - ylo); % 5% hysteresis band
lo = mid - hys; hi = mid + hys;

% rising when crossing lo->hi, falling when crossing hi->lo
r = find(y(1:end-1) < lo & y(2:end) >= hi);
f = find(y(1:end-1) > hi & y(2:end) <= lo);

if isempty(r) || isempty(f)
    duty = NaN; skew = NaN; return;
end

% sub-sample refine crossing times
tr = t(r) + (hi - y(r)).*(t(r+1)-t(r))./(y(r+1)-y(r));
tf = t(f) + (hi - y(f)).*(t(f+1)-t(f))./(y(f+1)-y(f));

% align as (rise, fall) pairs within the same cycle
if tf(1) < tr(1), tf(1) = []; end
np = min(numel(tr), numel(tf));
if np < 2, duty = NaN; skew = NaN; return; end

T = diff(tr(1:np)); % period(s) rise->rise
Th = tf(1:np-1) - tr(1:np-1); % high-time(s) within each period

duty = median( max(min(Th./T,1),0), 'omitnan' ); % square @10% => ~0.10

% ----- Skew via derivative sign (rise-time / period) -----
% smooth a bit to kill quantization noise % ~0.2% of record
win = max(3, round(0.002*numel(y)));
ys = movmean(y, win, 'Endpoints','shrink');

dy = diff(ys)./diff(t); % slope
sgn = zeros(size(dy));
sgn(dy > 0) = 1; sgn(dy < 0) = -1;

% edges where slope sign flips (rise->fall or fall->rise)
e = find(sgn(1:end-1) ~= sgn(2:end)) + 1;
if numel(e) < 3, skew = NaN; return; end

te = t([1; e; end]); % segment boundaries (time)
segDur = diff(te); % alternating rise/fall durations

```

```

% we need per-cycle Tup and Tdown: pair consecutive segments
% Start from whichever segment is 'rising' at the beginning:
risingFirst = sgn(1) > 0;
if risingFirst
    Tup = segDur(1:2:end);
    Tdown = segDur(2:2:end);
else
    Tdown = segDur(1:2:end);
    Tup = segDur(2:2:end);
end
n = min(numel(Tup), numel(Tdown));
if n < 2, skew = NaN; return; end

frac = Tup(1:n) ./ (Tup(1:n) + Tdown(1:n));
skew = median( max(min(frac,1),0), 'omitnan' );    % triangle @10% => ~0.10
end

function T = local_estimate_period(t, y, f_hint)
% Robust period estimator:
% 1) if f_hint is provided and finite, use 1/f_hint
% 2) else estimate via FFT peak on a uniform grid (parabolic refine)
% Falls back gracefully if data are short or irregular.

if nargin>=3 && ~isempty(f_hint) && isfinite(f_hint) && f_hint>0
    T = 1/f_hint;
    return
end

% sanitize inputs
t = t(:); y = y(:);
good = isfinite(t) & isfinite(y);
t = t(good); y = y(good);
if numel(t) < 8
    T = (t(end)-t(1)) / max(numel(t)-1,1);
    return
end
[t,ord] = sort(t); y = y(ord);
y = y - mean(y, 'omitnan');

% uniform resample
dt = median(diff(t));
if ~isfinite(dt) || dt<=0
    T = (t(end)-t(1)) / max(numel(t)-1,1);
    return
end
tu = t(1):dt:t(end);
if numel(tu) < 8
    T = (t(end)-t(1)) / max(numel(t)-1,1);
    return
end
yu = interp1(t, y, tu, 'linear', 'extrap');
yu = yu - mean(yu, 'omitnan');

% FFT with Hann window
L = numel(yu);
N = 2^nextpow2(L);
n = 0:L-1; w = 0.5 - 0.5*cos(2*pi*n/L); w = w(:).';
X = fft((yu(:).').*w, N);
mag = abs(X); mag(1) = 0;

% find fundamental bin (scalar)

```

```

kvec = 2:floor(N/2);
[~,kidx] = max(mag(kvec));
if isempty(kidx)
    T = (t(end)-t(1)) / max(numel(t)-1,1);
    return
end
k = kvec(kidx); % scalar index

% parabolic refinement (guard denominator)
if k>1 && k<floor(N/2)
    a = mag(k-1); b = mag(k); c = mag(k+1);
    denom = (a - 2*b + c);
    d = 0;
    if abs(denom) > eps
        d = 0.5*(a - c)/denom;
    end
    f0 = (k + d - 1) * (1/(N*dt));
else
    f0 = (k - 1) * (1/(N*dt));
end

if ~isfinite(f0) || f0<=0
    T = (t(end)-t(1)) / max(numel(t)-1,1);
else
    T = 1/f0;
end
end

```

Above is matlab code for plots from the oscilloscope